2016 Suite

OCR
Oxford Cambridge and RSA



Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 1

Fundamentals of IT

M/507/4999

Guided learning hours: 90

Version 4 February2017

*changes indicated by black vertical line

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LEVEL 3

UNIT 1: Fundamentals of IT

M/507/4999

Guided learning hours: 90

Essential resources required for this unit: none

This unit is externally assessed by an OCR set and marked examination.

UNIT AIM

A sound understanding of IT technologies and practices is essential for IT professionals. Information learnt in this unit will provide a solid foundation in the fundamentals of hardware, networks, software, the ethical use of computers and how business uses IT.

After completing this unit, the knowledge, skills and understanding you have developed will underpin your study for the additional units.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as CompTIA A+, CompTIA Mobility+ and Cisco IT Essentials.

Amendments made to this document will be indicated with a black line.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.
Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative.

For externally assessed units, where the content contains i.e. and e.g. under specific areas of content, the following rules will be adhered to when we set questions for an exam:

- a direct question may be asked about unit content which follows an i.e.
- where unit content is shown as an e.g. a direct question will not be asked about that example.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
1. Understand computer hardware	1.1 Computer hardware, i.e.: • input devices • output devices • communications devices • benefits (e.g. integrated devices make portable devices simpler to use) • limitations (e.g. voice recognition performs poorly in noisy environments) • uses (e.g. membrane keyboard could be used in harsh physical environments) 1.2 Computer components, i.e.: • processors • motherboards • storage (i.e. hard drive, solid state, flash, internal, removable, SAS, SCSI, portable, Cloud) • ports (i.e. USB, Firewire, SATA, Network, Fibre Channel) • memory (i.e. RAM, ROM, cache)	Learners should know about the different types of computer hardware required for a variety of computer systems. This should lead into learners developing their understanding of their benefits, limitations and uses. Learners should know about the component parts of a computer system and their characteristics. This should lead into learners developing their understanding of the purpose of each component.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• expansion cards (i.e. sound, network, graphics, storage controller, fibre channel) • power supplies • characteristics • purpose 1.3 Types of computer system, i.e.: • desktop/server • tablet/hybrid • smartphone • embedded system/Internet of Things (e.g. cars, home appliances, etc.) • mainframe • quantum • uses (e.g. tablet device can be used when travelling due to physical properties) • benefits (e.g. desktop computer can have a large screen which can improve productivity) • limitations (e.g. mainframes can be expensive to purchase and maintain) 1.4 Connectivity methods, i.e.: • copper • fibre • wireless technologies (i.e. Bluetooth, WiFi, microwave, infrared, laser, Satellite, GSM, 3G/4G and future technologies) • characteristics • purpose	Learners should know about the different types of computer systems. This should lead to an understanding of where and how they are used, benefits and limitations of each type of computer system and a justification of a suitable system in a given context. Learners should be aware that some devices feature embedded systems, such as modern cars, washing machines and other home appliances. Learners should know about different connectivity methods and their characteristics. This should lead to an understanding of their purpose and justification of different methods within a given context.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	1.5 Communications hardware, i.e.: • hub • switch • router • modem • wireless access point • combined/hybrid devices • characteristics • purpose and use 1.6 Hardware troubleshooting, i.e.: • identifying hardware faults • troubleshooting tools • documentation/fault management 1.7 Units of measurement, i.e.: • bit, nibble, byte • metric (i.e. kilo, mega, giga, tera, peta) • binary (i.e. kibi, mebi, gibi, tebi, pebi**)** • comparison in sizes between metric and binary measurements. e.g. 1 kilobyte = 1000 bytes vs 1024 bytes 1.8 Number systems, i.e.: • binary • decimal • hexadecimal 1.9 Number conversion, i.e.: • converting between binary, decimal and hexadecimal	Learners should know about different communications hardware and their characteristics. This should lead to an understanding of their purpose and use. Learners should be aware of the difference between a combined or hybrid device, which often provides the functionality of a modem, router, switch and wireless access point in one device, and the individual devices. Learners should know about the process needed to troubleshoot common hardware problems as well as the documentation involved. Learners should know about the units of measurement used in IT. This should lead to an understanding of how to convert between the different number systems.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
2. Understand computer software	2.1 Types of software, i.e.: • open source • closed source • off the shelf • bespoke • shareware • freeware • embedded • characteristics • use	Learners should know about different types of software and their characteristics. This should lead to an understanding of the use and justification of different types of software within a given context.
	2.2 Applications software, i.e.: • productivity software (i.e. word processor, spreadsheet, database, email) • development tools (i.e. compiler, debugger, translator, integrated design environment) • business software (i.e. MIS, multimedia, collaboration, project management, manufacturing, CAD/CAM, publishing, expert systems, healthcare)	Learners should know about the different applications software available. This should lead to an understanding of the purpose and advantages and disadvantages of each application.
	2.3 Utility software (i.e. backup, anti-virus, compression) • purpose • advantages and disadvantages	
	2.4 Operating systems, i.e.: • single user/multiuser • single processor/multiprocessor • off the shelf/open source/bespoke • Functions • Benefits and limitations	Learners should know about different forms of operating systems and their key functions. This should lead to an understanding of the benefits and limitations of operating system types.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	2.5 Communication methods, i.e.: • SMS • email • messaging software • social networking/social media • VoIP • personal assistants (e.g. Siri, Cortana) • teleconference • video conference • cellular/satellite • instant messaging • characteristics • purpose • advantages and disadvantages 2.6 Software troubleshooting, i.e.: • common faults (i.e. unexpected software behaviour, software freeze, unexpected rebooting) • troubleshooting tools to investigate a problem (i.e. logs, installable tools, baselines) • documentation (i.e. types of documentation) 2.7 Protocols, i.e.: • popular protocols o IP (Internet Protocol) o TCP (Transmission Control Protocol) o UDP (User Data Protocol) o SMTP (Simple Mail Transfer Protocol) o FTP (File Transfer Protocol) o HTTP (Hyper Text Transfer Protocol)	Learners should know about different types of communication methods used within a business context and their characteristics. This should lead to an understanding of the purpose, advantages and disadvantages of each method. Learners should know about common software faults, the tools used to investigate them and the documentation involved. This should lead to an understanding of why different trouble shooting tools are used in different contexts and a justification for their use. Learners should know about protocols and their features. This should lead to an understanding of the purpose of protocols and their common usage in given contexts.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	○ SNMP (Simple Network Management Protocol) ○ ICMP (Internet Control Message Protocol) ○ POP (Post Office Protocol) • features • purpose • common usage scenarios	Learners should be aware of the TCP/IP protocol stack including common functions in each layer.
3. Understand business IT systems	3.1 Types of servers, i.e.: • file/print • application • database • web • mail • hypervisor 3.2 Virtualisation, i.e.: • server • client • storage • cloud • hybrid • benefits and limitations 3.3 Networking characteristics, i.e.: • peer to peer • client server (i.e. DNS) • bus/star/ring/mesh • addressing (i.e. default gateway, IP address, subnet mask) • diagrammatical representation	Learners should know about different types of server. Learners should know about different forms of virtualisation. This should lead to an understanding of the benefits and limitations to a business of using virtualisation technology. Learners should know about different networking topologies and their characteristics. This should lead to an understanding of the use and justification of a particular topology in a given context.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• linking to given context 3.4 Connectivity methods, i.e.: • LAN (i.e. Ethernet, Token Ring) • WAN (i.e. ADSL, leased line, ISDN) • MAN • voice (i.e. PSTN, cellular) • satellite (i.e. voice, data) • characteristics • purpose 3.5 Business systems, i.e.: • MIS (Management Information System) • CRM (Customer Relationship Management) • SOP (Sales Ordering Process, Standard Operating Procedures) • helpdesk • purpose • benefits and limitations	Learners should know about different connectivity methods and their characteristics. This should lead to an understanding of the purpose of the different methods and how these would be used in a given context. Learners should know about different business systems. This should lead to an understanding of the purpose, and of the benefits and limitations of these systems in a given context.
4. Understand employability and communication skills used in an IT environment	4.1 Communication skills, i.e.: • interpersonal skills (i.e. eye contact, body language) • questioning techniques • verbal (i.e. meetings, telephone, group discussions) • written (i.e. reports, letters, emails, social networking) • non-verbal (i.e. body language) • barriers (i.e. language, distraction, noise, lack of concentration) • appropriate use of language (i.e. formal, informal,	Learners should know about the different communication skills used in the IT environment and the potential barriers involved. This should lead to an understanding of the different skills used for different audiences and situations.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	technical, non-technical) 4.2 Communication technology, i.e.: • presentation software • word processing • email • web • blogs/vlogs • instant messaging • use 4.3 Personal attributes (i.e. self-motivation, leadership, respect, dependability, punctuality, problem solving, determination, independence, time management, team working, written numerical and verbal skills, planning and organisation skills) 4.4 Ready for work, i.e.: • dress (i.e. appropriate clothing depending on situation) • presentation (i.e. personal grooming, appearance etc.) • attitude (i.e. can do attitude, responsive) 4.5 Job roles, i.e.: • Network manager • IT technician • Programmer • Web designer • Animator • Key skills required for each (i.e. technical and non-	Learners should know about different communication technologies available. This should lead to an understanding of the use and justification of different communication technologies in a given context. Learners should know about different personal attributes. This should lead to an understanding of why these attributes are important for certain job roles and valued by an employer. Learners should know about being ready for work. This should lead to an understanding of why this is important for themselves as well as the organisation. Learners should know about different job roles in the IT industry. This should lead to an understanding of the skills required.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	technical) 4.6 Professional bodies (e.g. BCS) • purpose • benefits and limitations 4.7 Industry certification • benefits to individual and employer • current vendors (e.g. CompTia ®, Cisco ®)	Learners should know about different professional bodies and industry certification. This should lead to an understanding of the purpose of professional bodies, and the benefits and limitations of membership to themselves and an employer. This should also lead to an understanding of why it is useful to gain industry certification and the benefits to themselves and an employer.
5. Understand ethical and operational issues and threats to computer systems	5.1 Ethical issues, i.e.: • whistle blowing • disability/gender/sexuality discrimination • use of information • codes of practice • staying safe online • bias 5.2 Operational issues, i.e.: • security of information • health and safety • disaster planning and recovery • organisational policies (i.e. acceptable use policy, code of conduct, etc.) • change management • scale of change: o drivers (i.e. change in business practice, legislation, competition)	Learners should know about different ethical and operational issues. This should lead to an understanding of how these issues can be addressed.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	o needs (i.e. improved networking, remote access for employees) 5.3 Threats, i.e.: • phishing • hacking • virus • Trojan • interception • eavesdropping • data theft • social engineering 5.4 Physical security, i.e.: • locks • biometrics • RFID • tokens • privacy screens • shredding • characteristics 5.5 Digital security, i.e.: • anti-virus • firewalls • anti-spyware • username/passwords • permissions • encryption • characteristics	Learners should know about different threats to computer systems. Learners should know about physical and digital security methods and their characteristics. This should lead to an understanding of why different security methods are used in different contexts and a justification for their use

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	5.6 Safe disposal of data and computer equipment, i.e.: • legislation • overwrite data • electromagnetic wipe • physical destruction	

LEARNING OUTCOME (LO) WEIGHTINGS

Each learning outcome in this unit has been given a percentage weighting. This reflects the size and demand of the content you need to cover and its contribution to the overall understanding of this unit. See table below:

LO1	10-20%
LO2	10-20%
LO3	25-35%
LO4	5-15%
LO5	10-20%

ASSESSMENT GUIDANCE

All LOs are assessed through externally set written examination papers, worth a maximum of 80 marks, and 1 hour 30 minutes in duration.

This unit provides the underpinning knowledge and understanding to support other units in the qualification. Learners should study the basics of computer hardware and software, and information systems used by businesses. It is important that learners understand what is required from IT specialists in the working environment, and the importance of good communication skills along with the technology available to them. The use of computer systems brings issues and it is therefore important that learners study the threats involved when using computer systems, and the ethical and operational issues that have to be considered to support their use.

Exam papers for this unit will include a short answer section and a section of extended questions. Questions will provide sufficient information to support the application and interpretation of the taught content of the unit. During the external assessment, learners will be expected to demonstrate their understanding through questions that require analysis and evaluation skills in particular contexts.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org
www.**cisco**.com/UK

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	LO4
Critical thinking	LO1, LO2, LO3, LO5
Decision making	LO1, LO2, LO3, LO5

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory content. This unit is a mandatory unit in all the specialist pathways.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience in an organisation with an IT function. The work experience should be structured to cover all or some of the learning outcomes in this unit.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	An IT Manager could give guest lecture how their business addresses ethical, operational issues and/or threats to their computer systems (LO4).

To find out more
ocr.org.uk/it
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 2

Global information



R/507/5000

Guided learning hours: 90

Version 2 - revised May 2016

***changes indicated by black vertical line**

LEVEL 3

UNIT 2: Global information

R/507/5000

Guided learning hours: 90

Essential resources required for this unit: none

This unit is externally assessed by an OCR set and marked examination.

UNIT AIM

The purpose of this unit is to demonstrate the uses of information in the public domain, globally, in the cloud and across the internet, by individuals and organisations. You will discover that good management of both data and information is essential, and that it can give any organisation a competitive edge.

This unit will provide you with a greater understanding of how organisations use information sources both internally and externally and the types of information you will encounter. The skills gained by completing this unit will give you knowledge of the functionality of information and how data is stored and processed by organisations. You will also learn about how individuals use information of various types.

This unit will help you to understand the legislation and regulation governing information that flows into and out of an organisation and the constraints and limitations that apply to it. You will also learn the relationship between data and information.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as VM Ware.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.
Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative.

For externally assessed units, where the content contains i.e. and e.g. under specific areas of content, the following rules will be adhered to when we set questions for an exam:

- a direct question may be asked about unit content which follows an i.e.
- where unit content is shown as an e.g. a direct question will not be asked about that example.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
1. Understand where information is held globally and how it is transmitted	1.1 Holders of information, i.e.: • categories of holders (individual citizens, businesses, educational institutions, governments, charities, healthcare services and community organisations) • location (e.g. developing country, developed country, urban, rural, home, workplace) • comparison of technologies available and access issues across the global divide (e.g. between developed and developing countries) 1.2 Types of information storage media, i.e.: • paper (e.g. forms, handwritten notes, maps, telephone directories) • optical media (e.g. CD and DVD) • magnetic media (e.g. magnetic hard drives and tapes) • solid state media (e.g. SSD hard drives, memory	Learners should know about different holders of information, their category and locations. This should lead to an understanding of the access issues to information across the global divide. Learners should know about different types of information storage media and devices and their characteristics. This should lead to an understanding of the use and advantages/disadvantages of both media and devices in a given context.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	cards) • characteristics • purpose • advantages and disadvantages 1.3 Types of information access and storage devices, i.e.: • handheld device (e.g. small tablet, smart phone, wearable device, eBook readers) • portable devices (e.g. laptop, large tablet) • fixed devices (e.g. desktop computer, smart TV, games consoles) • shared devices (e.g. database server, data centre, cloud storage devices) • characteristics • purpose • advantages and disadvantages 1.4 The internet, i.e.: • a network of interconnected networks, spanning the world • internet connections o type (e.g. copper-cable, optical-fibre, satellite, microwave, mobile data networks) o characteristics (e.g. speed, range (distance), storage capacity) 1.5 World Wide Web (www) technologies, i.e.: • types of networks that use www software: o internet (e.g. public, open access) o intranet (e.g. private, closed access) o extranet (e.g. private, part shared access)	Learners should know about what the internet is and the characteristics of internet connections. Learners should know about the different types of www network technologies and their characteristics. This should lead to an understanding of the purpose of the different technologies in a variety of contexts.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• comparison of networks (e.g. suitability for given uses, issues related to access to the network) • characteristics of networks • purpose of networks 1.6 Information formats, i.e.: • webpages (static and dynamic) • blogs • podcasts • streamed audio and video (e.g. internet radio, catch-up TV) • social media channels (e.g. Twitter, LinkedIn, discussion boards) • document stores (upload and download) • RSS feeds: o purpose o accessibility 1.7 Advantages: • for individuals (e.g. speed of personal communication, easy access to large amounts of information for research, access to internet banking 24/7) • for organisations (e.g. share large amounts of information quickly between different countries; charity websites accepting donations 24/7) 1.8 Disadvantages: • for individuals (e.g. potential for identity theft, cost of data connection) • for organisations (e.g. threats caused by malicious	Learners should know about different information formats used on the World Wide Web. This should lead to an understanding of the purpose of each format as well as a justification of the accessibility of each format to meet the needs of different holders of information. This should also lead to an understanding of the advantages and disadvantages of the World Wide Web to different holders of information.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	attacks, cost of maintaining websites and data stores)	
2. Understand the styles, classification and the management of global information	2.1 Information styles and their uses, i.e.: • text (different character sets, e.g. Western, Cyrillic, Arabic, etc.) • graphic (e.g. logo, photograph, diagram) • video (e.g. instructions on how to carry out a software update, live broadcast of a music festival) • animated graphic (e.g. pop-up book character, operation of the human heart) • audio (e.g. spoken instructions, music track) • numerical (e.g. profit, date and time) • Braille text (e.g. written report printed on a Braille printer) • tactile images (e.g. NASA's Hubble Space Telescope images converted into tactile images for people who cannot explore the images by sight) • subtitles (e.g. translated speech for a film in a foreign language) • boolean (e.g. yes or no answer on a form) • tables and spreadsheets (e.g. simple database tables and spreadsheets) • charts and graphs (e.g. identifying trends, making comparisons) 2.2 Information classification, i.e.: • sensitive • non-sensitive • private • public • personal	Learners should know about different information styles. This should lead to an understanding that different styles of information are used for different purposes. Learners should know about different classifications of information. This should lead to an understanding of the impact on holders of information of different types of information.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• business • confidential • classified • partially anonymised • completely anonymised • impacts on different stakeholders 2.3 Quality of information, i.e.: • characteristics (e.g. valid, bias, reliable, comparable) • importance of good quality information to stakeholders (e.g. innovation, agility, improved strategic decision making and focus) • consequences of poor quality information on stakeholders (e.g. misinformation, reputational damage) 2.4 Information management, i.e.: • collecting, storing and retrieving (e.g. adding a new member to a cycling club membership database) • manipulating and processing (e.g. producing a graph from a table of data) • analysing (e.g. looking for patterns in annual rainfall in an area) • securing (e.g. storing patient records on an encrypted hard drive) • transmitting (e.g. posting a printed school report to a parent) • impact on individuals and organisations (e.g. additional costs associated with keeping sensitive information secure)	Learners should know about the characteristics of information. This should lead to an understanding of why it is important holders of information have access to good quality information and the impacts when information quality is poor. Learners should know about the steps involved in the management of information. This should lead to an understanding of how the management of information impacts on holders of information in different ways.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
3. Understand the use of global information and the benefits to individuals and organisations	3.1 Data versus information, i.e.: • data-raw, unorganised facts that needs to be processed information-data which is processed, organised and structured into a meaningful context. 3.2 Categories of information used by individuals, i.e.: • communication (e.g. to send an email to a relation living overseas) • education and training (e.g. by a student to check their current grades on a hand written feedback sheet from their teacher) • entertainment (e.g. to read a film review in a magazine) • planning (e.g. to use a shared electronic diary to arrange meeting dates) • financial (e.g. to use a bank statement to help plan saving for a holiday) • research (e.g. to look up a recipe online) • location dependent (e.g. to search for emergency dental care when on holiday) • benefits and limitations 3.3 Categories of information used by organisations, i.e.: • knowledge management and creation (e.g. to create an accurate model of key markets) • management information systems (MIS) (e.g. to monitor staff training in a hospital; the location and contact details of each charity worker in a disaster area; personnel record of all staff) • marketing, promotion and sales (e.g. to identify patterns or trends in sales figures) • financial analysis and modelling (e.g. to determine	Learners need to know the difference between data and information. Learners need to know about categories of information and how they can be used by different holders of information. This should lead to an understanding of the benefits and limitations of the use of different categories of information by holders of information.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	the top selling products; cash flow each week over a year) • contact management (e.g. to keep track of appointments at a doctor's surgery) • decision making (e.g. to decide the number of tents to be sent to a disaster area by a charity; the percentage of faulty items made each month by a manufacturer) • internal and external communication (e.g. to inform all staff of office closures over the Christmas period) • big data, i.e.: o any data that is either too large or too complex for traditional data analysis techniques to be used, e.g. the annual web clicks of a major online retailer, health data on the population of an entire country • benefits and limitations 3.4 Stages of data analysis, i.e.: • identify the need (e.g. what information is needed? what do we want to find out?) • define scope (e.g. content, detail, timescales, constraints) • identify potential sources (e.g. sales figures, customer surveys) • source and select information (e.g. determine accuracy and reliability of sources, selecting the best) • select the most appropriate tools (e.g. charts, graphs, regression, trend analysis) • process and analyse data (e.g. set up a	Learners should know about the different stages of data analysis.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	spreadsheet to produce a graph) • record and store information (e.g. write a report based on the results of the processing) • share results (e.g. send the report to stakeholders) 3.5 Data analysis tools, i.e.: • data tables (e.g. a database table of patients) • visualisation of data (e.g. a pie chart showing sales of five leading trainers) • trend and pattern identification (e.g. a line graph of last year's sales per month) • data cleaning (e.g. removing customers who have not made a purchase in the last two years) • geographic information system/location mapping (e.g. tracking the movement of shipping containers around the world) 3.6 Information system structure, i.e.: • open systems • closed systems • characteristics • benefits and limitations	Learners should know about different analysis tools. This should lead to an understanding of the use and justification of different data analysis tools in a given context. Learners should know about the structure of information systems and their characteristics. This should lead to an understanding of the benefits and limitations of each system structure.
4. Understand the legal and regulatory framework governing the storage and use of global information	4.1 UK legislation and regulation relating to the storage and use of information, i.e.: • current UK legislation and regulation: o Data Protection Act (DPA) 1998 o Regulation of Investigatory Powers Act (RIPA) 2000 o Protection of Freedoms Act 2012 o Privacy and Electronic Communications	Learners should know about the different legislation and regulation that relates to the storage and use of information. This should lead to an understanding of the impact and consequences of legislation and regulation on holders of information.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	Regulations 2003 (amended 2011) o Freedom of Information Act 2000 o Computer Misuse Act 1990 o Information Commissioner's Office (ICO) codes of practice o Copyright, Designs and Patents Act 1988 o Equality Act (EQA) 2011 • impact and consequences of UK legislation and regulation on organisations operating in the UK and the way they handle information and individuals' personal data • actions that can be taken by organisations to comply with legislation and regulatory requirements 4.2 Global information protection legislation and regulation, i.e.: • regulation relating to data protection outside the UK (e.g. USA, France, Far East and Africa) • comparison between data protection legislation and regulation in different countries (e.g. similar legislation in many countries, but not all) • UN Convention on the Rights of Persons with Disabilities (UNCRPD): o (e.g. the UNCRPD specifically recognises (under articles 9 and 21) that access to information, communications and services, including the internet, is a human right) 4.3 Green IT, i.e.: • global requirements on organisations and individuals • United Nations Climate Change Summits	This should also lead to an understanding of the actions holders of information can take to comply with legal and regulatory requirements. It is important that learners are aware of the most recent legislation when studying this unit. Learners should know about the global requirements for Green IT.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• UK Government policy (e.g. Greening Government ICT Strategy (2011)) • reducing carbon footprint • purpose (e.g. sustainability) • benefits (e.g. enhanced brand image, reduced energy costs)	This should lead to an understanding of the rationale for Green IT and the global benefits.
5. Understand the process flow of information	5.1 Information sources and data types, i.e.: • internal source (e.g. internal financial reports, market analysis) • external source (e.g. supplier price lists, financial report from a third party) • primary data (e.g. reports direct from employees, foot measurements taken in a shoe shop) • secondary data (e.g. survey results received from a market research organisation, interest rate charged on a loan from a bank) • qualitative data (e.g. the colour of products, the names of employees) • quantitative data (e.g. expiry date of medicines, the number of staff working in an organisation) • purpose 5.2 Data flow diagrams (DFDs), i.e.: • external entities • processes • data stores • data flows • standard symbols used • connectivity rules for drawing Level 1 DFDs o at least one input or output for each external	Learners should know about different information sources and data types. This should lead to an understanding of the use and justification of information sources and data types within a given context. Learners should know about Level 0 and 1 DFDs. This should lead to an understanding of the components of a DFD and how they are used to show the flow of information. This should also lead to learners having an understanding of the impacts affecting the flow of information in information systems.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	entity o data flows only in one direction o every data flow is labelled o every data flow connects to at least one process o at least one input data flow and/or at least one output data flow for each process • impacts affecting the flow of information in information systems	Learners will not be expected to draw DFDs for this unit.
6. Understand the principles of information security	6.1 Principles of information security, i.e.: • confidentiality – information can only be accessed by individuals, groups or processes authorised to do so • integrity – information is maintained, so that it is up to date, accurate, complete and fit for purpose • availability – information is always available to and usable by the individuals, groups or processes that need to use it 6.2 Risks, i.e.: • unauthorised or unintended access to data (e.g. espionage, poor information security policy) • accidental loss of data (e.g. human error, equipment failure) • intentional destruction of data (e.g. computer virus, targeted malicious attack) • intentional tampering with data (e.g. fraudulent activity, hacking) 6.3 Impacts,, i.e.: • loss of intellectual property	Learners should know about the aims of information security for holders of information. This should lead to an understanding of the risks of breaches in information security and their impact on holders of information.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• loss of service and access • failure in security of confidential information • loss of information belonging to a third party • loss of reputation • threat to national security • recent cases of failures of information security 6.4 Protection measures, i.e.: • Policies, e.g.: o staff access rights to information o responsibilities of staff for security of information o disaster recovery o information security risk assessment o effectiveness of protection measures o training of staff to handle information 6.5 Physical protection, i.e.: • locks, keypads and biometrics used on: o workstations o server room access • placing computers above known flood levels • backup systems in other locations • security staff • shredding old paper based records 6.6 Logical protection, i.e.: • tiered levels of access to data • firewalls (hardware and software) • anti-malware applications • obfuscation	Learners should know about the different protection measures to mitigate breaches in information security. This should lead to an understanding and justification of different measures that can be used in a given context.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• encryption of data at rest • encryption of data in transit • password protection	

LEARNING OUTCOME (LO) WEIGHTINGS

Each learning outcome in this unit has been given a percentage weighting. This reflects the size and demand of the content you need to cover and its contribution to the overall understanding of this unit. See table below:

LO1	10-20%
LO2	25-35 %
LO3	10-20%
LO4	15-25%
LO5	0-10%
LO 6	10-20

ASSESSMENT GUIDANCE

All LOs are assessed through externally set written examination papers, worth a maximum of 80 marks, and 1 hour 30 minutes in duration.

Learners should study where information is held in a global context, including the different styles and classification, as well as how it is managed and transmitted. With any gathering, processing, storing and transmitting of information, legislation and regulation considerations need to be addressed. It is important, therefore, that learners study the relevant legislation and regulation not only within the UK but globally, and assess its impact on how organisations and individuals handle information. Learners study the process of how information flows internally and externally within and between organisations and individuals as well as how the information is used and the benefits it provides.

Exam papers for this unit will include a pre-released case study which will be the focus for the questions for Part A of the paper, while Part B will consist of questions of a more general nature. Questions will provide sufficient information to support the application and interpretation of the taught content of the unit. During the external assessment, learners will be expected to demonstrate their understanding through questions that require the skills of analysis and evaluation in particular contexts.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**vmware**.com/uk

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	LO2, LO3, LO5
Critical thinking	LO1, LO2, LO3, LO4, LO5, LO6
Decision making	LO2, LO3, LO4, LO5, LO6

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory content. This unit is a mandatory unit in all the specialist pathways.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could carry out research about aspects of the teaching content when on work-placements.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Industrial practitioners could present lectures or master classes on areas such as information security and the legal frameworks.

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 3

Cyber security

Cambridge TECHNICALS 2016

Y/507/5001

Guided learning hours: 60

Version 3 - revised September 2016



ocr.org.uk/it

LEVEL 3

UNIT 3: Cyber security

Y/507/5001

Guided learning hours: 60

Essential resources required for this unit: none

This unit is externally assessed by an OCR set and marked examination.

UNIT AIM

The need for secure digital systems is more crucial than ever before. We rely on computerised systems and networks to collect, process, store and transfer vast amounts of data and to control critical systems such as water and power supplies. Business and e-commerce can be undertaken twenty four hours a day, seven days a week and telecommunications enable us to keep in touch with family and friends and collaborate with colleagues at any time. Mobile devices offer us freedom and flexibility of where and how we learn and work. However, for all the advantages that these systems offer us, some people have found ways to exploit them and this poses a threat to our safety and security in the real world, as much as in the cyber world. To deal with this problem the cyber security industry is expanding at a rapid rate.

This unit has been designed to enable you to gain knowledge and understanding of the range of threats, vulnerabilities and risks that impact on both individuals and organisations. You will learn about the solutions that can be used to prevent or deal with cyber security incidents resulting from these challenges. You will be able to apply your knowledge and understanding of cyber security issues and solutions by reviewing and making recommendations for ways to best protect digital systems and information.

Learning within this unit will also support the delivery of the Cisco Cyber Security and CompTIA A+, CompTIA Security+, CompTIA Mobility+ qualifications. The unit also makes reference to UK government cyber security initiatives, for example, the UK government's The UK Cyber Security Strategy, Cyber Essentials Scheme, 10 Steps Strategy, and Cyber Streetwise.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.
Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative.

For externally assessed units, where the content contains i.e. and e.g. under specific areas of content, the following rules will be adhered to when we set questions for an exam:

- a direct question may be asked about unit content which follows an i.e.
- where unit content is shown as an e.g. a direct question will not be asked about that example.

Learners are expected to keep up-to-date with the latest developments, innovations and new approaches in cyber security when acquiring knowledge and understanding of this unit content.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
1. Understand what is meant by cyber security	1.1 Cyber security aims to protect information, i.e.: • confidentiality • integrity • availability 1.2 Types of cyber security incidents, i.e.: • unauthorised access including hacking, escalation of privileges • information disclosure including personal information, government information • modification of data • inaccessible data including account lockout, denial of service • destruction including using malware, deliberate erasure • theft including identity, finance, military secrets	Learners should know what is meant by the term cyber security. They should know about digital systems and understand why the information stored on them needs to be kept secure at all times. Leaners should know about the types and nature of cyber security incidents that affect individuals, states and organisations.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	1.3 The importance of cyber security, i.e.: • the need to protect personal data (e.g. health, financial, national insurance) • the need to protect an organisation's data (e.g. financial, research, development plans) • the need to protect a state's data (e.g. economic data, national security)	
2. Understand the issues surrounding cyber security	2.1 Threats to cyber security, i.e. • vulnerabilities o system attacks o physical threats o environmental • accidental • intentional • organised crime • state sponsored 2.2 Types of attackers, i.e.: • hacktivist • cyber-criminal • insider • script kiddie • vulnerability broker • scammers • phishers • cyber-terrorists • characteristics including age, location, social group 2.3 Motivation for attackers, i.e.: • espionage • righting perceived wrongs	Learners should know about the wide range of threats to cyber security including those threats that are accidental or intentional. Learners should know about the types of attacker, their characteristics and their motivations.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• publicity • fraud • score settling • public good • thrill • income generation 2.4 Targets for cyber security threats, i.e.: • people • organisations • equipment • information • methods that can be used during an attack 2.5 Impacts of cyber security incidents, i.e.: • global problem, individuals, organisations and states • loss including confidentiality, integrity, availability, data, finance, business, identity, reputation, customer confidence • disruption including people's lives, business, industry, transport, industry, the media, utilities • safety including identity theft, oil installations, traffic control 2.6 Other considerations of cyber security, i.e.: • ethical • legal • operational • implications for stakeholders	Learner should know about the different targets for cyber security threats and how these threats might manifest themselves. This should lead to an understanding of the possible impacts from cyber security incidents and how these affect different stakeholders in a variety of different ways. Learners should know about other cyber security considerations. This should lead to an understanding of the implications for different stakeholders in this wider context. Learners should be aware of the latest or most up-to-

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
		date versions of legislation
3. Understand measures used to protect against cyber security incidents	3.1 Cyber security risk management, i.e.: • identify assets and analyse risks • mitigate risks by: o testing for potential vulnerabilities • monitoring and controlling systems • protect vulnerabilities • cost/benefit 3.2 Testing and monitoring measures, i.e.: • vulnerability testing including penetration testing, fuzzing, security functionality, sandboxing • intrusion detection systems (IDS) including network intrusion detection systems (NIDS), host intrusion detection systems (HIDS), distributed intrusion detection system (DIDS), anomaly-based, signature-based, honeypots • intrusion prevention systems (IPS) • emerging technologies • effectiveness 3.3 Cyber security controls (access controls), i.e.: • physical including biometric access, swipe cards, alarms • hardware including cable locks, safes • software including firewalls, anti-malware, operating system updates, patch management • data including in use, at rest, in-transit, in the cloud • encryption including disks, databases, files, removable media, mobile devices • cryptography	Learners should know about the various measures that should be taken to manage cyber security. This should lead to an understanding of, and justification for, different measures that can be taken in a given context. Learners should know about different testing and monitoring measures that can be used to test for vulnerabilities. This should lead to an understanding and justification of the effectiveness of different measures in a given context. Learners should know about the different security controls and their characteristics. This should lead to an understanding and justification of the effectiveness of different controls in a given context.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• devices including. hard drives, external drives, USBs • procedures including access management, data backup, remote working, device management, user accounts and permissions, awareness and training • emerging technologies • characteristics	
4. Understand how to manage cyber security incidents.	4.1 Responding to an incident, i.e.: • know responsibilities • know who to contact • know procedures • know the extent of the incident • contain the incident • eradicate the incident • reduce the impact of the incident • recover from the incident • confirm the system is functioning normally 4.2 Cyber security incident report, i.e.: • incident title and date of incident • target of the incident • incident category, i.e.: o critical o significant o minor o negligible • description of the incident • type of attacker(s) • purpose of incident • techniques used by the attacker(s) • capability of attacker(s)	Learners should know about different procedures that should be followed in the event of a cyber security incident. This may include conducting investigations or being subject to an investigation. This should lead into an understanding and justification of why certain procedures should be taken in a given context. Learners should know the various stages of investigation that should be undertaken should a cyber security incident occur. This should lead to an understanding of, and justification for decisions that must be taken in a given context. It is possible learners will be asked to complete sections of a cyber security report as part of the examination for this unit.

Learning outcomes The Learner will:	Teaching content Learners must be taught:	Exemplification
	• impact of the incident on business, data, recovery time • cost of the incident • responses needed • future management o review (of incident) o evaluation to include identification of trends o update of documentation, key information, procedures and controls o recommendations of changes	

LEARNING OUTCOME (LO) WEIGHTINGS

Each learning outcome in this unit has been given a percentage weighting. This reflects the size and demand of the content you need to cover and its contribution to the overall understanding of this unit. See table below:

LO1	5-15%
LO2	35-45%
LO3	20-30%
LO4	10-20%

ASSESSMENT GUIDANCE

All LOs are assessed through externally set written examination papers, worth a maximum of 60, marks and 1 hour in duration.

Learners should study the meaning of cyber security and gain an understanding of its overall purpose. They should study the wide variety of issues surrounding cyber security and the measures that are used to protect against cyber security incidents. Breaches in cyber security can cause serious issues to individuals and organisations and, therefore, learners should have a good understanding of how to manage cyber security incidents.

Exam papers for this unit will include a pre-released case study. The paper will include questions associated with the pre-released case study as well as questions to demonstrate a more general understanding of the subject. Questions will provide sufficient information to support the application and interpretation of the taught content of the unit. During the external assessment, learners will be expected to demonstrate their understanding through questions that require the skills of analysis and evaluation in particular contexts.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org
www.**cisco**.com/UK

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	LO4
Critical thinking	LO1, LO2, LO3, LO4
Decision making	LO1, LO2, LO3, LO4

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory content. This unit is a mandatory unit in all specialist pathways in the Level 3 Cambridge Technical Diploma in IT (720 GLH) and the Level 3 Cambridge Technical Extended Diploma in IT (1080 GLH).

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	As part of a learners work experience they could find out what procedures the business has in place to manage cyber security incidents and how the business protects itself against cyber security incidents (LO3/LO4)
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	An Industry Practitioner could be used to present a guest lecture on how they manage cyber security in their company (LO3).

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 4

Computer networks

H/507/5003

Guided learning hours: 60

Version 1 September 2015

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LEVEL 3

UNIT 4: Computer networks

H/507/5003

Guided learning hours: 60

Essential resources required for this unit: Learners should have access to a private network that they will be able to work on without restrictions. They must be able to modify the infrastructure, install, update and remove software and configure the components. The composition of the network, as well as available resources and tools, should be sufficient to ensure that learners are able to achieve the highest grades for this unit.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Computer networks form a key part of the information economy. They are the foundation of the World Wide Web on which eBay, Amazon, Facebook and a multitude of other companies depend for their success. The demand for networking capability is enormous and increasing daily. The business world demands network administrators, engineers and technicians who can set up, manage and maintain its networks.

The emphasis of this unit is to give you the practical ability to plan, implement and maintain computer networks. The approach adopted by this unit is 'bottom up' where you begin with a solid set of components, cables and connectors of a network and then progressively build a networking capability. The range of protocols has been deliberately limited to those which are used in the vast majority of computer networks, TCP/IP and Ethernet.

The teaching content is designed to support networking qualifications for certifications such as Microsoft, Cisco and CompTIA Network+.

This unit is mandatory in the IT Infrastructure Technician specialist pathway in the Level 3 Diploma suite of qualifications due to its relevance in an IT technical environment. The unit supports the development of skills, knowledge and understanding relevant to a technical support or network technician job role.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the concept of networks	1.1 Network interfaces 1.2 Network types, i.e.: • local area network (LAN) • wireless local area network (WLAN) • wide area network (WAN) 1.3 Network components, i.e.: • servers • workstations • other hosts (e.g. printer, scanner) • wireless access point • hubs • routers 1.4 Network topologies (layout) (e.g. bus, wireless, segments, backbones) 1.5 Network protocols, i.e.: • Ethernet • IP • TCP • UDP • FTP • SMTP 1.6 Networking models, i.e.: • OSI 7 layer model • TCP/IP model 1.7 IP versions (e.g. IPv4, IPv6)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	1.8 Network addressing, i.e.: • MAC Addressing • APIPA • Class addressing • Classless addressing • Private addressing • Loopback addressing • TCP ports • Default gateway 1.9 Network data units, i.e.: • Ethernet frames • IP packets • TCP packets 1.10 Network security, i.e.: • threats (e.g. hacking, virus, spyware, theft) • security systems (e.g. firewall, routers, WEP) • risk assessment 1.11 Network virtualisation, i.e.: • concept overview • use for system testing • examples e.g. VMware, VirtualBox
2. Be able to plan computer networks to meet client requirements	2.1 Network configuration, i.e.: • network components (e.g. servers, workstations, routers) • network operating systems (e.g. Windows, Linux) 2.2 Network services, i.e.: • DHCP • routing • internet connection sharing • Wi-Fi access • other services (e.g. printing, email, web, DNS) 2.3 IP configuration, i.e.: • manual • automatic • routing • masking 2.4 Testing tools, i.e.: • ping • ipconfig • pathping • tracert • route

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	2.5 Network specification, i.e.: • stakeholder requirements i.e.: client, network user • applications • services • constraints (e.g. cost, environment, time) • security o risk assessment o Wi-Fi security o network security (e.g. firewall, MAC filtering) • purpose (e.g. file sharing, internet access, network printing) 2.6 Network plan, i.e.: • user requirements • components (e.g. servers, workstations, routers) • services (e.g. DHCP, printing, email) • software (e.g. operating system, firewall) • configuration requirements • security plan • test plans: o configuration test plan o user acceptance testing plan: ▪ purpose ▪ plan a test ▪ design a test ▪ execute a test ▪ defect log ▪ plan update
3. Be able to present network solutions to clients	3.1 Solution proposal, i.e.: • client requirements • physical design o devices and technologies • logical design: o topology o protocols o addressing (e.g. public, private) o security o configuration • network design test plan • network map (e.g. servers, workstations, routers) • installation schedule • component configurations • configuration test plan • user acceptance test plan 3.2 Performance benchmarking tools (e.g. LANBench, NetIO-GUI, NetStress)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
4. Be able to plan maintenance activities for computer networks	4.1 Maintenance plan (e.g. network map, procedures, policies, security permissions) 4.2 Maintenance issues, i.e.: • virus: o sources (e.g. email, hacking, software install) o treatment (e.g. virus protection, firewall, security update) • hardware: o faulty hardware o adding new hardware (e.g. workstation) o replacing hardware (e.g. network server, switch) • software: o update software o add software • slow speed • security: o security update o security breach o security settings (e.g. router, firewall) 4.3 Troubleshooting, i.e.: • diagnostic tools: o hardware (e.g. cable tester, protocol analyser) o software (e.g. event logs, microsoft message analyzer) o ip utilities (e.g. ping, ipconfig, tracert, loopback) • diagnose the problem (e.g. gather symptoms) o hardware (e.g. slow, intermittent) o software (e.g. unexpected quit, load fail) o data transmission (e.g. congestion, fail connection) • determine cause • document the process 4.4 Disaster recovery (e.g. backup, restore) 4.5 Software/hardware updates, i.e.: • security updates • performance updates • risks (e.g. application failure) • minimise downtime • configuration changes 4.6 Performance management (benchmarking), i.e.: • establish a baseline • performance measurements (e.g. response time, workload)

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the concept of networks	P1: Explain how network addressing is used	M1: Compare and contrast the OSI and TCP/IP networking models	D1: Discuss the role of TCP/IP in networks
	P2*: Explain security considerations for computer networks *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		
2. Be able to plan computer networks to meet client requirements	P3: Create a network specification to meet an identified client requirement	M2: Justify security measures for inclusion in an identified network solution	
	P4: Produce planning documentation for the implementation of an identified network solution		
3. Be able to present network solutions to clients	P5: Communicate the network solution to the identified client	M3: Recommend performance tools to benchmark network solution	
4. Be able to plan maintenance activities for computer networks	P6: Create a maintenance plan for the network solution		D2: Evaluate the selection of maintenance activities for the network solution

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the concept of networks

P1: Learners are required to explain the network addressing and how it is used. The explanations must include how the IP address identifies a specific network, how to determine different hosts and networks that can be addressed and the function of the MAC address in transporting data between two IP hosts. The evidence could be presented as a report or a technical guide and include examples of network types.
M1: Learners are required to compare and contrast the OSI and TCP/IP networking models. The comparisons must include the purpose as well as the similarities and differences between the two models. The comparison should identify each of the listed network protocols with each model and explain the role each protocol plays. The evidence could be in the form a technical guide or detailed report.
D1: Learners are required to discuss the role of TCP/IP in networks for different topologies as listed in the teaching content. Learners should discuss how TCP/IP routes data between the listed network components and across different types of network. The discussions will also include the purpose of TCP ports and the use of subnet masking. They should also consider the role played by virtual networks when working with TCP/IP. The evidence could be in the form of a report or a technical guide supported by illustrated examples showing connectivity.
P2: Learners are required to explain security mechanisms in computer networks. They must explain different threats to the security of a network and then explain security systems that can be used to minimise the threat. A risk assessment could be included in a report and the solution presented as a technical guide.

LO2 Be able to plan computer networks to meet client requirements

P3: Learners are required to create a network specification to meet an identified client's requirement. They should be given a detailed scenario set of requirements that gives them the opportunity to use different components and topologies. Learners will use the scenario to create a network specification. The specification may have to be modified iteratively until it meets the client's requirements. The evidence could be in the form of a report that will indicate client agreement.
M2: Learners are required to justify security measures for inclusion in an identified network solution. This criterion extends the specification of P3. They will carry out a security risk assessment for the given scenario. The evidence could be a clearly marked addendum to the network specification of P3.
P4: Learners are required to produce planning documentation for the implementation of an identified network solution. They will use the network specification to create a complete network plan. The evidence will be the planning documentation which will include any modifications required by the client.

LO3 Be able to present network solutions to clients

P5: Learners will be able to propose the computer network solution to the client. They will use the network plan to prepare a visualisation of the network. The evidence will be the network plans and visualisation and may be supported by a videoed presentation of learners presenting the solution.
M3: Learners are required to recommend performance tools to benchmark their network solution. The recommendation should give an explanation of what these tools do. The evidence could be in the form of a report or a technical guide.

LO4 Be able to plan maintenance activities for computer networks

P6: Learners are required to create a maintenance plan for an identified computer solution. The maintenance plan should relate to the proposed solution from P5. The evidence will be the actual maintenance plan, which should consider issues as listed in the teaching content.
D2: Learners are required to evaluate the selection of maintenance activities identified for the chosen computer solution in P6. They should review the maintenance plan created in P6 and evaluate how the activities will support the ongoing functionality of the network.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org
www.**cisco**.com/UK
https://www.**microsoft**.com/learning/en-gb/default.aspx

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, P4, P5, P6, M1, M3, D1,
Problem solving	P3, P4, P6, M2, M3, D2
Time management	P3, P4, P5, P6, M2, M3
Critical thinking	D1, M2, D2
Negotiation	P5, M3
Decision making	P3, M2, M3, P6, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units. This unit is a mandatory unit in the IT Infrastructure Technician pathway and in the Digital Technician pathway.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Work placements in IT departments supporting the network manager to see how some or all of the learning outcomes in this unit in a work-based context.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Learners could be provided with a scenario based on an employer requirements where they are required to interview the client and develop a network solution for them.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A network manager could be asked to fulfil certain activities that lead to the securing of criteria e.g. how to establish client requirements for a network solution.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	This could be linked to the work-placement opportunity highlighted above.

To find out more
ocr.org.uk/it
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 5

Virtual and augmented reality



K/507/5004

Guided learning hours: 60

Version 2 - revised May 2016

*changes indicated by black vertical line

ocr.org.uk/it

LEVEL 3

UNIT 5: Virtual and augmented reality

K/507/5004

Guided learning hours: 60

Essential resources required for this unit: Learners need to be given access to the software and tools needed to create both virtual and augmented reality products. In order to assess examples of virtual and augmented reality products, learners will need access to completed products and the designs that have led to their creation.

If a virtual reality headset e.g. Oculus Rift is provided or used to create, develop and test learner's work, then the Centre must be fully aware of the issues connected with its use. The Centre is advised to undertake its own research into the health and safety issues. A starting point is offered here:
http://static.oculus.com/documents/gear-vr-health-and-safety-warnings.pdf
http://www.livescience.com/49669-virtual-reality-health-effects.html

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Virtual reality is a simulated environment that is intended to replicate the physical experience of being in places in the real or imagined worlds by giving the user sensory experiences that match those which would be experienced were the user actually in that environment. Augmented reality is the process of changing the user's view of the real world in order to give them an improved, or more detailed, view of what they are seeing.

You will learn about both technologies and how they are used. You will research both technologies and design both a virtual and an augmented reality resource. Finally, you will use your research and skills learnt whilst designing and creating resources to suggest future applications for virtual and augmented reality.

This unit is mandatory to the emerging digital technology practitioner specialist pathway in the Level 3 Diploma suite of qualifications due to its relevance to emerging digital technologies. The unit supports the development of skills, knowledge and understanding relevant to a job role in the areas of 3D modelling, digital transformation and even the film and games industry.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand virtual and augmented reality and how they may be used	1.1. Virtual reality as a concept i.e.: • pioneers of virtual and augmented reality e.g.: o Douglas Engelbart o Ivan Sutherland o Tom Caudell and David Mizell • uses of virtual and augmented reality e.g.: o US Military Nuclear Defence systems o pilot training o Mattel “data glove” o personal guidance system for visually impaired o chameleon 1.2. Areas of use, e.g.: • architecture • business (marketing, service and planned maintenance) • education (e.g. textbooks, skills development, remote collaboration) • entertainment, leisure and the media (tourism, games, museums) • health care and surgery (training, simulations) • military (training, simulations) • sport (live streaming of scores and other statistics, sponsorship images) 1.3. Possible impacts, i.e.: • visualisation of designs • simulations • training • demonstrations of concepts • virtual tours

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to design virtual and augmented reality resources	2.1. Technologies, i.e.: • hardware o processor o display (e.g. handheld device, head mounted display, eyeglasses, head up display) o sound (e.g. speakers, headphones) o sensors (e.g. optical, accelerometer, GPS, compass, RFID) o input devices (e.g. camera, microphone) • software o range of products available o features of the software o image registration o Augmented Reality Mark-up Language 2.2. Design, i.e.: • aims of the product (e.g. intended outcome, success criteria, information to be delivered, where the product is to be used) • financial plan • quality plan • target audience (e.g. age, gender, income) • nature of the product (e.g. single user product, multiuser product) • content including resource plan • design tools (e.g. storyboarding, mind mapping, mood boards) • trigger image(s) and the stage(s) that follow on from the trigger being accessed. • hardware and software requirements
3. Be able to create a virtual **or** augmented reality resource	3.1. Develop, i.e.: • create the trigger point of interest • create the layer(s)/overlay(s) 3.2. Testing, i.e.: • create a test plan • testing during development • end user testing • review against original success criteria 3.3. Evaluation, i.e.: • design stage: o has the project: ▪ identified suitable success criteria • project management stage: o has the project: ▪ deviated from the original scope ▪ deviated from the budget as defined in the financial plan

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• creation stage: o has the project: ▪ delivered business benefits identified in the business case ▪ achieved the objectives in the terms of reference ▪ deviated from forecast resource levels as per the resource plan ▪ conformed to the management process as per the execution phase • identify potential improvements for similar future projects
4. Be able to predict future applications for virtual and augmented reality	4.1. Future uses, i.e.: • possible developments of virtual and augmented reality and how these may impact on society. (e.g. advances in treating injuries or disease, leisure activities, the environment, the home and education). 4.2. Re-purposing, i.e.: • how existing products may be re-purposed and used in wholly new ways • benefits of repurposing using current examples of resources in new ways **(**e.g. medical uses in the field of animal welfare, training uses in the field of education) • heads up display used to augmented learning in schools

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand virtual and augmented reality and how they may be used	P1*: Describe the uses of virtual and augmented reality by organisations *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M1: Explain the impact that an identified virtual reality resource has had on society	D1: Assess the impact that an identified augmented reality resource has had on society
2. Be able to design virtual and augmented reality resources	P2: Produce a design specification for a virtual reality resource for an identified purpose		
	P3: Produce a design specification for an augmented reality resource for an identified purpose		
3. Be able to create a virtual or augmented reality resource	P4: Develop a virtual reality or an augmented reality resource for an identified purpose		
	P5: Test the product during creation and once complete	M2: Make adjustments to the design based on outcomes of testing	D2: Evaluate the development stages during the creation of the resource
4. Be able to predict future applications for virtual and augmented reality	P6: Suggest possible future roles of virtual and augmented reality in future applications	M3: Evaluate the specific benefits to be gained by repurposing current examples of virtual and augmented reality into identified roles	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand virtual and augmented reality and how they may be used.

P1: Learners are required to describe the use of virtual and augmented reality by organisations. Learners should discuss at a wide range of uses and include examples of where and how they are used. The learner must ensure that they clearly identify whether they are describing virtual or augmented reality applications and their uses. The evidence could be presented as a report, part of a technical guide or a presentation (either videoed or with detailed speaker notes).
M1: Learners are required to select one example of a virtual reality resource and explain the impact that the use of the identified technology has had on society. The evidence could be in the form of a presentation (either videoed or with detailed speaker notes), report, or information sheet.
D1: Learners are required to select one example of an augmented reality resource and assess the impact that the use of the identified technology has had on society. The evidence could be in the form of a presentation (either videoed or with detailed speaker notes), report or information sheet.

LO2 Be able to design virtual and augmented reality resources.

It is important that learners are provided with a scenario that will enable them design a virtual and an augmented reality resource.
P2: Learners are required to produce a design specification for an identified purpose for a virtual reality resource. The design specification must include the items to be created in LO3. This evidence should be the design specification which includes the actual designs, e.g. storyboard, mood board etc.
P3: Learners are required to produce a design specification for an identified purpose for an augmented reality resource. The design specification must include the items to be created in LO3. This evidence should be the design specification which includes the actual designs, e.g. storyboard, mood board etc.

LO3 Be able to create a virtual or augmented reality resource.

Learners need to select and develop **one** of the design specifications from LO2.

P4: Learners are required to develop either a virtual reality resource **or** an augmented reality resource for an identified purpose. The evidence for this assessment criterion should be the completed product, along with evidence that the product works as intended. This evidence for functionality of the product could be a witness statement or video evidence of the completed product in use.
P5: Learners must test the product **both** during the creation **and** once it has been created. The evidence will include the test plan, confirming the testing of the functionality during creation and at the end of the final testing stage.
M2: Learners must review the results of the testing and make adjustments to their design. The expectation is that in any development and testing process there are always aspects of the design that need further consideration. This is why it is important that iterative testing is carried out. The evidence will be the test results and supporting documentation of any changes made to the design.
D2: Learners are required to evaluate the development stages of the creation process. The evaluation must consider the design, project management and creation stages and include the bullet points for each section within the teaching content from the section headed **"Evaluation"**. The evidence will be the documented evaluation. This could be in the form of a report or a presentation with detailed speaker notes or a video of learners presenting the results of their evaluation.

LO4 Be able to predict future applications for virtual or augmented reality.

P6: Learners are required to suggest possible future roles for virtual and augmented reality in future applications. Both virtual and augmented reality should be considered and a range (at least three) of predictions should be made. The evidence could be a written report, a journalistic report (to camera or as a written piece for inclusion in a magazine), or a presentation with detailed speakers notes.
M3: Learners are required to use the research into current uses of virtual and augmented reality applications in LO1 to identify and describe the benefits to be gained by utilising specified applications in new and creative ways. Both virtual and augmented reality should be considered and a range (at least three) of possible instances of repurposing should be made. The evidence could be a written report, a journalistic report (to camera or as a written piece for inclusion in a magazine), or a presentation with detailed speakers notes.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, M1, M2, M3, D1,D2
Problem solving/decision making	P1, P2, P3, P6, M43, D2
Time management	P4,P5, M2
Critical thinking	P2, P3, P6, M1, M3

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units. This unit is a mandatory unit in the Emerging Digital Technology Practitioner pathway and in the Digital Technician pathway.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners should be given the opportunity to carry out work experience visits and placements in organisations that are involved in the creation and use of virtual and augmented reality. It is likely you will have limited opportunities to offer practical experience of the creation of such resources other than on a fairly basic level. Whilst this is sufficient for this qualification, the option of exploring these concepts in a more focussed manner should be taken if offered.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	You could contact local organisations that would benefit from the use of virtual or augmented reality and offer them the opportunity to set their own assignment. A range of such assignments could be offered to learners (possibly with learners expected to pitch their ideas before they are given the project). Once completed, these organisations could be involved in the assessment and feedback provided to learners.

To find out more
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Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 6
Application design

M/507/5005

Guided learning hours: 60

Version 1 September 2015

LEVEL 3

UNIT 6: Application design

M/507/5005

Guided learning hours: 60

Essential resources required for this unit: none

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The world is increasingly reliant on applications that help individuals, businesses and organisations achieve specific activities or purposes. In this unit you will explore potential ideas for a new application and develop the fundamental design for it. You will then develop the designs for an application and how users will interact with it. The application that you design can be for any sector and for any purpose. You will have the opportunity to present your ideas, prototype them, and gain feedback before refining your design.

Besides the technical knowledge that you will gain about designing an application, you will also learn key transferable skills such as liaising with clients, questioning people effectively to gain the information you need to develop successful designs, and presenting your ideas to an audience and getting feedback from them.

This unit is mandatory to the application developer specialist pathway in the Level 3 Diploma suite of qualifications as it supports the development of skills, knowledge and understanding appropriate to a wide range of job roles requiring the development of applications in mobile technology, business software, graphics, game and web design.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand how applications are designed	1.1 Application development models divide the process of development into distinct phases. These vary depending on the model but all include, in some form stages, i.e.: • requirements analysis • design • implementation/coding • testing • deployment • maintenance 1.2 Characteristics and features of application development models, e.g.: • waterfall model • iterative model • agile development model • rapid application development (RAD) model • spiral model • prototype model
2. Be able to investigate potential solutions for application developments	2.1 Methods of gathering user requirements, i.e.: • client and user interviews e.g. o closed and open questions o leading questions o funnelling o structure to interviews o allowing thinking time for respondents o encouraging further detail or thought • observation of tasks • analysis of existing documents and systems

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	2.2 User requirements, i.e.: • functional requirements, e.g.: o what the application should do o data and information collected and used in the existing approach o data and information to be collected and used in the new application o functions or processing that the application should perform o outputs from the application o core functional requirements o optional functional requirements o user interface requirements including accessibility requirements Functional requirements may be divided into core requirements and optional requirements. • Constraints, e.g.: o hardware or platform constraints o software constraints o development constraints e.g. development software • limitations (e.g. scope of solution, aspects that will not be developed) 2.3 Possible solutions (e.g. different ways to address the identified user need) 2.4 Feasibility study, i.e.: • technological requirements • economic or financial costs of development and potential benefits • legal issues • operational impact • scheduling and resources (e.g. time scale for development, resources needed for development)
3. Be able to generate designs for application solutions	3.1 The use of diagrams to represent aspects of the design of an application, i.e.: • functional requirements (e.g. use case diagrams) • processing and data handling (e.g. flowcharts, data flow diagrams, class diagrams, object diagrams, entity relationship diagrams) • user interface designs (e.g. wireframe diagrams and graphical mock-ups)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.2 Standardisation of design, e.g.: • standard algorithms or processes • modularisation • cross-platform standards • standard protocols • standard interface widgets (appearance of buttons, dropdown menus, hyperlinks) • common user interface layouts, icons and labels throughout application 3.3 Advantages of proposed solution, e.g.: • automation • operational efficiency • cost-effectiveness • globalisation • improved communication • customisation and adaptability • increased markets • ease of access for customers • new marketing opportunities • customer or user information • real-time information • new employment 3.4 Disadvantages of proposed solution, e.g.: • financial cost • changeover costs and risks • training needs • lack of job security and job losses • security issues • privacy issues • potential customer concerns • loss of personal contact
4. Be able to present application solutions to meet client and user requirements	4.1 Pitch content, e.g.: • what is the proposed design solution? • who would be interested in it? • why is it a valuable idea? • what makes it effective? 4.2 Effective pitch delivery, e.g.: • courtesy • speak clearly and concisely • be aware of body language • accurate spelling, punctuation and grammar • engage the audience • be honest • be positive

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	4.3 Effective responses to questions, e.g.: • anticipating likely questions • giving a positive response • seeking clarification where necessary • recognising improvements and responding in a way that suggests how these can be incorporated 4.4 Prototyping, i.e.: • purpose of prototyping • features of prototypes • interviewing and questioning techniques • development formats 4.5 Aspects for user feedback, i.e.: • meeting core requirements and any optional requirements • effectiveness (e.g. how well the design meets each requirement) • usability (e.g. how easy it is to carry out actions, readability and clarity of displays or output to user, navigability) • learnability (e.g. how easy it is to learn how to use the application, clarity of the function of different components or elements) 4.6 Analysis of client feedback and discussion, i.e.: • identify distinct points in feedback • identify required changes • identification and implementation of improvements based on feedback

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand how applications are designed	P1: Describe the key stages in application development	M1: Compare and contrast different application development models	
2. Be able to investigate potential solutions for application developments	P2*: Gather client requirements for an application solution *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M2: Conduct a feasibility study of different solutions for the client requirements	
3. Be able to generate designs for application solutions	P3: Illustrate the requirements, functioning, and designs of an application solution, using diagrams		D1: Justify design choices identifying the advantages and disadvantages of each
4. Be able to present application solutions to meet client and user requirements	P4: Present a proposed design solution to the identified client	M3: Negotiate adaptations with the identified client to refine the design solution	
	P5: Create a prototype based on the design solution		
	P6: Gather client and/or user feedback on the prototype		D2: Implement improvements based on the analysed client and/or user feedback

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand how applications are designed

P1: Learners must describe the key activities of different stages in application development. They should describe these for a named application development model. This can be in the form of a report, presentation with detailed speaker notes, user guide or tutor resource.
M1 Learners should consider similarities and differences between different application development models. They should also explain how the different application development models are appropriate for different types of project. This can be an extension of P1 or presented as a hand out for learners, presentation or report.

LO2 Be able to investigate potential solutions for application developments

P2: Learners are required to establish the requirements of the client for an application development. Learners must document the specific requirements, together with constraints and limitations on the solution or its development.
M2: Based on the client requirements established in P2, learners should identify potential solutions and undertake a feasibility study. The feasibility study should consider the client and/or user need for which the application solution is to be designed. The study must consider whether the proposed solutions are technologically possible, estimates of the likely costs of development, any laws that will apply to the proposed solutions or their development, how well each of the proposed solutions meets the identified needs of the client, the impact that each will have, and the resources required for development of each solution and likely timescales. The evidence will be the document feasibility study and proposed solutions.

LO3 Be able to generate designs for application solutions

P3: Learners must provide a series of annotated diagrams that describe their proposed application design. These should include diagrams to show the scenarios in which the application will be used with functional requirements, process and data handling and user interface designs. The evidence will be the annotated diagrams.
D1: Learners should justify their design choices, outlining the advantages and disadvantages of each. The evidence for this could be a report, presentations, or an extension to P3.

LO4 Be able to present application solutions to meet client and user requirements

P4: Learners must present their proposed design solution to the client. The presentation format must be 'fit for purpose' and they should have undertaken appropriate quality checks. It should be sufficiently detailed to enable the client to understand the key features of their proposed solution. Evidence could be in the form of a formal presentation that is either videoed or with detailed speaker notes, or could be a formal report.
M3: Learners must provide evidence of their meeting, discussion, or communication with the client. They should identify clearly the adaptations that the client wanted and what was agreed, with clear statements of the refinements to the solution design and possible implications.
Learners should make changes to their designs following the client review and negotiations. Changes should be clearly documented.
P5: Learners should create a prototype of the chosen design solution. Evidence will be the actual prototype of the design.
P6: Learners must collect client and/or user feedback. They should provide evidence of collecting feedback together with a summary of the outcomes from the feedback.
D2: Learners should analyse client and/or user feedback to identify any improvements required. They must provide evidence of implementing improvements to the design based on their analysis of user feedback. The evidence will be the analysis and implemented improvements to the prototype and/or design documentation.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, P4, P6, M1, M2, M3, D1
Problem solving	P2, P3, P4, P5, M2, M3, D1, D2
Time management	P2, P3, P4, P5, P6, M2, M3, D2
Critical thinking	M2, M3, D1, D2
Negotiation	P2, P6
Decision making	P3, P4, P5, M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units. This unit is a mandatory unit in the Application Developer pathway and the Application Data Technician pathway.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners on work experience placements could investigate the possibility of designing a solution for a user need in their work placement.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Employers could be involved in setting the scenario and context for the design of a solution to specified user need. Local businesses could be involved in reviewing proposed designs for individual students or groups of students. If this is prior to finalisation of designs, feedback from the businesses could then form the basis for learners to carry out an appropriate evaluation and modification of the application designs.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Industry practitioners could provide input into sessions on: • gathering client and user requirements • feasibility studies • effective questioning of clients • making a business 'pitch' • prototyping and gathering user feedback Local businesses could provide input into sessions on how to make a successful pitch of an idea to business leaders.

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 7

Data analysis and design

Cambridge TECHNICALS 2016

A/507/5007

Guided learning hours: 60

Version 2 - revised May 2016

*changes indicated by black vertical line



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LEVEL 3

UNIT 7: Data analysis and design

A/507/5007

Guided learning hours: 60

Essential resources required for this unit: Learners must have access to detailed case studies or current involvement that provides a data analysis and design opportunity.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

This unit will enable you to develop the skills and knowledge required to actively use data analysis techniques to provide evidence and interpretation for decision making for a range of organisational needs. Organisations and individuals collect both quantitative and qualitative data and store it for current or future use. The data analyst examines, cleanses, transforms and models data in order to support decision making and understanding.

This unit is mandatory to the Data Analyst specialist pathway in the Level 3 Diploma suite of qualifications due to its relevance in conducting data analysis and design solutions to meet business requirements. The unit supports the development of skills, knowledge and understanding relevant to the role of a data analyst and the techniques required.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the purpose and stages of data analysis and design	1.1 Data types, i.e.: • qualitative • quantitative • structured • unstructured 1.2 Stages of data analysis, i.e.: • investigate information requirements (e.g. market share, particulates in the air, testing of new drugs) • data collection (e.g. observations, interviews, review of existing data) • data organisation (e.g. digitalisation, transcription, sorting, data mining) • data storage (e.g. in–house, external) • data cleansing (e.g. errors, missing elements, duplicates) • data manipulation (e.g. arranging, collating, aggregating, interpreting, correlation) • presentation of findings (e.g. tables, charts, graphs, dashboard, reports) 1.3 Importance of accurately defining information requirements, e.g.: • prevents time wasting • aids planning of: o data capture o data organisation and storage o cleaning and manipulation o presentation of results

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to investigate client requirements for data analysis	2.1 Investigate information requirements (e.g. business intelligence, scientific research, medical research, political) 2.2 Techniques (e.g. holding focus groups, preparing and distributing questionnaires and surveys to stakeholders, analysing current documents) 2.3 Qualitative data analysis, i.e.: • typology (e.g. activities, actions, relationships) • event analysis • logical analysis (e.g. flow charts, flow diagrams) 2.4 Quantitative data analysis i.e.: • mean • median • standard deviation • range
3. Be able to develop data design solutions to meet business requirements	3.1 Levels of data model design, i.e.: • conceptual: relationship between entities • enterprise: unique business requirements • logical: specific entities, attributes and relationship in a business function • physical: application and database specific implementation of the logical data model 3.2 Phases of logical data modelling, i.e.: • structure – set of rules (e.g. entities, attributes, relationships, queries) • manipulating (e.g. updating, retrieving, editing or deletion of content) • integrity – validation of accuracy

Learning outcomes The Learner will:	Teaching content Learners must be taught:
4. Be able to present data analysis and design solutions to stakeholders	4.1 Data design documentation, e.g.: • data flow diagrams (DFDs) • information flow charts • entity attribute relationship diagram (EARD) • hierarchical tree diagram • events • entity life history (ELH) 4.2 Presentation of solution, e.g.: • reflects all aspects of design • can be understood by audience (e.g. level of technical jargon used) • format (e.g. report or presentation supported by diagrams or models) 4.3 Evaluation of design solution (e.g. meets business requirements, achievable, manageable, extendable)

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the purpose and stages of data analysis and design	P1: Explain the types of data that can be analysed	M1: Explain the importance of accurately identifying information requirements prior to data collection	
	P2*: Summarise the stages of data analysis *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information)*		
2. Be able to investigate client requirements for data analysis	P3: Establish the data analysis and design requirements for a specified business requirement		
	P4: Gather data for the specified business requirement using quantitative and qualitative techniques	M2: Develop the data requirements for the specified business requirement using different qualitative and quantitative data analysis methods	
3. Be able to develop data design solutions to meet business requirements.	P5*: Create the outline scope of the data design model for the specified business requirement. *(*Synoptic assessment from Unit 3 Cyber security)*		D1: Construct the logical data model for the specified business requirement
4. Be able to present data analysis and design solutions to stakeholders	P6: Prepare the data design documentation for a presentation to stakeholders	M3: Present the data design documentation to stakeholders	D2: Evaluate the logical data model against the original specified business requirement

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the purpose and stages of data analysis and design

P1: Learners should explain the four different data types identified in the teaching content, using appropriate examples they have researched. This can be in the form of a guide for new IT learners, a presentation with speaker notes or a report.
P2: Learners must summarise all of the stages of data analysis to include purpose and relevance of each stage to the production of useful and cost effective data. This could be presented as a presentation with speaker notes, a report or a guide to data analysts.
M1: Learners could use actual examples to highlight the importance of recognising the information requirements of a specific business need prior to data collection. Research may identify only issues arising from poor requirements planning but learners would be able to use such examples to emphasise the importance of accurately identifying requirements. This may take the form of a report or a presentation with speaker notes or a video presentation.

LO2 Be able to investigate client requirements for data analysis

P3: Learners should be given a scenario that will enable them to investigate the data requirements for a specified business need. If available, learners could provide evidence from a real current project. The evidence should include the business need, the qualitative and quantitative data to be gathered and their potential sources. The evidence could be presented in the form of a report, a set of recording documentation and summary or an oral presentation.
P4: Learners should select different qualitative and quantitative techniques that could be used to obtain the necessary data requirements for a specified business need. They could produce a brief report or presentation confirming selection with a brief rationale.
M2: Learners should use the quantitative and qualitative techniques selected in P4 to develop the data requirements for the specified business need. The outcomes may be presented as annotated diagrams, a report, or a set of completed analysis forms.

LO3 Be able to develop data design solutions to meet business requirements

P5: Learners should create content relating to all phases of the data design model. Evidence could include notes of meetings, notes of observations, diagrams, a data dictionary, a summary report or presentation with speaker notes.
D1: Learners must create a logical data model for the specified business requirement using appropriate terminology and techniques to include EARD, data dictionary, entity relationships and queries based on their design model.

LO4 Be able to present data analysis and design solutions to stakeholders

P6: Learners should create data design documentation to reflect the data analysis and design solution to meet a specified business requirement. They must produce a report or presentation that includes relevant diagrams to support and justify their explanations, ideally, using the design from P5 as supporting evidence and potentially the logical data model in D1.
M3: Learners must present their findings to the stakeholders. This could be as formal written report with relevant diagrams and a set of clear conclusions and recommendations but as it is being presented there should be some narrative, either verbal or written, to underpin the delivery documentation.
D2: Learners should evaluate their logical data design model against the original business requirements and justify any changes from the original specification, making further recommendations in the light of the presentation in M3 and any potential feedback they may have received as a result.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P4, P5, P6, M1, M2, M3, D1, D2
Problem solving	P1, M1, P2, P3, M2, P5, P6, D2
Time management	P2, P3, P4, P5, P6, M2, M3, D1, D2
Critical thinking	P1, M1, P2, P3, M2
Negotiation	P2, D2
Decision making	P1, P2, P3, P4, P5, P6, M1, M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units. This unit is a mandatory unit in the Data Analyst pathway and the Application Data Technician pathway.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners may join a team engaged in data analysis and design, contributing to customer requirements, the collection of data, recording the data requirements, working alongside experienced analysts to construct elements of the logical data model and contributing to the preparation of materials for presentation to the stakeholders.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Data analysts from a business could provide ideas for projects, short, self-contained projects would enable learners to carry out real life activities in specified learning outcomes.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A data analyst from a local business could give a guest lecture on: the purpose of data analysis and the stages which they undertake during the process and why. The reality of gathering client requirements using a case study as a basis, especially where there are conflicting requirements, or limited financing would all be of value. Examples of documentation could confirm the depth, range and quality of the content.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	If you worked with a data analyst to set a project in point 2 above, they could be asked to be an 'expert witness' and contribute to the assessment of the learners' projects.

To find out more
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or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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Cambridge **TECHNICALS LEVEL 3**

IT

Cambridge TECHNICALS 2016

Unit 8

Project management

F/507/5008

Guided learning hours: 60

Version 2 - revised July 2016

*changes indicated by black vertical line



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LEVEL 3

UNIT 8: Project management

F/507/5008

Guided learning hours: 60

Essential resources required for this unit: none

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

This unit will provide you with the opportunity to understand and use various project planning skills and techniques, thereby enabling you to become more effective in the workplace.

The key to any project being a success is the planning that takes place. Project management skills are essential transferrable skills that can be used for all IT related projects whether it's traditional methodologies or more recently adapted agile approaches within the IT development environment. These skills can be adapted and used even on the smallest 'tasks' during the planning and implementation stages.

Regardless of your job role, you will often be called upon to participate in projects for a variety of reasons; consequently this unit is optional within all four of the specialist pathways. This unit will assist you in developing your skills, knowledge and understanding of different project methodologies and the key factors that can influence the success or failure of a project.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as CompTIA Project+.

Note: Learners must work on an IT project that would reflect the type of project carried out within the IT industry. For those learners following the Diploma qualifications their chosen project will ideally reflect their chosen specialist pathway.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the project life cycle	1.1 Project methodologies (e.g. Prince2, waterfall, DMAIC methodology, critical path method (CPM), agile development, individual methodologies as required by the client) 1.2 Project life cycle, i.e.: • initiation phase • planning phase • execution phase • evaluation phase 1.3 Project issues, i.e.: • communication (e.g. between team members, project manager, stakeholders, clients) • external factors (e.g. finance, team members, etc.) • conflicts (e.g. between project team members and/or client) • lack of management/leadership • poor planning • legislation/regulation 1.4 Documentation, i.e.: • main documentation o project brief/project mandate (overview of project/scope/objectives) o project initiation document (PID)/project definitions document (project outline/scope/objectives) o contract (agreement from project sponsor/budget holder to start project) o business case (project justification, cost versus benefits) o client acceptance form (to obtain agreement from project sponsor/budget holder that project is complete) o work breakdown structure o project progress report (status updates while project is in progress) o project closure report (indication of outcomes) o lessons learned report (also survey)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• main control registers and plans: o project planner (visual aid to show the project timelines and activities to be completed) o risk register (to record details of project risks, monitoring and management of project risks) o issues register (to record details of project issues, monitoring and management of project issues) o lessons learned register (to capture details of things when they go wrong to inform report at end of project)
2. Be able to initiate and plan projects	2.1 Initiation phase, i.e.: • stakeholders, clients, target audience • scope definition • purpose • objectives • resources (e.g. fast tracking, crashing, delaying, optimising) • Tools (e.g. Gantt charts, Network diagram (ADM, PDM, CPM,CCM)) • deliverables • timescales • structure • business case o several possible solutions o costs/benefits of each • feasibility study (evaluation and analysis of a proposed project) • terms of reference o accurate description of: ▪ what the project aims to achieve ▪ criteria and flexibilities involved ▪ project parameters, scope, range, outputs, sources, participants, budgets and timescales • phase review 2.2 Planning phase, i.e.: • project plan consisting of: o resource plan (e.g. identification of staff, equipment and materials needed) o financial plan (e.g. quantify the financial expenditure required) o quality plan (e.g. quality targets and quality control methods) o risk plan (identify risks and plan actions need to minimise risks) o acceptance plan (e.g. specify criteria for accepting deliverables) o phase review

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	▪ is the project under schedule and within budget? ▪ have the deliverables been produced and approved? ▪ have the risks been controlled and mitigated? ▪ have issues been resolved? ▪ is the project on track? ▪ should the project continue to the execution phase?
3. Be able to execute projects	3.1 Execution phase, i.e.: • time (e.g. tracking and recording time spent on tasks as per project plan) • cost (e.g. identifying and recording costs against budget plan • quality (e.g. review quality of deliverables and management process) • change (e.g. reviewing and implementing requests for changes to project) • risk (e.g. assessing level of project risks and taking action to minimise the risks) • issue (e.g. identifying and resolving project issues) • communication (e.g. communication with stakeholders, end users, clients, project team members) • acceptance (e.g. identifying completion of deliverables and gaining client's acceptance) • phase review o is the project under schedule and within budget? o have the deliverables been produced and approved? o have the risks been controlled and mitigated? o have issues been resolved? o is the project on track? o should the project continue? • project closure report (final document produced to enable formal closure of the project) o project summary o reason for project closure (has the project reached its planned end date or is it being closed prematurely for some reason?) o assessment of project performance (how well did the project perform against the business case?) ▪ achievement of the project objectives, outputs and activities (where was the project successful or not successful?) ▪ performance against planned time, resources and costs (explanation of variance from original planned milestones, costs or allocated resources) ▪ performance against planned savings and benefits (explanation of any variance from original planned benefits)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	o lessons learned o celebrating success (what were the biggest successes for the project?) o next steps (description of any outstanding actions and handover arrangements) ▪ outstanding activities ▪ remaining risks and issues ▪ on-going dependencies ▪ costs of on-going support ▪ stakeholder communications ▪ handover of assets and contracts
4. Be able to carry out project evaluations	4.1 Evaluation phase, i.e.: Obtain feedback from: • clients • stakeholders • users • team o has the project: ▪ delivered business benefits identified in the business case? ▪ achieved the objectives in the terms of reference? ▪ deviated from the original scope? ▪ met the quality targets identified in the quality plan? ▪ proceeded according to the delivery schedule? ▪ deviated from the budget as defined in the financial plan? ▪ deviated from forecast resource levels as per the resource plan? ▪ conformed to the management process as per the execution phase? o the effect the selected tools and methods have had on the outcome of the project o identify major achievements and the positive effect on the client's business o identify any project failures and the effects on the client's business o identify lessons learned o identify potential improvements for similar future projects

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the project life cycle	P1: Explain the different phases within an identified project life cycle		D1: Evaluate the importance of each phase of the identified project life cycle
	P2: Describe different project methodologies	M1: Compare the features and benefits of different project methodologies	
2. Be able to initiate and plan projects	P3*: Complete the documentation for the initiation phase for an identified project *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		D2: Create a Business Case to support an identified project
	P4: Develop a project plan for the identified project	M2: Carry out and document a phase review of the project plan	
3. Be able to execute projects	P5: Follow the project plan and conduct a phase review of the identified project		D3: Prepare a project closure report based on the execution phase review of the identified project
4. Be able to carry out project evaluations	P6: Evaluate feedback from client and team in relation to the project	M3: Recommend potential improvements for future projects based on the outcome of the project evaluation	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the project life cycle

P1: Learners must explain the different phases within an identified project life cycle. Their descriptions must provide an account of each phase including their position within the project life cycle, and the supporting documentation that is used. Evidence can be in the form of a written report, presentation with detailed speaker notes, a video of presenting the information to an audience, or a detailed information guide on the project life cycle.
D1: Learners are required to evaluate the importance of each phase of the identified project life cycle. The evaluation must show that learners have determined the significance of each phase of the project life cycle as well as the documentation involved. The evidence can be presented in the form of a written report, presentation with detailed speaker notes or comprehensive information guide. The evidence produced for P1 can be used as a useful starting point for the evaluation.
P2: Learners must describe project methodologies. The summary should be concise and include the main points of their chosen project methodologies written/spoken in their own words. The evidence can be in a written or verbal format eg written report, presentation, audio or video recording of a presentation, etc.
M1: Learners are required to compare the features and benefits of different project methodologies. It is suggested that learners use their evidence from P2 as a starting point for their comparison. The comparison should show that learners have examined the similarities and differences between the different project methodologies and must not just be a list of the features and benefits of each in isolation. The evidence can be in the form of a written report, a presentation with detailed speaker notes or comprehensive information guide.
Learners could present evidence for the entire learning outcome as a tutor guide for learners wanting to study project management.

LO2 Be able to initiate and plan projects

This learning outcome is about learners carrying out the initiation phase and project planning phase for an identified project. It is important that the scenario and/or project that learners are provided with allows them to meet the assessment criteria in a manner that reflects working practices.
P3: Learners must carry out the initiation phase for an identified project taking into consideration the bullet list in the teaching content under initiation phase. They should then prepare a PID which must include, as a minimum, the stakeholders, clients and target audience (end user), scope definition, purpose, objectives, deliverables, timescales and structure. The evidence will be the completed PID.

D2: Learners are required to create a business case for an identified project. It can be linked to the project for which they carried out the initiation phase. The business case must give clear justification as to why the project should be carried out i.e. the reasons and why. They could present the evidence as a report or as a presentation with detailed speaker notes or a presentation of them delivering their business case to an audience.
P4: Learners must develop a project plan for an identified project. The project plan can be a follow on from the evidence presented for P3 and D2. The project plan must include the items in the bullet list in the teaching content. Although learners may not be "working with budgets", they should still take into consideration that 'time is money' and therefore cost in potential value of time of those involved in the project as well as the cost of other physical resources that may be used.
M2: Learners must carry out a phase review of the project plan. The phase review must show that learners have reviewed every section of the project plan to ensure that the project can progress to the execution phase. They will need to show what they have reviewed and the results of their review. This will then be concluded by including the justification for the continuation of the project. This may also include where learners have identified that changes have to be made, etc. The evidence will be the documented review of the planning phase of the project.

LO3 Be able to execute projects

This learning outcome is about learners carrying out the execution phase of a specified project. It is important that the scenario and/or project with which learners are provided, allows them to meet the assessment criteria in a manner that reflects working practices. Learners could follow through with the project that they have used for LO2 Be able to initiate and plan projects.
P5: Learners are required to provide evidence of following a project plan for an identified project and carry out a phase review. They should be reviewing each phase of the project plan to check whether the project has achieved its planned objectives and if not, what are the reasons and how should these be addressed. The evidence will be the documented phase review. The evidence for this assessment criterion should include, as a minimum, the following: considerations of time, quality, change, risk, issue, communications and acceptance. The evidence will be the documented records of the different aspects of the execution phase.
D3: Learners are required to prepare a project closure report based on the execution phase review carried out for M3. The project closure report must include the actions identified in the phase review which must be carried out prior to the project being closed. The bullet points within the teaching content should be included in the project closure report.

LO4 Be able to carry out project evaluations

P6: Learners are required to evaluate the feedback from the client and team of the identified project. They should evaluate the project that they have carried out for LO3. The evaluation must include the bullet points in the teaching content from the section "Has the project" to "Identify lessons learned". The evidence will be the documented evaluation. This could be in the form of a report or a presentation with detailed speaker notes or a video of learners presenting the results of their evaluation.
M3: Learners are required to recommend potential improvements for future projects based on the outcome of their project evaluation. They should be considering what issues have occurred and what could be done differently to alleviate/prevent the occurrence of these issues in future projects. Learners should be able to state why these recommendations would improve the delivery of future similar projects.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P3, P4, P6, M2, M3, D2, D3
Problem solving	P3, P4 ,P6, M2, M3, D2, D3
Time management	P3, P4, P6, M2, M3, D2, D3
Critical thinking	P3, P4, P6, M2, M3, D2, D3
Team working	P3, P4, P6, M2, M3, D2, D3
Negotiation	P3, P4, P6, M2, M3, D2, D3
Decision making	P3, P4, P6, M2, M3, D2, D3

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners may be able to participate in projects while on work placement. The centre would need the support of the work placement for learners to practice working through the various phases and to complete appropriate documentation (even if only for their own personal use).
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Learners could be given IT projects that have been developed with the aid of local employers. The employers could be asked to identify potential IT project scenarios similar to what they would work on in their particular environment.

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 9

Product development

Cambridge TECHNICALS 2016

A/507/5010

Guided learning hours: 60

Version 1 September 2015

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LEVEL 3

UNIT 9: Product development

A/507/5010

Guided learning hours: 60

Essential resources required for this unit: none

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The purpose of this unit is to prepare you to undertake product development activities. You will learn about different product design methodologies and the role of the product development life cycle. In addition, you will discover the factors that influence product developments.

The key to any product development being a success is the analysis, client review, design, testing and final acceptance that takes place. The skills that you will learn can be applied to the development of any product, large or small. You will use product development skills and work through the product development life cycle.

Whether you are building a network, developing a website, developing a system for data analytics or creating an augmented or virtual reality resource, they are all products. It is therefore important that you understand the processes required for the development of products and that you can apply them to a variety of situations.

It is recommended that you develop a product alongside the other units you are studying so that you can explore the units holistically as a wider project. If this unit is being taken as part of a specialist pathway, the product developed could align to your chosen pathway and support progression into your chosen field within the IT industry.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand the product development life cycle	1.1 Product development methodologies (e.g. Waterfall, Incremental, Spiral, Agile development) 1.2 Phases of the product development life cycle, i.e.: • requirements analysis • design • implementation • testing (e.g. unit, integration, product, acceptance) • deployment • maintenance 1.3 Constraints, i.e.: • time • financial • social (e.g. language, text types) • regulation (e.g. company codes of practice, governmental regulation, national and international law) • standards organisations (e.g. ISO, EN, BS, IEEE) • standards (e.g. safety, design, quality management, risk analysis, reliability, requirements analysis) • the need for compromise (e.g. cost/benefit trade-off, complexity/error trade-off,)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to design products that meet identified client requirements	2.1 Requirements analysis phase, i.e.: • market analysis (e.g. client needs and wants, end user needs and wants, competing products on the market) • limitations (e.g. target platform, bandwidth, development resources, human resources, etc.) • prototyping • product requirements specification: o introduction o purpose (e.g. what it needs to do, where it needs to be used) o context (e.g. stand-alone product or part of a larger whole, a replacement for an existing product, a totally new product) o assumptions (e.g. hardware or software to be used, user experience or training) o constraints (e.g. timescale, interoperability with existing products, design tools to be used) <u>Functional Requirements</u> • main functional requirements – a set of clearly defined, unambiguous needs of the product, with priorities • target user profiles (e.g. end user tasks, end user knowledge and level of experience, client goals) <u>Non-functional Requirements</u> • usability (e.g. easy to learn, clear operating procedures, appropriate for the level of knowledge of each target user profile) • performance – stated in an easily measurable form (e.g. capacity, availability, response times, environmental issues • safety (e.g. to prevent possible loss, damage or harm, specific safety standards to be adhered to) • security (e.g. standards compliance, privacy issues, certification) • maintainability (e.g. define consumables, regular calibration activities, updates, bug fixes, error recovery) • interfaces (e.g. hardware, software, communications, user) • appendices (e.g. glossary of terms, diagrams describing processing, a numbered list of TBD (to be determined) items with clear responsibilities and timescales defined)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	2.2 Design phase, i.e.: • outline solutions: o hardware and software o functional details (e.g. inputs, processing, outputs, user interface workflow) o non-functional details (e.g. usability, performance, safety, security, maintainability and interfaces) o prototyping • Presentation of outline solutions to the client: o meeting (actual or virtual) between client and product developer o presentation of outline solutions in an appropriate format o review of outline solutions by the client o agreement on design specification to be implemented: ▪ minor modifications and correcting errors ▪ choice of best features and functionality from the outline solutions ▪ extended features and functionality requested by client ▪ extended features and functionality suggested o record decisions made, follow up actions and deadline dates • final design solution (e.g. after obtaining agreement from client), i.e.: o implementation plan, including deadline dates o hardware and software o functional details o non-functional details o additional resources o design documentation o proposed acceptance test plan
3. Be able to implement and test products	3.1 Implementation, i.e.: • creation of product • unit testing (e.g. on-going testing of each separate function/process of the system) • integration testing (e.g. testing that the units work together as expected) • use results of unit testing and integration testing to improve and enhance product during implementation • implementation log • revisit the implementation plan

Learning outcomes The Learner will:	Teaching content Learners must be taught:
4. Be able to carry out acceptance testing with clients	4.1 Acceptance testing with target users, i.e.: • acceptance testing against the agreed design specification: o acceptance test plan created (e.g. based on proposed acceptance test plan from the agreed design specification): ▪ functional tests (e.g. complete and correct operation) ▪ non-functional tests (e.g. usability, performance, reliability, maintainability, safety, security) o acceptance testing with target users (e.g. based on the user profiles listed in the agreed design specification) o deployment decision by client (e.g. acceptance, non-acceptance or acceptance after modifications and retest) 4.2 Maintenance phase, i.e.: • routine maintenance • enhancements to product features and functions (e.g. bug fixes, minor alterations to input ranges or processing carried out on inputs, minor changes to target user requirements) • product upgrades (e.g. due to major changes to target user requirements, additional functionality needed)

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the product development life cycle	P1: Outline the phases of the product development life cycle	M1: Compare and contrast different product development methodologies	D1: Assess the potential impact of constraints upon product development
2. Be able to design products that meet identified client requirements	P2*: Develop a product requirements specification to meet an identified client's requirements *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		
	P3: Present an outline of the design solutions to the identified client and obtain feedback	M2: Agree the inclusion of features that extend or enhance the functionality of the chosen design solution with the identified client	
3. Be able to implement and test products	P4: Develop the product in line with the agreed design solution		
	P5: Conduct product testing	M3: Analyse the results of testing and recommend improvements and enhancements to the design solution	
4. Be able to carry out acceptance testing with clients	P6: Carry out acceptance testing for users in line with the agreed design solution		D2 : Discuss with the identified client potential enhancements, upgrades and maintenance of the final product

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the product development life cycle

P1: Learners must outline the phases of the product development life cycle. They must include the main characteristics of each phase including its position within the product development life cycle. Evidence could be in the form of a written report, presentation with detailed speaker notes, a video of presenting the information to an audience, or an information guide on the product development life cycle.
M1: Learners are required to identify similarities and differences between a variety of different product development methodologies. Learners should clearly demonstrate their understanding through the comparison of the different methodologies. It is suggested that they include a summary of their findings in their evidence. The evidence could be in the form of a written report, a presentation or an information guide.
D1: Learners are required to make an informed judgement about the potential impact of constraints upon the product development life cycle. It is suggested that learners consider the potential impact that each constraint could have on the phases of the product development life cycle, highlighting where the major impacts could occur, together with possible knock on effects to subsequent phases. The evidence could be in the form of a written report or a presentation with very detailed speaker notes.
Learners could present evidence for the entire learning outcome as a tutor guide for learners wanting to study product development.

LO2 Be able to design products that meet identified client requirements

This learning outcome (LO) is about learners carrying out the requirements analysis phase and the design phase for an identified client and product. The scenario could be supplied by the centre or by learners. In all cases, it is important that the scenario, which learners are provided with, is sufficiently detailed to allow them to meet the assessment criteria fully.
P2: Learners must produce a product requirements specification that describes a product that meets the client requirements as given in the scenario. To achieve this, learners will need to carry out a requirements analysis, including prototyping if appropriate. The evidence will be a completed product requirements specification that covers all areas of the 'product requirements specification' section of the teaching content.
P3: For the first part of P3, learners must present outline solutions (two or more) to an identified client (i.e. from the scenario). Each outline solution must include hardware and software details functional and non-functional details and could include a prototype. Each outline solution must be presented in an

appropriate format. The presentation can be at a face to face meeting, via email exchange or another suitable method. However, there must be clear evidence that this has taken place.
The second part of P3 focuses on receiving feedback on the outline solutions, from the client. Learners must record the decisions made, any follow up actions, the implementation plan and deadline dates agreed with the client, the proposed acceptance test plan, and any additional resources that will be needed. The record, together with the outline solutions and product requirements specification, will be used to form the agreed design solution. The evidence of the decisions made could include annotated copies of the outline solutions and product requirements specification. The evidence produced for P3 should cover all items listed under 'Final design solution' in the teaching content.
M2: Learners are required to agree the inclusion of features that extend or enhance the functionality of the chosen solution with the client. It is expected that this will be an extension to P3. The extended or enhanced features and functionality may be asked for by the client or suggested by learners and will not have been included in the outline solutions presented in P3. Learners must record the full details of the extended or enhanced features and functionality, including any follow up actions and additional deadline dates agreed with the client. Learners are required to add these details to the agreed design solution. Learners should clearly identify the evidence pertaining to the extended or enhanced features and functionality.

LO3 Be able to implement and test products

This LO is about learners carrying out the implementation phase and the first part of the testing phase, using the design solution agreed with the client in the previous LO and including an implementation plan and proposed acceptance test plan.
P4: Learners must develop a product in line with the agreed design solution, following the implementation plan and keeping an implementation log. It is expected that the product will be complete and that it will meet all aspects of the agreed design solution. Evidence of development will consist of the completed implementation log (e.g. the annotated implementation plan); other documentation (e.g. photos, videos, and screenshots) and the actual product. In addition, learners must test the product as it is being developed; evidence of this unit testing and integration testing could be included in the implementation log. The implementation log should show that unit testing and integration testing have taken place during implementation.
P5: Learners must create a product test plan and use this to carry out testing of the completed product at the end of implementation. The product test plan must be sufficiently detailed to enable full testing against the agreed design specification. Learners must clearly evidence that a test plan has been created and followed.
M3: Learners are required to analyse the results of testing. This could be the results of any testing carried out on the product, i.e. unit testing and integration testing during implementation or product testing against the agreed design solution. Furthermore, learners are required to recommend improvements and enhancements (at least one of each) to the product. The evidence should give details and a justification of all improvements and enhancements. Learners are not required to implement the improvements and enhancements; the focus of M3 is on the analysis of the test results and the further development ideas that flow from that.

LO4 Be able to carry out acceptance testing with clients

P6: Learners must create an acceptance test plan, as detailed in the teaching content. This should be based on the proposed acceptance test plan, agreed with the client, as part of P3. The acceptance test plan must be used to carry out acceptance testing with a variety of target users, and as detailed in the agreed design specification). The evidence should be the completed acceptance test plan showing test outcomes, details of the deployment decision by the client and a hand-over document. This acceptance testing must cover functional and non-functional tests.

D2: Learners are required to discuss potential enhancements, upgrades and maintenance of the final product. Learners must discuss these changes with the original, identified client. The rationale for each potential enhancement, upgrade and maintenance activity must be clearly identified. Evidence will be a record of the discussions, together with full details of each potential enhancement, upgrade and maintenance activity. This could be in the form of meeting notes, audio/video recording of the meeting, or documentary evidence of an email exchange. **N.B.** Learners are **not** required to implement these changes.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	M1, D1, P2, P3, M2, P4, P5, M3, P6, D2
Problem solving	P2, M2, P4, P5, M3, P6, D2
Time management	P2, M2, P4, P5, P6, D2
Critical thinking	M1, D1, P2, M2, P4, P5, M3, P6, D2
Team working	P2, P3, M2, P6, D2
Negotiation	M2, D2
Decision making	P2, M2, P4, P5, M3, P6, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners may be able to participate in product developments while on work placement. You would need the support of the work placement for learners to practise working through the various development phases and complete appropriate documentation (even if only for their own personal use)
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Learners could be given product development scenarios which have been produced with the aid of local employers. The employers could also be involved by acting as the client and target users during the product development.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	If you have worked with employers to produce projects in point 2 above, they could be invited to act as an 'expert witness' and contribute to the assessment of those projects.

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 10

Business computing



F/507/5011

Guided learning hours: 60

Version 1 September 2015

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LEVEL 3

UNIT 10: Business computing

F/507/5011

Guided learning hours: 60

Essential resources required for this unit: none

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Businesses and organisations are driven by the information that they gather, process and provide. Often, this involves computers, networks and databases. In this unit, you will gain an understanding of how information technology and computer-based systems facilitate the needs of business and how businesses use information. You will learn about the skills and attributes needed by people working in data analysis and will gain practical experience of editing and manipulating a variety of different forms of information before applying these skills to solve a specific problem.

This unit is in the Data Analyst specialist pathway due to its relevance to a business and data analytical environment. The unit supports the development of skills, knowledge and understanding relevant to a data analyst role, and can be used as a starting point to further develop their understanding of working with and analysing data, regardless of size and complexity.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Know the attributes required for data analyst job roles	1.1 Data analyst jobs, e.g.: • retail marketing data analyst • software licensing data analyst • banking reference data analyst • DBA data analyst • technical business analyst • systems analyst • business data analyst • SQL developer • financial data analyst • HR data analyst • senior data analyst • voice data analyst • gaming data analyst 1.2 Personal attributes for employment in data analyst roles, e.g. interpersonal skills and team-working • time management • mathematical ability • written communication • spoken communication • IT skills • accuracy and attention to detail • determination • adaptability • decision-making skills • problem-solving skills • commercial awareness 1.3 Qualifications

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to capture and store data for analysis	2.1 Data suitability, e.g.: • relevance • availability • timeliness • reliability • accuracy • appropriate format • suitability for processing 2.2 Data capture techniques, e.g.: • web data capture form • video recording • audio recording • sensors and data logs • website visitor tracking • mobile and wearable technology • image and document scanning • machine-readable media and codes • legacy data import • optical character recognition • hand-writing recognition 2.3 Automation of data capture, e.g.: • spreadsheet forms auto-linked to data sheets, e.g. use of macros or Google forms • web forms and server-side processing on form submission • macros for download of web data that frequently updates • automated download of sensor data • event-triggered capture, e.g. motion or noise triggered • web analytic tools • data virtualisation 2.4 Storage considerations, e.g.: • data volumes • data access requirements • local, networked and cloud storage
3. Be able to use tools to edit and analyse data	3.1 Types of data, i.e.: • text • numbers • video • audio • images/graphics 3.2 Applications for manipulating data, e.g.: • desktop publishing and text manipulation software • spreadsheets, financial packages, statistical analysis software • video analysis and editing applications • audio analysis and editing applications • photo editing software • graphics packages

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.3 Data compatibility and transformation including data format, data output for compatibility with other applications or platforms and data import 3.4 Techniques in selected software applications to correct errors in captured data, e.g.: • numbers stored as text, empty fields, commas in text fields of CSV files • colour correction, perspective, distortion, retouching, image noise reduction • audio noise reduction, removal of unwanted sounds 3.5 Core functions of automated processing tools, e.g.: • natural language processing • speech synthesis and analysis • video image processing (e.g. Prozone, Google car pedestrian recognition) • song recognition (e.g. Shazam, Soundhound) • face recognition • fingerprint analysis • functions, formulae • environmental sensors (e.g. heat, light, humidity) 3.6 Basic processing and manipulation techniques used in applications 3.7 Advanced processing and manipulation techniques used in applications 3.8 Combining techniques and applications to process data 3.9 Analysis of processed data, i.e.: • key question(s) for analysis • data selection • identification of patterns and trends • limitations of data and impact on outcomes from analysis
4. Be able to present data analysis outcomes	4.1 Presentation content, e.g.: • what was the problem? • what information or data were required to address it? • how were these captured? • what processing was necessary? • what are the outcomes from processing of information? • how does it meet the client's needs? • how might it be used by the client? • why is this useful? • what makes it effective? 4.2 Effective presentation delivery, e.g.: • courtesy and integrity • speaking clearly and concisely • awareness of body language • accurate spelling, punctuation and grammar

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• use of appropriate specialist vocabulary • engaging the audience • positive attitude and language • providing clarification as necessary 4.3 Laws applying to collection, storage, processing and use of data and information, i.e.: • Data Protection Act 1998 • Computer Misuse Act 1998 • Freedom of Information Act 2000 • Copyright, Designs and Patents Act 1988 4.4 Organisation procedures for compliance

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Know the attributes required for data analyst job roles	P1: Describe the knowledge, skills and qualifications required for data analyst job roles		
2. Be able to capture and store data for analysis	P2*: Plan and capture data for an identified purpose *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M1: Explain storage considerations required for storing captured data for an identified purpose	
3. Be able to use tools to edit and analyse data	P3: Use IT tools to correct errors and transform data for processing		D1: Use core functions to automate the processing of captured information, justifying choices made
	P4: Process data for the identified purpose	M2: Analyse the processed data for the identified purpose	D2: Justify decisions in the processing of information for the identified purpose
4. Be able to present data analysis outcomes	P5: Present results of data analysis to a specified audience	M3: Explain the legal issues associated with the presentation and use of processed information	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Know the attributes required for data analyst job roles

P1: Learners should make it clear how the knowledge, skills or qualifications that are required for a data analyst job role are related to the role itself and the tasks it may involve. The evidence could be in the form of a report based on learners' research into different data analyst job roles advertised by employers online or in newspapers, but could be demonstrated by learners creating job advertisements for specified data analyst roles.

LO2 Be able to capture and store data for analysis

P2: Learners must show that they understand what outputs are required for a given scenario, what information will be required and how this could be captured in a suitable format for later processing. The evidence will be plans for capturing the data and the actual data captured.
M1: Learners are required to explain storage considerations required for storing captured data for an identified purpose. The explanations should include a brief description of the different storage considerations they have made and how these would meet the identified purpose. The evidence could be in the form of a report or presentation with detailed speaker notes or a technical guide. It could also be an extension of the evidence for P2.

LO3 Be able to use tools to edit and analyse data

P3: Learners must use a wide range of tools to manipulate types of data, at least one of which must be numerical. The use of tools must be purposeful and should be realistic data processing tasks. Learners must provide evidence of transforming data for subsequent processing. They should provide evidence of how errors can be identified in provided or imported data and how they have corrected these errors.
D1: Learners must use core functions to automate the processing of data. The evidence will be the core functions selected and the resultant processing of the transformed data from P3. Learners must justify the choices made for the core functions.
P4: The assessment evidence is the outcome from the processing of data. Learners must provide evidence of processing the data and the outcomes from that processing, together with any necessary annotation. The processing should include basic techniques in relevant applications or using a combination of applications in order to provide the necessary processing.

M2: Learners should present their findings from analysing their processed data, making specific reference to the evidence as appropriate. Evidence could be in the form of a formal report, or may be a presentation with detailed speaker notes. If learners support the analysis with graphs and/or charts they should be labelled correctly.
D2: Learners must justify the choices that they made in the processing of the data for the identified purpose. For each aspect they should explain the reasons for each of their decisions. They should include consideration of the constraints on their solution, what was feasible and how the solution could be addressed more effectively with fewer constraints.

LO4 Be able to present data analysis outcomes

P5: Learners are required to present outcomes from their data analysis. The presentation may take the form of a formal presentation that is video recorded or has detailed speaker notes, or may be a formal report. The audience will be defined within the scenario given.
M3: Learners must describe the laws relevant to the collection, storage, processing and use of processed information. They could also describe the actions that an organisation could take or the policies they could put in place in order to comply with each law. They could explain how these actions or policies fulfil the legal requirements and outline what is required of employees as part of this. They must describe the consequences of failing to meet legal requirements, both for organisations and individuals. The evidence could be a presentation for the appropriate audience that could include the client (or tutor in that role) and should be fit for purpose. The presentation may be face-to-face, but could be in another format such as video conferencing, video presentation, or written format.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P5, M1, M2, M3, D1, D2
Problem solving	P2, P3, P4, P5, M2
Time management	P2, P3, P4, P5, M2
Critical thinking	M1, M2, D1, D2
Decision making	P2, P3, P4, P5, M2, D2

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The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could see this unit in context though a work placement in a team that process and manage data.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	For the processing of data, industry practitioners could provide a scenario for investigation or develop a scenario with you which will enable the learners to use the skills and techniques expected in the working environment.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Industry practitioners could provide sessions on the nature of data analysts' work, techniques used to process information in selected applications and how to remove errors from erroneous data.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Industry practitioners could review the evidence produced by learners as a result of a scenario which they developed with the centre. The industry practitioner could provide feedback on how the learners' evidence meets industry standards and reflects the requirements and standards expected in a working environment.

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 11

Systems analysis and design

Cambridge TECHNICALS 2016

J/507/5012

Guided learning hours: 60

Version 1 September 2015

ocr.org.uk/it

LEVEL 3

UNIT 11: Systems analysis and design

J/507/5012

Guided learning hours: 60

Essential resources required for this unit: Learners must have access to detailed case studies or current involvement with a systems development opportunity.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

All organisations have information systems. This unit will provide you with the knowledge and skills to support the design of formal information systems. These systems provide useful reliable, validated information through the integration of data, hardware, software and humans into processes which enable the organisation to meet its internal goals and its external obligations.

IT technicians are involved in the implementation of systems that have been designed and in many instances assist in their design. Emerging technology practitioners also have to have a knowledge and understanding of how to analyse and design systems that are supported by emerging technologies such as virtual and augmented reality, mobile technologies and the Internet of Everything. In order to effectively develop systems for analysis, it's important to have an understanding of the role of the data analyst and the tasks that they carry out. Data analysts also benefit from having an overview of systems analysis and design in order to support the development of suitable systems on which to work.

This unit is in the IT Infrastructure Technician specialist pathway as well as in the Emerging Digital Practitioner and Data Analyst Specialist pathways.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the role of systems analysis and design in relation to the systems development lifecycle	1.1 The components of the systems development lifecycle, i.e.: • feasibility study: o financial aspects including cost benefit analysis o business aspects o technical aspects o outcomes e.g. constraints, functional and non-functional requirements • requirements engineering (i.e. business and user): o requirements gathering (e.g. ideas, concerns) o requirements analysis (i.e. sensible, achievable, affordable) o requirements sign off o requirements monitoring (i.e. avoiding mission creep) • system design including virtualisation: o capacity o performance o throughput • software development • testing of the software • systems testing • implementation 1.2 Life cycles, i.e.: • linear (e.g. waterfall, v model, incremental) • evolutionary (e.g. iterative, spiral) • agile (e.g. scrum, Agile Unified Process (Agile (UP)))

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to use investigative techniques to establish requirements for business systems	2.1 Business requirements, i.e.: • business systems (e.g. human resources, finance, catering, production) • business analysis processes and models • identify business requirements • deliverables from business analysis 2.2 Techniques, i.e.: • interviewing stakeholders • conducting workshops (e.g. facilitation, goal definition, selection of appropriate stakeholders) • holding focus groups • observing worker activities • shadowing workers • preparing and distributing questionnaires and surveys to stakeholders • analysing current documents • identifying deliverables
3. Be able to develop and document models for business systems	3.1 The three view approach, i.e.: • functionality: o data flow diagrams (DFDs) o flow charts o Jackson Structure Chart • static data: o Entity Relationship Diagram (ERD) o hierarchical tree diagram o bubble diagram • events: o Entity Life History (ELH) 3.2 Unified Modelling Language (UML) i.e.: • use case diagram • activity diagram • interaction diagram: o sequence diagram o communication diagram o timing diagram o interaction overview diagram
4. Be able to create logical and physical designs for specified business systems	4.1 Logical design, i.e.: • platform independent • inputs • outputs • processes • data 4.2 Physical design, i.e.: • platform dependent • hardware, e.g. Intel, AMD, HP, Sun, Apple, IBM • software, e.g. operating systems, Java, .Net Framework • implementation environment, e.g. web browser

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	4.3 Refining designs based on stakeholder feedback, i.e.: • analyse the feedback: o identify types of problems o determine consistency of comments • decision on whether the refinements are viable • make changes to design in line with feedback and viability considerations

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the role of systems analysis and design in relation to the systems development lifecycle	P1: Summarise the main components of the systems development life cycle	M1: Compare and contrast a range of systems development life cycles	
2. Be able to use investigative techniques to establish requirements for business systems	P2: Explore the business requirements for an identified business system		
	P3: Use different techniques to support the analysis of the identified business system		
3. Be able to develop and document models for business systems	P4: Use the three view approach to document the design model for the identified business system	M2: Develop a set of Unified Modelling Language (UML) diagrams for the identified business system	D1: Evaluate the design model for the identified business system against original business requirements
4. Be able to create logical and physical designs for specified business systems	P5*: Create a logical design for the identified business system *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M3: Present the logical and physical designs for the identified business system to relevant stakeholders	D2: Refine the logical and physical designs for the identified business system incorporating any stakeholder feedback
	P6: Create a physical design for the identified business system		

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

Wherever possible the use of case studies or real work opportunities should be used to provide the opportunity for assessment. It is possible to use local examples such as a centre's functional area's information systems and requirements e.g. a new canteen information system or improvements to the library system.

A report should be presented with a title, section headings e.g. introduction, methodology, findings, conclusions and recommendations and page numbers. A report template could be provided as in most organisations, at this level, technicians would be expected to follow the organisational paperwork.

Learners are encouraged to use the Command Verb Definition Document to help them understand the requirements of the assessment. This document may be downloaded from the OCR website.

LO1 Understand the role of systems analysis and design in relation to the systems development lifecycle

P1: Learners must be able to summarise the main elements of the systems development life cycle and how they are linked. The position of analysis and design within the life cycle must be clearly described. The evidence could take the form of a recorded presentation by learners to a group, a presentation with detailed speaker notes or a written report. An essay is not appropriate.
M1: Learners should compare and contrast at least three different life cycles, only one of which should be from the linear range. They must consider the differences and similarities of the methods and how they influence the choice of lifecycle for a given business system. The evidence could be presented as a report or a presentation with detailed speaker notes. Learners could also record their presentation.

LO2 Be able to use investigative techniques to establish requirements for business systems

P2: Learners must have access to a live systems development or a detailed scenario or case study which will enable to them to carry out a thorough study of the business requirements for the system. They must have sufficient information to enable them to explore a range of business documentation and stakeholder views. This may include oral or written statements and questions from stakeholders, financial reports, press cuttings, incident reports, organisational goals and objectives, interview notes or live interviews of interested parties. The evidence must be formally presented either as a report or a detailed presentation, with speaker notes.
P3: This should be conducted alongside the study of the scenario in P2, as the activities should be recorded using at least three different examples of the techniques identified in the teaching content.

LO3 Be able to develop and document models for business systems.

P4: Learners must use the three view approach covering functionality, static data and events to provide a suitable design model of the requirements for a specified business system which may be that used for P2 and P3.
M2: Learners should convert the design model from P4 into a set of UML diagrams which must include at least one Use case diagram, one activity diagram and two interaction diagrams.
D1: Learners must communicate their design model to at least two stakeholders, one internal and one external. The communication can take the form of a report, a detailed presentation with comprehensive speaker notes or a video presentation.

LO4: Be able to create logical and physical designs for specified business systems

P5: Learners must create a logical design using appropriate diagramming techniques which are suitably annotated to provide a rationale for the stakeholders to consider.
P6: Learners must create a physical design using suitable diagramming techniques; diagrams should be annotated to provide a rationale for the stakeholders to consider.
M3: Learners are required to present their designs from P5 and P6 to the stakeholders for consideration. The evidence could take the form of a detailed report including the diagrams created, a presentation with detailed notes which also includes the diagrams, or a recording of learners presenting their findings to the stakeholders.
D2: Learners are required to analyse their logical and physical designs, including any stakeholder feedback. The learners will need to show how they have considered the feedback and evaluated the designs themselves, and will need to confirm on what basis they have refined their designs. The evidence will be the record of any feedback, the outcomes of the analysis of the feedback along with learners' own evaluation of the designs. This will be supported by the revised logical and physical designs and the rationale for the refinements made.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P4, P5, P6, M1, M2, M3, D1, D2
Problem solving	P1, M1, P2, P3, M2, P5, P6, D2
Time management	P2, P3, P4, P5, P6, M2, M3, D1, D2
Critical thinking	P1, M1, P2, P3, M2
Negotiation	P2, D2
Decision making	P1, P2, P3, P4,P5, P6, M1, M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners' work-experience could be in an experienced team; they could contribute to, or observe activities, such as documentation scrutiny, observation and questioning used to elicit the business requirements for a system. Learners could also contribute to the development of the systems analysis documentation, following the organisation's approach. They could also contribute to the development of the elements of the logical and physical design and provide some input to the presentation.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Learners could engage in a project, set by the analyst, where they would carry out a realistic range of activities, such as designing questionnaires for stakeholders, identifying relevant issues from the current documentation to identify business requirements and analyse their findings. Using the three view model, learners could provide the necessary design and communicate it to a client, possibly the systems analyst. From this, learners could create the logical and physical design, present it to the client and refine it in the light of the feedback received.

3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	The systems analyst could provide insight into the importance of the various elements of the systems development and their strengths and weaknesses. Learners could be given examples of documentation, questions and answers from real projects, and be shown how they are interpreted. It is important learners are taught about the positive, where it worked and the disaster where they did not. This helps learners to understand that this is not an exact science but each project has its own difficulties and triumphs. The systems analyst could also provide examples of the use of documentation to record the findings of the investigations and to create the logical and physical design, including stories of successes and failures.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	The systems analyst could observe elements of a centre based project or observe learners working on real projects in the work place. It is important that the expert witnesses knows to include ephemeral communication skills and methods as well as demonstration of skills such as questioning, document scrutiny and the creation of designs.

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 12

Mobile technology

R/507/5014

Guided learning hours: 60

Version 1 September 2015

ocr.org.uk/it

LEVEL 3

UNIT 12: Mobile technology

R/507/5014

Guided learning hours: 60

Essential resources required for this unit: Learners will need access to multiple mobile devices.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

You may come to this unit as a proficient user of a mobile phone but you may be less familiar with other mobile technologies and their operating systems. The aim of this unit is to broaden your knowledge and understanding of the wider potential of mobile technologies and its consequences to people and businesses. This unit is as much about new technologies as it is about promoting critical analysis of existing situations and proposing better solutions.

Technical developments move quickly and legislation usually lags behind. Tutors are encouraged to incorporate relevant new developments and laws into this unit.

This unit is an optional unit to the Emerging Digital Technology Practitioner pathway due to the strong links between mobile technology and emerging digital technologies. In addition it is also in the Application Developer pathway due to the increase in the use of mobile technology within business, as application developers need to have a good understanding of the uses of mobile technology and the technologies involved and as mobile technology is now in wide use by individuals and businesses. It is also in the IT Infrastructure Technician specialist pathway as it is important that IT technicians have an understanding of the hardware, software and technology involved. This unit will provide them with an insight into mobile technology providing them with the foundation to develop their skills, knowledge and understanding of this important area further in the future.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as CompTIA Mobility+.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand mobile technologies	1.1 Devices, i.e.: • Sizes (e.g. phone, phablet, tablet, sat-nav, smart watch) • embedded (e.g. laptop, watch, car for insurance purposes) • active versus passive • input (e.g. mini keyboard, voice, touchscreen) • output (e.g. print, audio and vibration alerts) 1.2 Connectivity, i.e.: • wired vs wireless (e.g. number of users, range, mobility, security, ease of access) • cellular technologies (e.g. CDMA, TDMA, GSM, CDS) • routing, e.g.: o IP address o IPv4 and IPv6 o DHCP vs static IP o Subnet Mask and Default Gateway o MAC address • network characteristics, e.g.: o bandwidth o latency o jitter o reliability • standards, e.g.: o GPS o 3G/4G/5G o Wi-fi o Blue-tooth o USB • limitations, ranges and uses of different standards, e.g.: o signal strength o effects of distance and intervening objects o IEEE 802.11 o frequencies e.g. 2.4 GHz, 5 GHz o channels and interference o RF characteristics (e.g. modulation, bandwidth, wavelength, amplitude, phase) o Preparation theory (e.g. absorption, refraction, reflection, interference) o Antennas (e.g. Omni-directional, semi-directional, bi-directional)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	1.3 Mobile device operating systems, i.e.: • Android • Windows • iOS • Linux based, e.g. Sailfish OS, Ubuntu Touch • future operating systems • advantages/disadvantages • characteristics • features 1.4 Current and potential future uses, e.g.: • Internet of Things • smart city • tagging endangered species • robotic devices e.g. iRobot vacuum cleaner • sat-nav • consumer internet e.g. price check in supermarket, QR codes • GPS • health monitors • CCTV • drones • ATC and TCAS • stock control and theft reduction with RFID • car trackers • electronic tagging of offenders • cars, e.g. self drive, traffic update
2. Be able to investigate how businesses use mobile technologies	2.1 Uses of mobile technologies, e.g.: • access to information from remote locations • maintain contact between staff • collaborative working • using cloud based solutions to reduce up-front technology costs • increase productivity and profitability • using social media for digital marketing 2.2 Ethical, e.g.: • child tracking • grooming and secrecy • personal data, e.g. location • employees always contactable • haves and have-nots
3. Be able to determine solutions for the use of mobile technologies	3.1 Investigating business requirements, i.e.: • client needs and wants • end user needs and wants • limitations (e.g. target platform, bandwidth, development resources, human resources, etc.) • prototyping • interfaces (e.g. hardware, software, communications, user)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.2 Planning, i.e.: • cost: o cost of ownership o frequent upgrades due to cutting edge o multiple device sizes and incompatible devices • risks e.g. data security • constraints, e.g.: o legal (e.g. tracking a lorry is also the tracking of the driver, countries have different laws) o technological (e.g. signal coverage areas, cutting edge technology implications) o compatibility between competing devices o power requirements (e.g. passive or active, charging of a battery powered device) • staff training 3.3 Technology business plan, i.e.: • current and future use (e.g. how is technology used now within the business and how will it be used in the future) • identifying appropriate technology (e.g. future hardware and software needs, integration with other technology and systems) • allow for advances in technology
4. Be able to present solutions for the use of mobile technologies	4.1 Promoting the mobile technological solution, i.e.: • summarising technical details to a non-technical audience • verbal versus written communication • making a convincing case • projecting confidence 4.2 Improvements to mobile technological solution, i.e.: • obtain feedback from stakeholders • analyse feedback • confirm viability of changes • implement improvements to solution based on viability of feedback 4.3 Predicting consequences of change, i.e.: • future actions or events which are measurable (e.g. increased profits, customer satisfaction, response time)

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand mobile technologies	P1: Describe different methods of mobile device connectivity	M1: Compare and contrast different operating systems used in mobile technology	D1: Evaluate the suitability of mobile technologies for different situations
2. Be able to investigate how businesses use mobile technologies	P2*: Carry out research on the impact of mobile technologies on businesses *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M2: Examine the ethical implications of the use of mobile technologies	
3. Be able to determine solutions for the use of mobile technologies	P3: Investigate the mobile technological requirements for an identified business need	M3: Prepare a technology business plan to support the implementation of the mobile technological solution for the business	
	P4: Plan a mobile technological solution for an identified business need		
4. Be able to present solutions for the use of mobile technologies	P5: Promote the mobile technological solution to relevant stakeholders		D2: Predict the effectiveness of the mobile technological solution for the identified business need
	P6: Improve the proposed mobile technological solution based on stakeholder feedback		

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

Passes are expected to be limited to factual information. Merits are expected to show comparison, analysis and justifications. Distinctions are expected to be written reports with balancing arguments and a conclusion.

LO1 Understand mobile technologies

P1: Learners are required to describe different methods of mobile device connectivity. They should not concentrate only on the connectivity methods of mobile phones, although the technologies used by mobile phones should be included. Descriptions should include characteristics and limitations as well as their uses. The evidence can be in the form of a report, a presentation with detailed speaker notes, a tutor resource or a technical guide.
M1: Learners are required to compare and contrast different operating systems used in mobile technology. The comparisons should include the features and characteristics of each system as well as the advantages/disadvantages of each. Evidence can be presented as a technical guide, a report or a presentation with detailed speaker notes.
D1: Learners are required to evaluate the suitability of mobile technologies for different contexts. Learners will draw on their knowledge and understanding for P1 and M1 when carrying out their evaluation of the different mobile technologies. Evidence can be in the form of a presentation with detailed speaker notes or a video of learners delivering the presentation to a 'business' to promote the concept of using mobile technology in their business. The evidence could also be in a form of a report or technical guide.

LO2 Be able to investigate how businesses use mobile technologies

P2: Learners are required to carry out research into the impact of mobile technologies on businesses. It is important that they consider the positive and negative impacts that mobile technologies have had. The evidence could be presented in the form of a report, a presentation with detailed speaker notes or a video of learners presenting to an audience, or a technical guide to help businesses make decisions with respect to implementing mobile technology in their business.
M2: Learners are required to examine the ethical implications of the use of mobile technologies. They could consider the ethical issues around monitoring locations, data sharing and costs. The evidence could be in the form of an information booklet, an article for a magazine or newspaper, a report or a presentation with detailed speaker notes.

LO3 Be able to determine solutions for the use of mobile technologies

P3: Learners are required to investigate the mobile technological requirements for an identified business need. Learners should be presented with a business scenario (although they could choose their own if they have identified a potential business). They should identify the business and give some background as to the functions of the business at the beginning of their evidence in order to 'set the scene'. The evidence can be in the form of a report or a presentation with detailed speaker notes (or a video of learners delivering the presentation), where learners present the results of their investigation into how the business could implement mobile technology, the types of technology that could be used, etc.
P4: Learners are required to plan a mobile technology solution for an identified business need. It is important that they document all aspects of planning as identified in the teaching content and consider the existing technologies within the identified business as part of their plan. The evidence could be in the form of a report to 'present' to the business for their consideration, or a planning document.
M3: Learners are required to prepare a technology business plan to support the implementation of the mobile technological solution for the business. The evidence will be the technology business plan.

LO4 Be able to present solutions for the use of mobile technologies

P5: Learners are required to promote the technological solution to relevant stakeholders and secure feedback. They will present their plan to the stakeholders for the business used in LO3. It is acceptable that there may not be an actual business available and therefore the teaching staff could undertake the role. Feedback should be recorded within their evidence. Learners could prepare a presentation with detailed speaker notes (or a video of them actually promoting the planned use to the stakeholder(s)). Evidence could also be presented as a formal report which would require accuracy with respect to spelling and grammar to ensure that it is fit for purpose.
P6: Learners are required to improve their proposed mobile technological solution based on stakeholder feedback. They must consider the feedback and identify how this feedback has influenced the improvements. Evidence should be the modified proposal and rationale for decisions made.
D2: Learners are required to predict the effectiveness of the mobile technological solution for the identified business need. The predictions should include some form of measurable value by which the effectiveness can be quantifiably judged. The evidence will be in the format of a report or presentation with detailed speaker notes.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P5, P6, M3
Problem solving/decision making	P4, M1
Time management	P4, P5, P6
Critical thinking	M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience with businesses who use mobile technology or who are considering the use of mobile technology.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	A local business can be invited to set the scenario for P2, to be the stakeholder for P5 and to provide the feedback for P6. Alternatively learners could visit businesses as part of their research.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A local shop could send a member of staff to speak about how they use RFID for stock control and to deter theft.

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Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**

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Cambridge **TECHNICALS LEVEL 3**

IT

Cambridge TECHNICALS 2016

Unit 13

Social media and digital marketing

D/507/5016

Guided learning hours: 60

Version 1 September 2015



LEVEL 3

UNIT 13: Social media and digital marketing

D/507/5016

Guided learning hours: 60

Essential resources required for this unit: Learners will need access to social media channels in order to assess their impact. Where learners are expected to evaluate social media campaigns, learners should be provided with quantifiable data that will enable them to draw informed conclusions.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The use of social media has increased massively over recent years and is now a world-wide phenomenon. Users of social media are able to share ideas and files, compare opinions and pass comment on the activities of their friends and contacts. In doing so, they are not only generating huge amounts of data about themselves, but also allowing others the opportunity to contact them and monitor some of their online activities. Social media also allows users to collaborate with others across the globe.

Digital marketing is part of the overall process of marketing and is the use of digital media to increase awareness of a product or service. As social media offers such a wealth of data and the ability to contact potential customers in their own homes across a range of media channels, it is only natural that digital marketing seeks to use social media as part of the marketing mix for goods and services.

This unit looks at digital marketing as a concept and then offers you the opportunity to explore the possible impacts, both positive and negative, that may be generated by the use of social media as a digital marketing tool.

This unit is in the Emerging Digital Practitioner, Application Developer and the Data Analyst Specialist pathways.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand digital marketing	1.1 The role of marketing within business, i.e.: • market research • raising awareness and affecting perception of need via promotion and advertising • selling 1.2 Digital marketing as a business tool, i.e.: • business establishment • business growth • business continuity 1.3 The tools of digital marketing, i.e.: • social media types (e.g. publishing - blogging and wikis, sharing - videos, music, images, discussing, networking - social, business) • email • landing page optimisation • banners and popups/unders • SEO (search engine optimisation) • paid and organic search results • channels (e.g. Facebook, paid advertisements on search engines, Twitter advertisements) 1.4 Digital marketing • strategies towards identified marketing goals, i.e.: o identifying potential customers and markets o setting short term and long term goals (e.g. raise awareness, increase sales, gaining information) o creating a marketing and sales funnel (e.g. awareness, interest, action) o developing a call to action o gathering data o creating traffic 1.5 Digital marketing life cycles • stages of the digital marketing life cycle (e.g. setup, traction, positioning, expansion, viral growth)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Understand the use of social media in a business	2.1 Research, i.e.: • primary data • secondary data 2.2 Data as a resource, i.e.: • data management • sources of data (e.g. relationship status, cookies) • collection of data • analysis of data (e.g. establishing patterns and trends) • sale of data 2.3 Use of data, i.e.: • identification of gaps in markets • identification of changes in customer habits and tastes • targeted marketing • planning campaigns 2.4 Communication, i.e.: • between staff • with customers (e.g. one way and two way) 2.5 Legislation and business policy and practice i.e.: • legislation (e.g. Data Protection Act 1998, other legal restrictions (e.g. specific restrictions due to the nature of a person's job), • business policy and practice (e.g. acceptable use policy, social media policy, recruitment policy) 2.6 Ethical and moral issues, i.e.: • bias (e.g. personal opinion versus fact, sponsorship, product placement, declaration of interest) • integrity (e.g. not disparaging the competition, honesty/misrepresentation, response to customers, misuse of free expertise (e.g. competitions))

Learning outcomes The Learner will:	Teaching content Learners must be taught:
3. Be able to plan content and propose appropriate social media channels for digital marketing campaigns	3.1 Social media channels (e.g.:Myspace, Twitter, Instagram, Facebook, Snapchat, blogs (e.g. including Wikipedia)) 3.2 Potential outcomes, i.e.: • planning techniques: o positive outcomes: ▪ increased market share (e.g. customer engagement, customer awareness, brand awareness) ▪ increased customer loyalty (e.g. impact of social media on the digital marketing life cycle) ▪ improved customer service (e.g. ease of access to customer data, speed and quality of advice and information to the customer via social media) ▪ effect on product life cycle (e.g. stages of the product life cycle and potential positive and negative impacts of social media) ▪ financial impacts • potential negative impact of social media on businesses and individuals: o work rate o bullying o threats o employability o business image 3.3 Possible restrictions, i.e.: • legislation • technological constraints • skills constraints 3.4 Target audience, i.e.: • market segmentation (e.g. age, gender, income, education levels)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
4. Be able to develop social media digital marketing campaigns	4.1 Features of a social media marketing campaign, i.e.: • appropriate staffing mix • clear objectives and shared targets • core message • unique selling point • company image • social media channels • timescales and review dates • social marketing funnel • social media tools e.g.: o social media measurement o social network aggregation o social media mining 4.2 Campaign considerations, i.e.: • business objectives • alignment with wider organisational marketing programme • format and restrictions on content • channels • frequency • image • bias • benefits • disadvantages 4.3 Effectiveness of digital marketing campaigns, i.e.: • comparison to original targets • feedback from the client or other suitable source • assessment against measurable outcomes e.g. o effect on sales and income (maintained, increased or decline slowed) o effects on customer loyalty (increased repeat sales) o effects on customer service (feedback to surveys, responses to popups) 4.4 Recommend improvements to business processes to support digital marketing campaigns, i.e.: • analyse assessment of measurable outcomes • review business processes in order to identify source of failure or short coming • make recommendations to identified business processes in order to improve effectiveness of digital marketing campaigns

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand digital marketing	P1: Outline the tools available for digital marketing		
	P2: Explain the stages of the digital marketing life cycle		D1: Assess the impact of digital marketing on an identified product
2. Understand the use of social media in business	P3: Describe how social media may be used to gather data	M1: Explain how data is used as part of social media digital marketing campaigns	
	P4*: Describe the legal and ethical restrictions on the use of social media as part of digital marketing campaigns *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		
3. Be able to plan content and propose appropriate social media channels for digital marketing campaigns	P5: Outline social media channels to be used in a digital marketing campaign	M2: Plan the social media content of the digital marketing campaign to meet identified business objectives	D2: Justify the use of identified social media channels in the digital marketing campaign
	P6: Describe the target audience for the identified social media channels		
4. Be able to develop social media digital marketing campaigns	P7: Propose a digital marketing campaign across different social media channels to meet identified business objectives	M3: Recommend adaptations to current business processes to support social media activities	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand digital marketing

P1: Learners must outline the tools available for a digital marketing campaign as outlined in the learning outcome (LO). The outline must provide a clear indication of what the tool is and how it is used within a digital marketing campaign. Evidence can be in the form of a written report, presentation with detailed speaker notes, a video of the information being presented to an audience, an information guide or an information-poster with detailed annotations.
P2: Learners are required to explain the purpose of each of the five stages of the digital marketing life cycle. Evidence can be in the form of a written report, presentation with detailed speaker notes, a video of the information being presented to an audience, an information guide or an information poster with detailed annotations.
D1: Learners will need to research into a digital marketing campaign and be able to assess the impact that the campaign had on the identified product. The identified impact may be positive, negative or a mixture of both. Evidence can be in the form of a written report, a presentation with detailed speaker notes or a video of the information being presented to an audience.

LO2 Understand the use of social media in business

P3: Learners are required to describe how social media may be used to gather data. In doing so, they should describe the range and types of data that may be garnered from social media and the relevance of the data that is gathered. Evidence may be in the form of a written report or an information guide.
M1: Learners are required to explain the use of data as part of a digital marketing campaign. This should concentrate on how data is gathered, analysed and applied to inform the planning and eventual outcome of the campaign. This should not concentrate on social media as the sole source of data, but on the role of data as a whole. Evidence may be in the form of a written report or an information guide.
P4: Learners need to research into the legal and ethical restrictions on the use of social media as part of a digital media campaign. The exploitation of data that users may supply with no awareness of its importance as a marketing tool should be explored, as well as the need to process such data fairly.
Learners may describe the legal and ethical considerations in general terms, or may be given a scenario, for which they then analyse the potential issues. The evidence may be in the form of a report, describing social and ethical considerations in general or, where a scenario is used, within the context of that scenario.

LO3 Be able to plan content and propose social media digital channels for digital marketing campaigns

It is important that learners are provided with a scenario for a digital marketing campaign that will enable them to propose suitable social media channels. This may be a wholly theoretical campaign produced by the teacher, or may be a campaign suggested by a business link.
P5: Learners are required to outline social media channels to be used in a digital marketing campaign. The intended channels should be related to the campaign and should reflect the full range of technologies available. Evidence could be in the form of a report or a presentation to an audience (videoed or with detailed speaker's notes).
P6: Learners are required to describe the target audience for the scenario with which they have been presented. The target audience may be described in many terms, but the use of standard market segmentation analysis would allow learners to link the description to the techniques they then suggest. However, consideration must be given to differing markets within identified social media channels. Evidence could be in the form of a report or a presentation to an audience (videoed or with detailed speaker's notes).
M2: Learners are required to plan the social media content that they will use as part of their digital marketing campaign. There is the opportunity for learners to take an innovative/creative approach to this task and the evidence could be a report, presentation to an audience (videoed with detailed speaker's notes) or a recording of an interview with the client or a visualisation.
D2: Learners are required to provide a clear and relevant rationale for their suggested social media channel(s). The rationale should consider arguments for and against each channel (where applicable) and come to a considered conclusion. There should be clear evidence that the specific needs of the project and the target audience have been considered in reaching the conclusion. Evidence could be in the form of a report, or a presentation to an audience (videoed or with detailed speaker's notes).

LO4 Be able to develop social media digital marketing campaigns

Learners need to develop the project on which they focused as part of LO3.
P7: Learners are required to create a proposal document for a social media marketing campaign, as outlined in the specifications. All features should be considered as part of one overall marketing campaign. This may be based on a scenario, or may be a preparatory report based on general understanding. Evidence could be in the form of a report, or a presentation to an audience (videoed or with detailed speaker's notes).
M3: Learners are required to suggest suitable adaptations to business processes so that any identified failure(s) and/or shortcoming(s) in their campaign may be reduced.
The evidence could be a report, a presentation to an audience (videoed with detailed speaker's notes) or a recording of an interview with the client, outlining how each feature is used as part of the solution to the project.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, M1, M2, M3, D1,D2
Problem solving/decision making	P7, M2, D2
Time management	P7, M3
Critical thinking	M2, D2, P7, M3

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could carry out work experience at a local business who is interested in using social media as a marketing tool. They could investigate what and how social media could be used and the benefits to the business concerned. They could develop a proposal for the business which they present to them to include how the business could monitor the success of the use of social media for marketing.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	You could set a project covering the content of this unit with a local business that is interested in using social media as a marketing channel.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A local business that uses digital marketing and/or social media could contribute to the delivery of this unit by co-delivering teaching on how they use digital marketing/social media in their business (LO1 & LO2).
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	If you set a project with a business in point 2 you could ask the business to be an 'expert witness' by contributing to the assessment of the learners' project.

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ocr.org.uk/it
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 14

Software engineering for business



H/507/5017

Guided learning hours: 60

Version 1 September 2015

ocr.org.uk/it

LEVEL 3

UNIT 14: Software engineering for business

H/507/5017

Guided learning hours: 60

Essential resources required for this unit: A programming language installed and available for all learners to use and a tutor that is proficient with this language.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The aim of this unit is to give you practical experience of writing computer programs for specific requirements, such as those found in a business. Programmers' first jobs tend to be coding against specific requirements or maintaining an existing program. You need only a basic appreciation of the full system life cycle at this level, freeing up more time for practical programming experience.

This unit focuses on developing code for a single customer with specific requirements. If you are on the application developer pathway, you could follow this unit with the games design and prototyping unit, which focuses on developing for a mass market, allowing you to further develop your coding skills and experience.

This unit is optional within the Applications Developer and Data Analyst specialist pathways. Application developers create, test and program applications software and therefore use the knowledge skills and understanding associated with software engineering. Data analysts require a good understanding of computers and software and an insight into software engineering can assist them in their design and analysis of systems and data.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand universal programming constructs	1.1 Programming constructs, i.e.: • encapsulation (e.g. modules, procedures, functions, classes, properties and methods) • predefined subroutines (e.g. string manipulation, screen output). • the use of parameters and return values (one technique is sufficient and depends on chosen language e.g. functions and return value, byref and byval parameters, getters and setters) • declaring and using variables with fundamental data types: o integer o floating point o string (or equivalent e.g. array of chars) o boolean • converting between data types • selection (if, then, elseif, else, endif) • iteration (fixed loop, conditional loop e.g. pre-condition, post condition) • GUI objects and their different applications (e.g. text box, label, button, check box, radio button, dropdown list, combo box, multi-line lists, menus)
2. Be able to investigate business requirements for programming solutions	2.1 A modern incremental system life cycle in which stakeholder feedback feeds back into the next round of development (e.g. Rapid application development (RAD), Agile, Spiral, Scrum) 2.2 Requirements specification, i.e.: • requirements (e.g. customer requirements of the new system) • investigation (e.g. existing system observations, data used, interview with the stakeholder) • analysis (e.g. tasks the existing system carries out, mission critical tasks, good and bad points about the existing system, integration with other systems, any specific processing known). • feasibility (e.g. scope, cost/benefit analysis) • phased development (which requirements will be in which versions)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
3. Be able to develop software solutions to meet business requirements	3.1 Design specification, i.e.: • what is it going to look like? (UI design) • what is it going to do? (e.g. flowchart, JSD, pseudocode) • processing (e.g. algorithms, mathematical formulae, input process output) • coding • methodology (e.g. top-down with stubs, bottom-up with reusable modules) • meaningful names (e.g. modules, procedures, variables) • house style (e.g. Hungarian notation) • debug tool (e.g. trace comments, profilers, breakpoints, stepping and watches) • testing • against requirements, to prove project completion and avoid requirements creep (stakeholders adding more requirements into the project as it progresses) • robustness (e.g. invalid data range, invalid data type, data extremes, data presence) • repeatable test data (data that is used in the tests) 3.2 Adaptations to design, i.e.: • obtain feedback from stakeholders • analyse feedback • confirm viability of changes • make viable changes to design based on analysed feedback 3.3 Design evaluation, i.e.: • how and why does the design meet the stakeholder requirements?
4. Be able to propose software solutions to meet business requirements	4.1 Presenting to an audience, i.e.: • personal skills (e.g. speech clarity and speed, appearance, gait, personal cleanliness) • interpersonal skills (e.g. facing audience, eye contact, verbal confidence, non-verbal signals such as smiling, hand gestures, nerves and rocking, handling difficult questions) • communication skills (e.g. summarising concepts, adapting content to audience composition, polite addressing of strangers, use of appropriate speech, the dangers of verbose slides, the dangers of auto-timed presentations) • preparation (e.g. prompt cards, practice, knowing what is wanted from a meeting before it starts, preparation for equipment failure, potential questions) 4.2 Adaptations to prototype, i.e.: • obtain feedback from stakeholders • analyse feedback • confirm viability of changes • make viable changes to prototype based on analysed feedback

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand universal programming constructs	P1: Explain the purpose of programming constructs when designing software for business		
2. Be able to investigate business requirements for programming solutions	P2*: Outline the requirements for a software solution to meet an identified business need *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M1: Propose a requirements specification for a software solution	D1: Discuss the feasibility of the proposed software solution
3. Be able to develop software solutions to meet business requirements	P3: Prepare a design specification for an identified software solution	M2: Make adaptations to final design specification following negotiations with stakeholders	D2: Evaluate the software solution prototype against stakeholder requirements
	P4: Create the software solution prototype based on the design specification		
	P5: Test the prototype software solution, rectifying issues		
4. Be able to propose software solutions to meet business requirements	P6: Demonstrate the software solution prototype to relevant stakeholders	M3: Adapt the software solution prototype based on stakeholder feedback	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand universal programming constructs

P1: Learners should explain the purpose of the different constructs in the teaching content. The explanations should include what they do and could be supported by examples. This could be in the format of a booklet, teaching resource, technical guide or report detailing and explaining the purpose.

LO2 Be able to investigate business requirements for programming solutions

P2: Learners are required to outline the requirements for a software solution to meet an identified business need. Learners should use the headings listed in the teaching content. Rather than describing generally what each section might contain, this must be related to the given business need. The evidence will be the outline requirements for the software solution.
M1: Learners are required to propose a requirements specification for a software solution. The outline requirements specification from P2 should be extended to clearly show where the new system will fit into the existing business, what it will replace and the existing processes with which it needs to integrate. The requirements will be achievable and provable. The evidence will be the proposed requirements specification.
D1: Learners are required to discuss the feasibility of the proposed software solution. The feasibility should be sufficiently detailed to confirm that all aspects of the business need and proposed software solution have been considered. The evidence could be presented as a report or as a presentation with detailed speaker notes.

LO3 Be able to develop software solutions to meet business requirements

P3: Learners are required to prepare a design specification based on the outline solution produced for P2. This could be a UI design, a form of flow design and described processing. The UI can be designed directly onto the IDE using the GUI design tools, or drawn either on paper or electronically. The flow can be pseudo code, flowchart, JSD or equivalent design method. There should also be a description of processing needed to turn the expected input into the desired output. This can be an algorithm or mathematical formula.
M2: Learners are required to make adaptations to their designs based on stakeholder feedback. Some requirements may change; some may be dropped or rephrased to avoid requirements creep. This may give learners an opportunity to identify with the stakeholders which requirements are critical, which are only desirable and in which release each requirement will appear. Any discussion should then be documented along with changes to the design specification.
P4: The program does not have to successfully meet all requirements but the prototype should follow the required specification. The main form of evidence will be the created prototype. The program code does not need to be printed out but must be available for inspection on request.

P5: There must be a test plan mapping every requirement against at least one test and it should include tests for robustness. There should be evidence of real data for repeatable tests, though not all tests are required to have actual data. Tests should be accompanied with evidence that the test was carried out, such as screen shots or a video of the program running them. Though not all tests need to be passed, in the case of a failed test, there should be evidence of how learners attempted to find the problem and rectify it.
D2: Learners will evaluate how each requirement was met or attempted. They should use their M1, M2 and D1 to feed into D2. For each requirement, learners will note how successful their solution was in satisfying the requirement. The evidence will be the documented evaluation and could be in the form of a report or a presentation with detailed speaker notes.

LO4 Be able to propose software solutions to meet business requirements

P6: Learners must demonstrate the software solution prototype to their stakeholders, formal panel, or a person unfamiliar to the project. Slides and hand-outs can be used if desired, but the presentation must feature a demonstration of the program running. The evidence could be in the form of a video of the learner delivering the presentation; presentation slides with detailed speaker notes should be supported by detailed witness statements and the actual prototype.
M3: Learners should analyse the feedback from the stakeholders and make adaptations to their software solution prototype based on the feedback. The evidence will be the feedback from the stakeholders, the adaptations to the software solution design and the prototype.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P6, M2, M3
Problem solving/decision making	P2, P3, P4, P5, M1, M2, M3
Time management	P4, P6
Critical thinking	M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work experience or work placements that develop skills and knowledge relevant to the qualification.	Learners could have work experience with local software development organisation where they job shadow and support industry practitioners in the development of software. This would provide them with an insight into the importance of the development lifecycle and the need to meet strict deadlines as well as the stages taken when developing software.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Local industry practitioners could assist you in developing realistic scenarios based on the types of development activities that are required within the sector.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Industry practitioners could deliver the technical elements of the unit for example, programming constructs, how to investigate business requirements or how to develop software solutions to meet business requirements.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	If an Industry practitioner worked with you on point 2, they could assist in the assessment of the learners' evidence based on the scenario that they assisted in developing.

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 15

Games design and prototyping

K/507/5018

Guided learning hours: 60

Version 1 September 2015

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LEVEL 3

UNIT 15: Games design and prototyping

K/507/5018

Guided learning hours: 60

Essential resources required for this unit: Learners should have access to a range of games for different audiences and purposes. They should also have access to suitable development software to allow them to build a prototype for a game.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Gaming is a continuously developing market. There are a number of platforms available for game developers to release games that they have designed and developed. This unit will help you develop skills in designing and developing a prototype for a simple game. It will enable you to consider the logic of the programming structures required, as well as the interface design. You will then build a prototype in order to demonstrate an element of your game.

This unit is optional within the Application Developer specialist pathway. Games designers and developers design and develop games across a wide range of applications and platforms.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand the principles of game design and prototyping	1.1 Game design principles, i.e.: • purpose of a game • audience of a game • common features in games • first/third person • player characters • non-player characters • sprites • achievement attainment • maintaining a player's interest • player interaction/controls • player immersion • progression • competition (e.g. highscore) 1.2 Game prototype, i.e.: • types of prototyping e.g. visual/representational (non-working) and proof of concept (working) • testing concepts • gauging player interest • skill level required • gauging difficulty of achievement • clarity of interface • clarity of understanding of gameplay/goals 1.3 Benefits of building a prototype, i.e.: • ensures the design does what it is supposed to do • helps to identify and address problems at an early stage • gives the client and end-user an appreciation of the final product • allows developer to explore ideas and obtain feedback with the client and end-user

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to develop game concepts	2.1 Game elements, i.e.: • navigation • scoring • movement • interaction/controls • conveying information • sound • levels • enemies • problem solving 2.2 Interface design, i.e.: • layout • colour palette • text styles • sound • stage/scene • actions (e.g. menus/buttons) 2.3 Stage design, i.e.: • perspective (e.g. 2D/3D) 2.4 Character generation, i.e.: • bitmaps • wireframe 2.5 Documentation i.e.: • requirement specification • design specification • project plan • system flowchart • top-down • JSP 2.6 Logic structure (e.g. clear definition of objectives of game, flow chart showing the 'flow' of the game through single or multiple layers with single or multiple players)
3. Be able to develop game prototypes	3.1 Use of bespoke software for game development, e.g. Gamemaker 3.2 Programming techniques, i.e.: • variables/constants • strings • inputs • outputs • sequence • selection • iteration (e.g. counting/conditional) • subroutines (e.g. functions/procedures) • conditions • counting • totalling

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• data structures (e.g. arrays/lists) • file handling • maintainable code • libraries 3.3 Testing, i.e.: • test plans • test data • black box • white box • alpha • beta • user testing
4. Be able to present and evaluate game concepts	4.1 Present the solution, i.e.: • choose a suitable method of presentation (e.g. live demonstration, use of presentation software, report) • plan a presentation to incorporate: o comparison of game concept against requirements o demonstration of functionality o demonstration of interactivity o demonstration of responsive design o justification of design choices o present the solution to the client 4.2 Evaluation, i.e.: • has it met the requirements? • does it reflect the design? • is it suitable for the identified audience and purpose? • suggest recommendations for full game concept • justify the continued development of the full game concept

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the principles of game design and prototyping	P1*: Outline key considerations that support games design *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M1: Compare and contrast the features of games for different audiences	
	P2: Explain the benefits of developing game prototypes		
2. Be able to develop game concepts	P3: Create a design for an identified game concept	M2: Prepare alternative interface designs for the identified game concept	D1: Justify the design rationale for the identified game concept
	P4: Produce a logic structure for the identified game concept		
3. Be able to develop game prototypes	P5: Build a prototype using core programming techniques and test for functionality		
4. Be able to present and evaluate game concepts	P6: Present the prototype to stakeholders to obtain feedback on the games concept	M3: Make changes to the games design and prototype based on stakeholder feedback	D2: Evaluate the game design and prototype against the identified game concept

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the principles of game design and prototyping

P1: Learners should outline key considerations that support games design. Learners are required to identify key features in a range of games and describe their purpose. This could be most effectively achieved by reviewing different types of games with a different focus, genre or purpose. Common key features are identified in the teaching content. The evidence for this could be presented as a written review or report, or a presentation.
M1: Learners are required to identify the features of games for different audiences. They are required to compare the use of features and comment on the suitability for different audiences and genres. This can be an extension of P1, and evidence can be presented as a written review or report.
P2: Learners are required to explain the benefits of developing a prototype for a game considering the criteria within the teaching content. The evidence for this could be presented as a report or a presentation (either videoed or with detailed speaker notes).

LO2 Be able to develop game concepts

P3: Learners are required to create a design for a game concept. They need to produce full design documentation that includes chosen game elements and relevant interface designs, stage designs and character generation, in addition to the core considerations and purpose for the game. Designs must contain enough detail to enable them to be understood by a third party.
M2: Learners are required to prepare alternative interface designs to the one identified in P3. The alternative designs must contain enough detail to enable them to be understood by a third party. Evidence can be an extension of P3 and be presented as additional visualisations or written explanatory designs, or a combination of both.
P4: Learners are required to produce a logic structure for their game concept using suitable documentation identified in the teaching content. Evidence would be the completed documentation, which may include diagrams, etc.
D1: Learners are required to justify their design choices and explain why they are suitable for the identified audience and purpose of the game concept. Evidence can be an extension of P3 and M2, and can be an addition to the design documentation, a presentation or a report, but should reference the designs submitted.

LO3 Be able to develop game prototypes

P5: Learners are required to understand different ways that a prototype can be produced. They are required to demonstrate the techniques identified in the teaching content to build a prototype for the game concept. The prototype could be a particular level or module of the full game concept. Evidence of the prototype could be presented in the form of annotated screenshots or as a video of a working prototype, or the prototype itself. Learners are required to plan and carry out testing of their prototype. Testing must be thorough enough to prove the prototype is functional and meets requirements.

LO4: Be able to present and evaluate game concepts

P6: Learners are required to present the prototype to stakeholders to obtain feedback on the games concept. The evidence could be in the form of a video of the learner demonstrating the prototype to the stakeholders and receiving feedback. Evidence could also be documented feedback from the stakeholders supported by a detailed witness testimony explaining how the learner delivered the presentation and the reaction from the stakeholders.
M3: Learners are required to analyse the results of the feedback from the stakeholders and make changes to the games design and prototype based on the feedback provided.
D2: Learners are required to evaluate the game design and prototype against the identified game concept. They should evaluate each stage of the design and provide a rationale as to how their game design and prototype reflects the original game concept.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P4, P6, M1, M3, D1, D2
Problem solving	P3, P4, P5, P6, M2, M3
Time management	P3, P4, P5, P6, M1, M3, D2
Critical thinking	M1, M2, M3, D1
Decision making	P3, P4, P5, P6, M2, M3, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience with a local game design/development business; the work experience could be structured to all or some of the learning outcomes in this unit.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	You could use an industry practitioner to develop a scenario/project for the development of a game.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A guest speaker from a local game design/development company could present a guest lecture on they develop game concepts and prototypes. An industry practitioner could co-deliver some of the more technical elements of the unit such as how to produce the logic structure design or alternative interface designs.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Industry practitioners could act as expert witnesses by observing the learners developing their prototype game designs and providing feedback to contribute to the assessment of the learners.

To find out more
ocr.org.uk/it
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 16

Developing a Smarter Planet



M/507/5019

Guided learning hours: 60

Version 1 September 2015

LEVEL 3

UNIT 16: Developing a Smarter Planet

M/507/5019

Guided learning hours: 60

Essential resources required for this unit: Access to research material and the internet

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Changes in technology over the last century now mean that we live in a Smarter Planet. You will consider how the evolution of technology has impacted on everyday life, and why the Smarter Planet is important for a global society. You will investigate the evolution of the Smarter Planet in a variety of contexts, including the impact it is having on society. You will consider potential Smarter Planet developments and put forward a business proposal for a Smarter Planet concept to potential stakeholders, revising the business proposal as necessary following their feedback.

This unit is optional within the Emerging Digital Technology Practitioner and IT Infrastructure Technician specialist pathways. The technology used within these two pathways can have a major impact on the sustainability of the Smarter Planet.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as Cisco IoE.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand what is meant by a Smarter Planet	1.1 Evolution of a Smarter Planet, i.e.: • improvements to original developments e.g.: o radio to DAB o telephones to mobile o manual to automated machinery o greener IT • purpose to, e.g.: o speed processes o improve efficiency o reduce waste and inefficiency o harness natural resources • human factors, e.g.: o reduce manpower requirements o improve quality of life 1.2 Importance for a global society, i.e.: • principles, e.g.: o information o instrumented o interconnected • focus/objectives (e.g. why do we need it?) • relevance 1.3 Impacts, i.e.: • environmental, e.g.: o pollution o food miles o increased energy consumption • ethical, e.g.: o health and transplants o the internet o data privacy • social, e.g.: o acceptance o communication o exposure to threat o reduced face-to-face communication

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• individuals, e.g.: o health o labour saving o time saving o flexibility o accessibility • life styles, e.g.: o health o comfort o travel o communication o social 1.4 Business sectors, e.g.: • healthcare • environmental • engineering • manufacturing • IT, retail • electronics • transport 1.5 Sectors with applications of a Smarter Planet, e.g.: • banking • construction • towns and cities • computing and data storage • education • energy • healthcare • infrastructures • oil • products • regulatory bodies • retail • telecoms • transport/traffic • water • e-commerce • environmental
2. Be able to propose ways to extend the scope of the Smarter Planet	2.1 Features for extension to Smarter Planet developments, i.e.: • use of information • instrumented, i.e.: o measure, sense and monitor conditions • interconnected, i.e.: o people, systems and objects can communicate in new ways • intelligent, i.e.: o infuse intelligence into systems and ways of working o models to manage massive amounts of data generated by the end-user devices o sensors

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• use of analytics, i.e.: o to translate data into making systems, processes and infrastructures more efficient, more productive and responsive and ultimately, making them smarter 2.2 Feasibility study, i.e.: • scope • cost/benefit analysis • viability • technology • stakeholder and global impact 2.3 Concept proposal, i.e.: • identification of concept, i.e.: o purpose o objectives • stakeholders, i.e.: o people involved o companies o manufacturers • requirements, i.e.: o technology o sectors • deliverables, i.e.: o what is achievable o benefits
3. Be able to present, refine and evaluate Smarter Planet concepts	3.1 Present concept proposal, i.e.: • audio/video • questionnaire • interview • pitch • focus group 3.2 Refining concept proposal, e.g.: • taking account of stakeholder feedback • consider viability of options • reduce costs • extend scope 3.3 Evaluation of concept proposal to sustainability of the Smarter Planet, i.e.: • speed processes • improve efficiency • reduce waste and inefficiency • harness natural resources • reduce manpower requirements • improve quality of life

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand what is meant by a Smarter Planet	P1*: Describe the evolution of the Smarter Planet in different global situations *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		D1: Evaluate why the Smarter Planet concept is important for a global society
	P2: Describe the impacts of the Smarter Planet on society	M1: Explain the impact of the Smarter Planet within a specified sector	
2. Be able to propose ways to extend the scope of the Smarter Planet	P3: Propose potential Smarter Planet developments	M2: Conduct a feasibility study for an identified Smarter Planet development	
	P4: Create a concept proposal document for the identified Smarter Planet development		
3. Be able to present, refine and evaluate Smarter Planet concepts	P5: Present the concept proposal document to potential stakeholders	M3: Refine concept proposal document based on stakeholder feedback	D2: Evaluate how the concept proposal will contribute to the sustainability of the Smarter Planet

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit, they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand what is meant by a Smarter Planet

P1: The learner will describe the evolution of the Smarter Planet for different global situations as identified in the teaching content for the unit. Each situation must be from a different category in the teaching content. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the learner presenting the information to an audience, or an information guide on the evolution of the Smarter Planet in different situations.
D1: The concept of the Smarter Planet will be evaluated in relation to its importance for a global society. A variety of concepts should be evaluated. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the learner presenting the information to an audience, or an information guide.
P2: The Smarter Planet has impacted on a number of areas of the global society. The learner should describe different impacts. Evidence for this criterion may link with criterion P3. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the learner presenting the information to an audience, or an information guide on areas of impact within a global society.
M1: The evidence for this assessment criterion will link with criterion P2 and will focus on the impact of the Smarter Planet within a specified sector. The selected sector does not need to be IT based, and learners should consider examples identified in the teaching content. The evidence can be presented as a newspaper article, report or presentation with detailed speaker notes.

LO2 Be able to propose ways to extend the scope of the Smarter Planet

P3: In order to identify potential developments using Smarter Planet advancements, it will be necessary for learners to explore existing developments. These developments could be an extension to an existing application, a new application or taking an existing application and identifying how it could be developed in a different context. There is an opportunity to link evidence for this criterion with LO1. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the learner presenting the information to an audience or an information guide identifying potential developments.
M2: The learner will be required to conduct a feasibility study for one of the developments identified in P3. The feasibility study should consider the items listed in the teaching content, but may incorporate other wider considerations if required. The evidence will be in the form of a report.
P4: Evidence for this criterion is likely to be an extension of that provided for criterion P3. Learners should produce documentation to enhance one of the developments into a concept proposal document. The evidence will be the formal concept proposal document.

LO3 Be able to present, refine and evaluate Smarter Planet concepts

P5: This criterion requires learners to prepare and deliver a concept proposal document to potential stakeholders. The concept proposal should cover the headings identified in the teaching content for the unit. The evidence could be in the form of a report, or a formal presentation with detailed speaker notes supported by the concept proposal document.
M3: Learners are required to consider the feedback from the potential stakeholders and revise the concept proposal. The concept proposal produced for criterion P5 will be updated to reflect the stakeholder feedback. The evidence will be the revised concept proposal and the learner's rationale for making the revisions based on the stakeholder feedback.
D2: Learners are required to evaluate how their concept proposal will contribute to the sustainability of the Smarter Planet. Some of the angles they may like to consider are listed in the teaching content. The evidence could be in the form of a report or a presentation with detailed speaker notes.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**cisco**.com/UK

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, P4, P5, M1, M2, M3, D1, D2
Problem solving	P3, P4, P5, M2, M3, D2
Time management	P1, P2, P3, P4, P5, M1, M2, M3, D1, D2
Critical thinking	P2, P4, P5, M1, M2, M3, D1, D2
Team working	P3, P4, P5, M2, M3, D2
Negotiation	P5, M3
Decision making	P4, P5, M2, M3, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake an investigation during work experience to establish how developments in technology have impacted on an application within a company and how it has affected the employee(s) and the company and identify how it affects wider society.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Industry practitioners could be used as 'expert witnesses'. Learners could deliver their concept proposal and receive feedback. In turn the learners would revise the business proposal as a result of the feedback. This would provide practitioners with the opportunity to be contribute to the assessment of LO3.

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Cambridge **TECHNICALS LEVEL 3**

IT

Cambridge TECHNICALS 2016

Unit 17

Internet of Everything



H/507/5020

Guided learning hours: 60

Version 2 - revised May 2016

***changes indicated by black vertical line**

LEVEL 3

UNIT 17: Internet of Everything

H/507/5020

Guided learning hours: 60

Essential resources required for this unit: Access to the internet and research materials

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

This unit is about the use of the internet and how it is impacting people and society. You will learn about the Internet of Everything (IoE) and how it is used. Using your knowledge you will carry out a feasibility study for a potential idea. You will pitch your idea to potential stakeholders and use their feedback to revise your proposal.

This unit is an optional unit in the Extended Certificate and all of the specialist pathways in the Diploma suite, as the Internet of Everything is everywhere. The Internet of Everything is expanding, appearing in all of the everyday devices found in homes, businesses and cities.

Knowledge gained in the study of this unit will also help prepare you for relevant industry qualifications such as Cisco IoE.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand what is meant by the Internet of Everything (IoE)	1.1 Things, i.e.: • physical objects • experiential interactions • aids to people • aids to society/community • machines 1.2 Where the IoE is used 1.3 Applications of the use of the IoE, i.e.: • body/health • home/garden • city/neighbourhood • industry • the environment 1.4 Global impacts, i.e.: • positive • negative • cost savings • increased productivity • new sources of revenue • enhanced citizen experiences 1.5 The four pillars of the IoE, i.e.: • people • data • process • things (devices and objects) 1.6 People, i.e.: • students • members of society • connecting people in relevant ways

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	1.7 People and how they connect, i.e.: • devices • social networks • wearables 1.8 Converting data into information to allow people to make decisions 1.9 Data, i.e.: • raw data • connected things (devices) • analysis • decisions • results 1.10 Information gathering devices, i.e.: • computers • smart phones • vehicles • other devices with sensors 1.11 Process, i.e.: • delivery of information • delivery to whom • timing of processing • methods to deliver processed information 1.12 Processing capabilities, i.e.: • local • server • big data technologies • cloud services 1.13 Connectivity, i.e.: • wired • Wi-Fi • Bluetooth • local • cloud-based • data centre-based • radio-frequency identification (RFID) 1.14 Networked connection, i.e.: • manageable • intelligent • secure • scalability • congestion 1.15 Security issues, i.e.: • ownership of personal information • unauthorised access and misuse of personal information

	• facilitating attacks on other systems • can be more challenging to secure than a home network
Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
2 Be able to repurpose technologies to extend the scope of the IoE	2.1 Developments, i.e.: • body/health, e.g.: o sensors, e.g. wearable thermometer o social safety wearables o Wi-Fi mattress cover o Bluetooth stethoscope o biometric patch o running analytics o Bluetooth weather sensor o Bluetooth maps for visually impaired o Bluetooth sunglasses • home/garden, e.g.: o smart air conditioner o Bluetooth tape measure o smart locks o smart lights o smart batteries o global location devices o Bluetooth measurement jars o Bluetooth flower pots o wireless water shutoff o Wi-Fi shopping lists o solar powered window blinds o Wi-Fi gas and carbon monoxide detectors • city/neighbourhood, e.g.: o real-time air traffic o smart signage o bicycle barometer o city dashboard o intelligent street lights o taxi locator o surveillance systems o wearable air quality sensor o smart urban furniture o connected car safety devices • industry, e.g.: o industrial smart helmet o smart glasses for warehouses o wireless pest monitoring o smart paving capturing kinetic energy o intelligent packaging o smart luggage/cargo o workforce driving monitors o connected e-paper displays o Wi-Fi cold storage monitoring o smart noise sensors o smart bottle labels

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	o wireless calving alert sensors o real-time remote excavation • the environment, e.g.: o environmental monitoring o wildlife tracking o flood detection network o illegal deforestation monitoring o landslide detection systems 2.2 Feasibility study, i.e.: • identify new opportunities through investigative process • evaluation and analysis of the proposed concept proposal • evaluation of alternative proposals • market assessment • results and conclusions (e.g. whether to proceed or not)
3 Be able to present concept ideas for repurposed developments	3.1 Business proposal, i.e.: • target audience • processing required • data to be exchanged • things • networking requirements • devices to be used • security issues 3.2 Pitch, e.g.: • report • presentation • website/multimedia 3.3 Feedback, e.g.: • stakeholders • developers • written • verbal 3.4 Stakeholder considerations, i.e.: • who benefits from the application? • how does society benefit? • how do companies benefit? • who will develop the application? 3.5 Revision of proposal, i.e.: • analyse the feedback o identify types of problems o determine consistency of comments • decision on whether the proposal is still viable • make changes to proposal in line with feedback and viability considerations

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.6 Possible success criteria (must be measurable), e.g.: • improved efficiency • increase in profits • increase in productivity • reduction in wasted time • reduction in overhead costs

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand what is meant by the Internet of Everything (IoE)	P1: Explain the concept of the IoE	M1: Analyse the global impacts of the IoE on society and the environment	
	P2*: Explain the four pillars of the IoE and how its innovations can transform businesses *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*		D1: Evaluate the potential negative impacts of these innovations on businesses
2. Be able to repurpose technologies to extend the scope of the IoE	P3: Outline potential development projects that could extend the scope of the IoE	M2: Conduct a feasibility study on one of these development projects	
3. Be able to present concept ideas for repurposed developments	P4: Prepare a business proposal for the chosen development project		
	P5: Deliver a business proposal pitch to potential stakeholders on the chosen development project	M3: Revise business proposal for the chosen development project incorporating stakeholder feedback	D2: Evaluate the success criteria that would be used to judge the sustainability of the chosen development project

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand what is meant by the Internet of Everything (IoE)

P1: Learners must explain what the concept of the IoE is. Consideration should be given to what the t*hings* are that are part of the IoE and where the IoE is used. Understanding of what the IoE is will be illustrated by examples taken from a variety of applications. The examples used by learners do not have to be confined to the categories listed in the teaching content. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the information being presented to an audience, or an information guide on the IoE.
M1: Learners are required to analyse the global impacts that the IoE has on individuals, society and the environment. This could be an extension to the evidence generated in criterion P2. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the information being presented to an audience, or an information guide on analysing the global impacts the IoE has on society and the environment.
P2: The interconnection of the four pillars of IoE (people, data, process and things) in the evolution of technology must be explained with an explanation of how these innovations can transform businesses. Evidence can be in the form of a written report, presentation with detailed speaker notes, a video of the information being presented to an audience or an information guide.
D1: Evidence for this criterion is likely to be an extension from criterion P2. As well as having positive impacts, there may be negative impacts of the innovations to a business. These should be evaluated, using examples to illustrate the points being made. Evidence can be in the form of a written report, a presentation with detailed speaker notes or a video of the information being presented to an audience.

LO2 Be able to repurpose technologies to extend the scope of the IoE

P3: In order to consider potential developments for the IoE it will be necessary for learners to explore existing developments. An extensive list of examples is provided in the teaching content. Learners will need to outline potential development projects that could extend the scope of the IoE. These developments could be an extension to an existing application, a new application or taking an existing application and identifying how it could be developed in a different context. Evidence can be in the form of a written report, a presentation with detailed speaker notes, a video of the information being presented to an audience or an information guide describing the different potential developments.
M2: This criterion is an extension from criterion P3. A feasibility study must be carried out on one of these development projects. Evidence presented must detail the nature of the feasibility study to be carried out as well as the analysis and evaluation of the findings. Evidence can be in the form of a written report, a presentation with detailed speaker notes or a video of conducting the feasibility study, analysing the outcomes and evaluating the concept proposal.

LO3 Be able to present concept ideas for repurposed developments

P4: This criterion requires learners to prepare a business proposal for the chosen development project. This naturally leads on from LO2. The business proposal should cover the headings identified in the teaching content for the unit. The evidence will be the completed business proposal.
P5: This criterion requires learners to deliver a business proposal pitch to potential stakeholders. This is an ideal opportunity to engage with representatives from industry. If suitable industry representatives are not available, other representatives from industry may be used for the purpose of the business proposal delivery pitch. Evidence can be in the form of a presentation with detailed speaker notes with an accompanying witness statement or a video of conducting the business proposal pitch. Evidence of feedback from stakeholders must also be included.
M3: Learners are required to consider the feedback from the potential stakeholders and revise the business proposal. The business proposal produced for criterion P4 will be updated to reflect the stakeholder feedback. The evidence will be the revised business proposal with a rationale for decisions made based on the feedback.
D2: Learners are required to evaluate the success criteria that would confirm the sustainability of the development project. The success criteria should provide a stakeholder with a good indication as to how to measure the success of the development and therefore should be measurable. The evidence could be an extension to the evidence presented for P5/M3 or a separate report.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**cisco**.com/UK

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, P4, P5, M1, M2, M3, D1, D2
Problem solving	P4, P5, M2, M3, D1, D2
Time management	P1, P2, P3, P4, P5, M1, M2, M3, D1, D2
Critical thinking	P2, P3, P4, P5, M1, M2, M3, D1, D2
Team working	P4, P5, M2, M3, D1, D2
Negotiation	P5, M3
Decision making	P4, P5, M2, M3, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake an investigation during work experience to establish how developments in technology have impacted on an application within a company and how it has affected the employee(s).
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Learners could produce proposal documents for a scenario and then carry out a feasibility study for these proposals. The industry practitioners would be able to identify strengths and weaknesses in the activities undertaken by the learners.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Industry practitioners would be engaged by centres so that learners could deliver a business pitch to the practitioners and receive feedback on the pitch. In turn learners would revise the business proposal as a result of the feedback. This would provide practitioners with the opportunity to contribute to the assessment of LO3.

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 18

Computer systems – hardware

K/507/5021

Guided learning hours: 60

Version 1 September 2015

LEVEL 3

UNIT 18: Computer systems - hardware

K/507/5021

Guided learning hours: 60

Essential resources required for this unit: Learners will require the use of PCs, plus various components of computer systems, with a view to updating and installing the components to the PC.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The aim of this unit is to enable you to understand how the components of computer systems work together. You will develop the skills needed to recommend appropriate hardware systems for various purposes. Using the skills developed you will build/upgrade a full computer system with a view to testing and considering preventive maintenance procedures.

This is an optional unit within the IT Infrastructure Technician and the Emerging Digital Practitioner specialist pathways. It is important that IT technicians and network technicians have in depth knowledge, skills and understanding associated with the installation, upgrade, troubleshooting and maintenance of hardware for computer systems. Computer systems and associated hardware is an important area within emerging digital technologies.

The learning within this unit will also support the delivery of the CompTIA A+ and Cisco ITE qualification objectives.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand the components of a computer system	1.1 Computer hardware components, i.e.: • internal system unit components o processors o motherboards o BIOS / UEFI o hard drive configuration and controllers (e.g. SATA, IDE, master, slave) o Thunderbolt o VGA, DVI, DisplayPort, HDMI o internal memory (e.g. RAM, ROM, cache) o specialised cards (e.g. network, graphic cards, sound). o power supply • peripheral devices o output devices (e.g. monitor, printer, speakers) o input devices (e.g. camera/webcam, scanner, microphone, mobile devices) 1.2 Storage, i.e.: • pen drives • optical media • flash memory cards • cloud • portable and fixed drives • DASD • SAS • SSD • enterprise storage • NAS • SAN • hybrid systems • virtual tape drives • characteristics • advantages • disadvantages • performance factors (e.g. security, capacity, transfer rate)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to propose a computer system for identified business requirements	2.1 Understanding business requirements, i.e.: • purpose • software • hardware • network sharing • maintenance • outputs • integration • accessibility including special requirements e.g. user has physical limitations) 2.2 Design considerations, i.e.: • single points of failure • recovery techniques i.e.: o clustering o replication • upgrade or renew • cost (e.g. financial, time, user) • services delivered • business requirements 2.3 Backup storage recommendations, i.e.: • advantages/disadvantages (e.g. cost, security, capacity, frequency of saving, transfer rate, redundancy, expansion) • DASD • SAS • SSD • enterprise storage • NAS • SAN • hybrid systems • virtual tape drives. • hard disks • cloud 2.4 Load balancing (e.g. optimise resource use, maximise throughput, minimise response time, and avoid overload of any single resource) 2.5 Methods of presentation of proposal (e.g. presentation, report, quote)
3. Be able to build or upgrade computers	3.1 Health and safety considerations, i.e.: • anti-static mats • wristbands • anti-static bags • appropriate use of tools

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.2 Configuration, i.e.: • BIOS configuration (e.g. setting a BIOS password, editing power management options) • anti-virus configurations • start-up options and logon/access permissions 3.3 Preventative maintenance, i.e.: • organisation, naming, deletion and archiving of files • back-up procedures (e.g. data prior to upgrading computer system, storage of backup, system backup for recovery) • defragmentation • deleting temporary files • cleaning hardware • partition hard drives • replacing consumables (e.g. printer paper, toner cartridges)
4. Be able to test and evaluate the functionality of computer systems	4.1 Test plan/table for accuracy and functionality to include: • test no • type of test • purpose of test • expected results • actual results • issue identified • resolution for identified issue • re-test no 4.2 Diagnostic software, i.e.: • memory diagnostics tools • device manager • control panel • administrative tools • event viewer • system information • command prompt • Regedit 4.3 Benchmarking, i.e.: • Windows system assessment tool • 3DMark • Prime95 • Novabench • PCMark • SiSoftware Sandra • HD Tune

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the components of a computer system	P1*: Explain the function of computer hardware components *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M1: Compare and contrast different hardware storage devices	
	P2: Outline different types of backup storage available		
2. Be able to propose computer systems for identified business requirements	P3: Select appropriate components to support identified business requirements		D1: Justify the components chosen and how they meet the identified business requirements
3. Be able to build or upgrade computers	P4: Install different hardware components on an identified computer system	M2: Recommend preventive maintenance activities for the identified computer system	
	P5: Implement preventative maintenance requirements for the identified computer system		
4. Be able to test and evaluate the functionality of computer systems	P6: Plan and implement test activities for the identified computer system, rectifying any errors	M3: Select and implement benchmarking activities for the identified computer system	D2: Analyse results from benchmarking activities for the identified computer system

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the components of a computer system

P1: Learners are required to explain the functions of the main internal and external components of a computer system. To evidence this learners could provide a detailed user guide that explains the functions of the computer components. The evidence could be presented as a report, part of a technical guide or a presentation (which can either be videoed or have detailed speaker notes).
P2: Learners are required to outline different types of backup storage as identified in the teaching content. The evidence can be in the form of a report, presentation (either videoed or with detailed speaker notes) or as a teacher resource.
M1: Learners are required to compare and contrast different types of hardware storage devices. The comparisons should include the similarities and differences between the devices including performance factors as outlined in the teaching content. The evidence could be presented as a report, part of a technical guide or presentation (either videoed or with detailed speaker notes).

LO2 Be able to propose computer systems for identified business requirements

P3: Learners are required to select appropriate components to support identified business requirements. They will need to be provided with a scenario that will enable them to consider each of the design considerations as per the teaching content. The evidence could be in the form of a report, presentation or work plan for the system.
D1: Learners are required to justify why they have chosen the components and backup storage from P3.The evidence could be an extension to P3 or a separate report or presentation with the justifications clearly provided.

LO3 Be able to build or upgrade computers

It is important that learners are provided with health and safety equipment, including hardware, that will enable them to build and install hardware safely. Once they have built or upgraded the system they can then carry out the benchmarking activities.
P4: Learners are required to install hardware components for a computer system. They could build or upgrade the system that they have designed in LO2. Learners must adhere to all health and safety considerations during the handling and installation of the components. The evidence could be presented as a report supported with annotated photographs, or presentation (either videoed or with detailed speaker notes). Witness testimonies can be used as another valid form of accompanying evidence.
M2: Learners are required to suggest preventative maintenance activities that are needed for the upkeep of the identified computer system. The evidence could be in the form of a report or a maintenance plan.

P5: Learners are required to implement preventative maintenance for the given system. The evidence could be presented as a report with annotated photographs, presentation (either videoed or with detailed speaker notes) or a completed work plan with printouts confirming the results of the activities.

LO4 Be able to test and to evaluate the functionality of computer systems

Learners will need to be provided with a system that will enable them to carry out testing, or ideally they would carry out testing on the system that they have built or upgraded.
P6: Learners are required to prepare and implement a test plan for a specified system. The test plan should be completed during and after any build, grade and/or maintenance activities and not completed retrospectively. The evidence will be the test plan with results of tests. Any errors should be clearly identified including resolution and re-testing.
M3 and D2: Learners will need to have access to a good range of programs that will enable them to benchmark the computer.
M3: Learners are required to select and implement benchmarking activities for an identified computer system. The evidence could be in the form of a report explaining the benchmarking activities selected and printouts showing the results of implementing the selected activities.
D2: Learners are required analyse the results of the benchmarking activities from M3. The evidence could be in the form of a report which may include statistical and/or graphical analysis of the results, a presentation with detailed speaker notes or a technical guide on how to analyse benchmarking test results.

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
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Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1,P3,D1,M2
Problem solving/decision making	M1,D1,P5,M2,P6,M3,D2
Time management	P6,M3,D2
Critical thinking	M1,P3,D1,M2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience to support the school's or college's IT technicians or with local computer businesses – the work experience should be structured so that learners can see in context all or some of the learning outcomes in this unit.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Centres could arrange for local IT technicians to deliver a session to learners which would provide an insight into an IT technician's role when: a) discussing with a client their individual requirements for a computer system b) formulating considerations when selecting and installing hardware components c) Implementing preventative maintenance activities for large and small computer systems.

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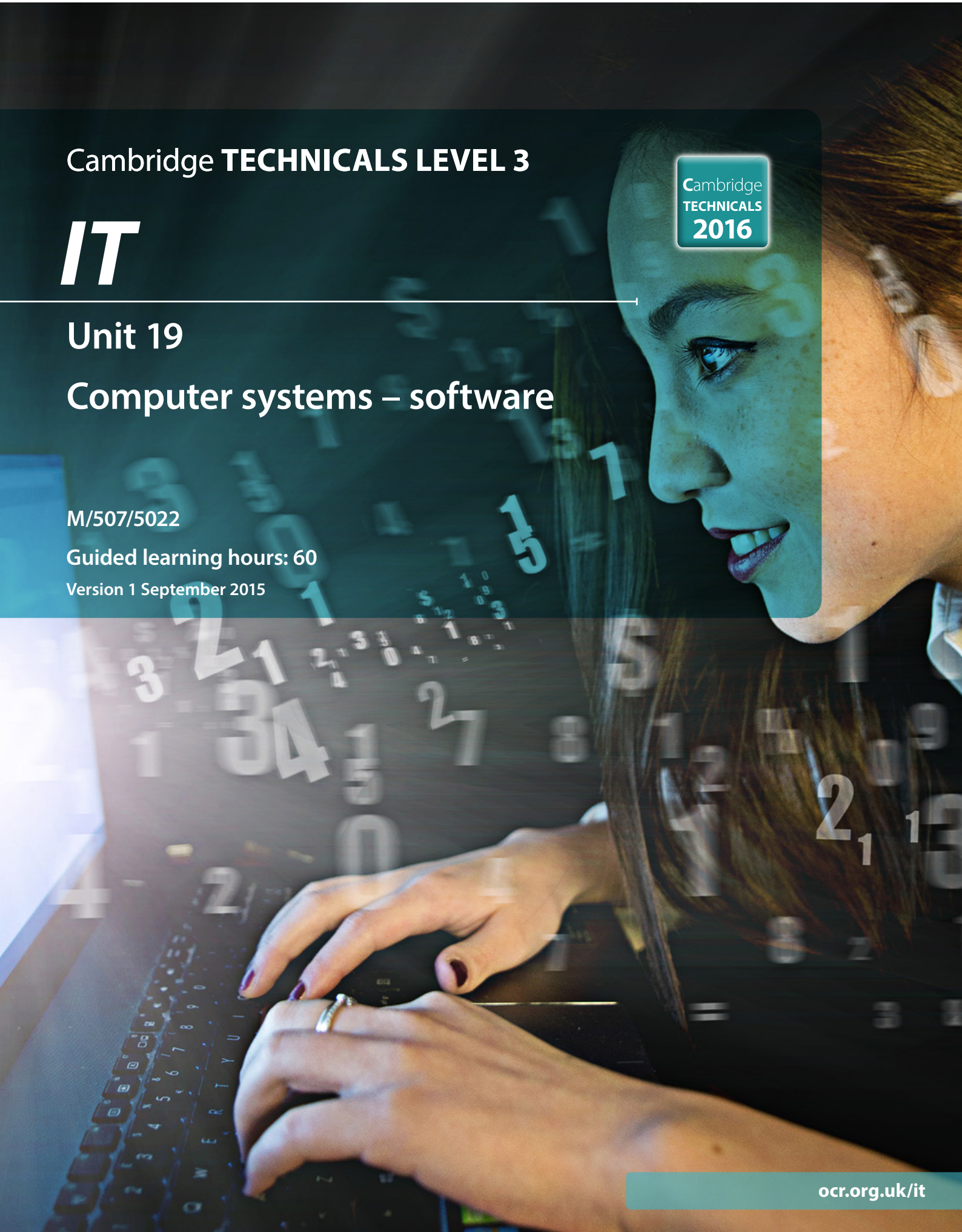
Cambridge **TECHNICALS LEVEL 3**

IT

Unit 19

Computer systems – software

Cambridge TECHNICALS 2016



M/507/5022

Guided learning hours: 60

Version 1 September 2015

ocr.org.uk/it

LEVEL 3

UNIT 19: Computer systems - software

M/507/5022

Guided learning hours: 60

Essential resources required for this unit: Learners should have access to a range of operating systems, application software and utility software. Systems should be made available, allowing learners to install and upgrade a wide range of software to include operating systems, application software and utility software. Maintenance activities will require learners to have access to utilities such as anti-virus, registry cleaners, defragmentation tools etc.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Computer systems rely on the combination of hardware and software to work effectively together to support the needs of all types of users, whether it is for business or an individual. Software comprising of operating systems and application software needs to be installed, configured and maintained correctly in order for the systems to remain efficient. This unit will help you to develop your knowledge, skills and understanding to successfully install or upgrade a wide range of software, including operating systems and application software. It will enable you to select and use appropriate utility software to carry out maintenance activities. In addition, you will be able to ascertain end user requirements in order to plan and carry out installation, upgrades and/or maintenance activities.

This unit is optional to the IT Infrastructure Technician specialist pathway due to its relevance in an IT technical environment. The unit supports the development of skills, knowledge and understanding relevant to a technical support or network technician job role. The unit is also an optional unit within the Emerging Digital Practitioner specialist pathway as the job roles in this field often require the development and installation of bespoke software applications such as creating software applications to accommodate the collection and manipulation of data or creation of software for use within mobile technology.

The learning in this unit will also support the delivery of the CompTIA A+, the CompTIA Mobility+ qualification objectives, as well as the Cisco ITE qualification.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand different software installations and their purpose	1.1 Systems software, i.e.: • operating systems: o open (e.g. Linux) o closed (e.g. Windows) • utility programs • library programs • translator software 1.2 Application software, i.e.: • general purpose (e.g. off the shelf software such as Microsoft Office) • special purpose (e.g. payroll software) • bespoke (e.g. written specifically to meet a particular client's needs) 1.3 Reasons for installation or upgrade, i.e.: • improvement to system (e.g. stability, performance, security, productivity) • resolve issues (e.g. viruses, conflicts etc.) • address risks (e.g. loss of data, loss of service, system downtime, costs) • security risks (e.g. prevention, rectification) • access to additional features/functions • support installation of new hardware • address end user requirements 1.4 Carry out maintenance activities

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	1.5 Types of installation, i.e.: • creating image/ghosting (e.g. make a copy of the hard drive configuration and software) • unattended installation • upgrade • clean install • repair installation • multi-boot • remote network installation • image deployment • Windows networking • Mobile device networking connectivity and email
2. Be able to implement software installations and upgrades to meet specified user requirements	2.1 Software installation/upgrade considerations, i.e.: • end user requirements • other stakeholder requirements (e.g. management) • costs • existing configuration and software • service level agreement (SLA) • software licensing: o end user license agreement (eula) o freeware o shareware o open source • system requirements: o minimum specification o recommended specification • timescales including permissible downtime o risks involved (e.g. incompatibility issues, system downtime, loss of data and/or service, additional costs) o risk minimisation actions (e.g. backups, testing, low risk downtimes for installation) • type of installation • methods and data related to mobile device synchronisation 2.2 Work plan, i.e.: • end user requirements • system specification • current system software • current system configuration • backup of system required • software to be installed • installation type • license requirements/legislative requirements • security risks • compatibility requirements • configuration requirements • timescales • test plan (i.e. test no, date, test, expected result, actual result, resolution, re-test no)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
3. Be able to conduct system maintenance using utility software	3.1 Maintenance plan, i.e.: • date • system description • maintenance activity • benchmark information (prior to maintenance activities and post maintenance activities) o benchmarking software e.g.: SiSoft Sandra Lite, Futuremark, PCMark 8, HyperPi, Phoronix Test Suite (Linux), Passmark Performance Test • licensing requirements/legislative requirements • security risks • backup requirements • software to be used • benefit to system • configuration requirements • testing • activities to be automated 3.2 Maintenance activities, i.e.: • removal of unwanted files • virtual memory settings • defragmentation of drives • registry cleaning • removal of unwanted: o data files o installation files o start-up programs o service settings • cleansing of internet history, temporary and cached files • automatic software updates • automatic anti-virus and spyware scanning • utilities (e.g. scan disk, clean disk) 3.3 Benefits of maintenance activities, i.e.: • improvements to system o boot time o processing speed o security o stability of system o storage capability o memory usage • benchmarking 3.4 Maintenance evaluation, i.e.: • analysis of results • comparison with identified benchmarks • identified issues and potential resolutions • future considerations • documentation of activities

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand different software installations and their purpose	P1: Explain the purposes of different systems software	M1: Compare the features and functions of different system software	
	P2: Outline the different application software available for end users		
	P3: Describe the reasons for carrying out software installations and upgrades		D1: Assess different types of software installations
2. Be able to implement software installations and upgrades to meet specified user requirements	P4*: Create work plans for software installations and upgrades to meet specified business requirements *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M2: Justify the considerations identified within the work plans for specified software installations and upgrades	
	P5: Carry out software installations and upgrades to meet specified business requirements		
	P6: Conduct system testing for software installations and upgrades		
3. Be able to conduct system maintenance using utility software	P7: Plan maintenance activities for a specified system	M3: Document the benefits of maintenance activities within the maintenance plan for the specified system	
	P8: Implement maintenance activities for the specified system		D2: Evaluate implemented maintenance activities for the specified system, post maintenance activities

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand different software installations and their purpose

P1: Learners are required to explain the purpose of the different systems software as identified in the teaching content. The evidence could be presented as a report, part of a technical guide or a presentation (either videoed or with detailed speaker notes).
M1: Learners are required to compare the features and functions of a range (three or more) of different systems software. This can be an extension of P1, but the comparisons must include any similarities and differences between the different software. The evidence could be presented as a report, part of a technical guide or a presentation (either videoed or with detailed speaker notes).
P2: Learners are required to outline the different application software available to end users. The outlines must cover general purpose, special purpose and bespoke, and include an account of the purpose of each application software category as well as the purpose for the example of each category type selected. The evidence can be in the form of a report, a presentation (either videoed or with detailed speaker notes) or as a teacher resource.
P3: Learners are required to describe the reasons for carrying out software installations and upgrades. The outline should include a description of each of the reasons (using examples), which includes the main characteristics/points. The evidence could be in the form of a presentation (either videoed or with detailed speaker notes), report, or information sheet.
D1: Learners are required to assess different types of software installation. The discussions should include a variety of ideas and arguments as to why certain types of software installation are more appropriate than others. Learners should discuss at least five of the types of software installation and include examples of where and how they are used. The evidence could be in the form of a technical guide, report or presentation (either videoed or with detailed speaker notes).

LO2 Be able to implement software installations and upgrades to meet specified user requirements

It is important that learners are provided with a scenario, equipment and software that will enable them to install a wide range of software which must include operating system, application software, and utility software and include a software upgrade (the upgrade may be as a result of installing the operating system and/or utility software).
P4: Learners are required to examine the given scenario and create a work plan for the installation and upgrading of software to meet the specified user requirements. It is important that the work plans include the items in the teaching content to ensure that all aspects of the installation process have been considered. The test plan should also include relevant tests that would be carried out during and/or after installation/upgrade has taken place. A system should be tested after each installation to test functionality prior to implementing further installations. The evidence will be the work plans for the installations and upgrades. Learners could link this to a project that they are carrying out for the project management unit.

M2: Learners are required to justify the considerations that they have taken into account when preparing their installation plans. The justification should present a reasoned case for the decisions made. The evidence could be in the form of a report or presentation (either videoed or with detailed speaker notes).
P5: Learners are required to carry out the software installations and upgrades by following the work plans they produced for P4. It is acceptable for learners to make adjustments to their work plans during the installation and/or upgrade process, should they identify any unforeseen issues. They must, however, document these changes and provide an explanation as to why these changes were made. Their evidence could be in the form of a detailed witness statement supported by any documentary evidence e.g. completed work plans, video of installations and/or upgrades taking place, photographs etc.
P6: Learners are required to carry out system testing during and following each installation and upgrade. These tests must test the functionality of the system and not just the functionality of the software itself. The evidence will be the completed test plans.

LO3 Be able to conduct system maintenance using utility software

It is important that learners are provided with scenarios, equipment and software that will enable them to meet the assessment criteria. Learners should have access to utility software which is "built in" to the operating system in use, as well as additional utility software which they may have installed for LO2. If learners are going to work towards the distinction criterion, they will need to benchmark the system prior to carrying out the maintenance activities in order to evaluate the improvements made.
P7: Learners are required to plan maintenance activities for a specified system. Learners must produce a maintenance plan to include the sections identified within the teaching content. The test plan should be completed during and after the maintenance activities and not completed retrospectively. The evidence will be the maintenance plan and test plan.
M3: Learners are required to document why the maintenance activities identified in the work plan produced for P7 will benefit the functionality of the specified system. The documentation should include the purpose or reason for each of the activities, as well as how this would benefit the system. The evidence could be in the form of a report, technical guide or presentation (either videoed or with detailed speaker notes).
P8: Learners are required to implement the maintenance activities for the system that they have identified within their maintenance plan created in P7. It is acceptable for learners to make adjustments to their maintenance plans during the maintenance activities, should they identify any unforeseen issues. They must, however, document these changes and provide an explanation as to why these changes were made. Their evidence could be in the form of a detailed witness statement supported by any documentary evidence e.g. completed maintenance plans. They must also include evidence of testing which must include the completed test plans. Additional evidence could be from photographs, printouts of tests etc.
D2: Learners are required to evaluate the effect that the implemented maintenance activities have had on the system after the maintenance activities have been completed. Learners should be able to carry out benchmark tests on the current system and compare them with the benchmark information taken prior to the maintenance activities occurred. Learners should make qualitative judgements, taking into account different factors and using the evidence that they have available to them. This evidence should include any documentary evidence from the benchmarking of the system prior to the maintenance activities, as well as after maintenance has taken place. This could be supported by a report, information sheet or presentation (videoed or with detailed speaker notes).

Some providers for the industry qualifications offer quizzes, tests and assessments. Reference to these websites may support knowledge and learning.
www.**comptia**.org
www.**cisco**.com/UK

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P4, M2, M3, D2
Problem solving/decision making	P4, P5, P6, P7, M3, D2
Time management	P4, P5, P6, P7, P8
Critical thinking	P4, M2, P6, P7, P8, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.
Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience to support the school's or college's IT technicians, or with local computer businesses – the work experience should be structured so that learners can see in context all or some of the learning outcomes in this unit.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Centres could arrange for local IT technicians to deliver a session to learners that would provide an insight into an IT technician's role when: a) discussing with a client their individual requirements for a computer system b) formulating considerations particularly when selecting and installing software components (unit 19) c) implementing maintenance activities for large and small computer systems.

To find out more
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Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 20

IT technical support

T/507/5023

Guided learning hours: 60

Version 1 September 2015

ocr.org.uk/it

LEVEL 3

UNIT 20: IT technical support

T/507/5023

Guided learning hours: 60

Essential resources required for this unit: Learners should have access to a variety of tools used by technicians to support clients; these tools may be software tools such as 'SiSoft Sandra' or hardware tools such as multi-meters.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

The aim of this unit is to explore the problems that are likely to be encountered by IT support professionals. You will learn about many of the tasks that they carry out.

You will develop an understanding of technical support tools and techniques. This will include looking at organisational policies and procedures to source technical information. You will also learn how to diagnose faults and communicate advice and guidance to clients to help resolve problems.

This unit is optional within the IT Infrastructure Technician specialist pathway as all IT support technicians require the skills, knowledge and understanding to troubleshoot a wide variety of hardware and software issues and provide advice and guidance to different stakeholders.

The learning in this unit will also support the delivery of the CompTIA A+, the CompTIA Mobility+ qualification objectives, as well as the Cisco ITE qualification.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the role of technical support	1.1 IT Support Roles (i.e. 1st line, 2nd line, 3rd line) 1.2 Support process, i.e.: • helpdesk activities (e.g. role, point of contact, personnel, fault logging) • IT support requests • analysis of previous trends (e.g. types of fault reports, support request documentation, analysis, reporting) • resources (e.g. staffing, availability, training, emerging technologies) • policies, e.g.: o service level agreements (SLAs) o computer use policy o standard, non-standard equipment policy 1.3 Client Types, i.e.: • environment (e.g. corporate, home) • skill level (e.g. novice, non-technical, technical) • software use (e.g. games, word processing, database) • features (e.g. basic, advanced) 1.4 Communicating advice, i.e.: • client type (e.g. technical, non-technical) • methods of communication (e.g. face-to-face, email, telephone) • interpersonal skills (e.g. listen, negotiate, value opinion, take criticism)
2. Be able to diagnose faults and solutions for computer systems	2.1 Diagnostic techniques, i.e.: • software tools (e.g. diagnostic software functions and sources) • sources of support (e.g. supplier, websites, technical documentation, fault logs) • gather information from customer • organisational constraints e.g. costs of resources required, time, client expertise • standard operational procedures (e.g. SOPs)

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	• testing to mirror fault • remote access, check bios, gather diagnostic information to solve problem 2.2 Types of faults, i.e.: • simple faults (e.g. no sound, display incorrect,) • complex faults (e.g. faults that require extra intervention, or remote access) • interlinked problems (e.g. a simple fault resolved can lead to further issues) 2.3 Documentation and record keeping to include, i.e.: • categories of information kept • format of the information i.e. digital, manual 2.4 Post fault testing, i.e.: • test for resolution of fault • benchmark testing against known specifications • follow up actions required
3. Be able to provide advice and guidance to specific customers	3.1 Communication techniques, i.e.: • Considerations, i.e.: o technical clients o non-technical clients o methods o documentation o use of technical language o visualisation o timescales o accuracy and relevance of information 3.2 Effectiveness of technical provision, i.e.: • feedback from clients (e.g. questionnaires, verbal , face to face, email, call loggers) • effectiveness in terms of, e.g.: o efficiency of support delivery o available tools, equipment, software o cost o training needs. • review (e.g. survey information, methods, outcomes, conclusions and decide what was effective or not) 3.3 Analyse trends and make recommendations, i.e.: • analysis of trends of problems and support requirements • reducing repeat occurrences

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the role of technical support	P1: Describe the activities within a technical support role	M1: Analyse the communication considerations used for differing client types	
2. Be able to diagnose faults and solutions for computer systems	P2*: Perform diagnosis of simple faults with an identified computer system and implement solutions *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M2: Investigate complex faults with an identified computer system and implement solutions	D1: Test computer system post fault resolution and interpret results
	P3: Document faults and solutions for computer systems		
3. Be able to provide advice and guidance to specific customers	P4: Communicate technical advice and guidance to an identified non-technical client		D2: Analyse trends from troubleshooting technical issues and make recommendations to reduce repeat occurrences
	P5: Present technical advice and guidance to an identified technical client	M3: Measure the effectiveness of technical support provision	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

Learners will be introduced in this unit to the work role of an IT support technician. Learners will need to gain experience and deal with common and complex problems.

Learners will need to be given scenarios in which they demonstrate that they can support their colleagues. This could be taught through role play with a wide variety of scenarios. An IT support technician will need to assist a wide range of clients with a wide range of knowledge. Hence the support provided will vary depending on the knowledge of the clients. Providing learners with experience working alongside technicians could enable them to encounter a range of issues, and to see how these issues are subsequently dealt with.

LO1 Understand the role of technical support

P1: Learners are required to describe the activities within the IT support roles of an organisation. Learners will need to look at how support is delivered to clients for each of the listed IT support roles in order to cover all the items listed in the IT support process. Evidence could be presented as a report, a presentation with detailed speaker notes or a career's guidance booklet.
M1: Learners are required to analyse communication techniques that can be used for differing client types. Learners should consider the listed client types in the teaching content. Their analysis should include the characteristics of the different client types and a reasoned comment as to why the communication technique used is appropriate. Evidence could be presented as a report, presentation with detailed speaker notes or a guide for new IT support technicians.

LO2 Be able to diagnose faults and solutions for computer systems

Learners will need to be provided with a variety of computer systems that have different faults for diagnosis. These faults should be a mixture of complex and simple.
P2: Learners are required to perform diagnostic activities on an identified client's computer system resulting in them establishing the faults on the system. They are then required to identify and implement appropriate solutions for the resolution of the faults. The evidence could be presented in the form of a report with annotated photographs, a video of the learner carrying out the activities, work log supported by printouts or screenshots of diagnostic activities and confirmation of resolution of faults.
M2: Learners are required to investigate complex faults for an identified client's computer system. Once they have investigated the faults, they are required to identify and implement appropriate solutions to resolve these faults. The faults should be complex with respect to challenging the learners' knowledge and understanding of fault diagnosis. Evidence could be in the form of a report with annotated photographs supported by printouts and/or

screen prints. Evidence could also be in the form of a presentation with detailed speaker notes and supporting evidence and/or work log with annotated photographs, printouts and/or screen prints.
P3: The evidence for this assessment criterion could be linked to the evidence for P2. Learners are required to document the simple faults and solutions that they have identified for a client's computer system. The evidence could be in the form of a report with annotated photographs and/or supported by printouts from tests or screen prints. Evidence could also be in the form of a presentation with detailed speaker notes or work log supported by annotated photographs, printouts and/or screenshots.
D1: Learners are required to show that they have interpreted the results from testing an identified client's computer system post the resolution of any faults. This can be an extension of M2 but learners will need to show they have tested the computer system and interpreted the results following the resolution of different simple and complex faults. The evidence could be in the form of detailed report or, alternatively, a presentation which includes detailed speaker notes. These would be supported by annotated photographs, printouts from testing or screen prints.

LO3 Be able to provide advice and guidance to specific customers

P4: Learners are required to show that they can communicate technical advice and guidance to a non-technical client. It is important that learners can demonstrate that they have selected a suitable form of communication as well as using appropriate language for the non-technical audience. The evidence could be in the form of an email providing technical advice and guidance to a customer, an information guide on how to resolve simple computer issues or a presentation with detailed speaker notes.
P5: Learners are required to show that they can present technical advice and guidance to a technical client. It is important that the learners can demonstrate that they have selected a suitable form of communication as well as using appropriate language for the technical audience. The evidence could be in the form of a technical guide, report, work logs, fault and resolution guide or a presentation with detailed speaker notes.
M3: Learners are required to measure the effectiveness of the technical support provided to different clients for different faults. The learner could be given a set of detailed reports from which these measurements could be made. The evidence could be in the form of a detailed report or a presentation with detailed speaker notes.
D2: Learners are required to analyse trends from troubleshooting technical issues and make recommendations to reduce repeat occurrences. This could be a report to management informing them of the trends analysed and the changes that need to be made to current working practices. The evidence could be in the form of a detailed report or presentation with detailed speaker notes supported.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, M1, P4, D2
Problem solving/decision making	P2, P3, M2,D2
Time management	P3
Critical thinking	M3, D1

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could undertake work experience to support the school's or college's IT technicians or in a local business IT support team. The work experience should be structured so the learner has the opportunity to see how businesses deal with technical support to support the learning outcomes in this unit.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	This could be setting up call loggers and students working with school or college technicians look at how the communication flows.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	You could arrange for local IT support technicians to deliver a session to your learners to support the learning in this unit for example: • Explaining the role of IT support within their business to cover learning in LO1 • How they provide advice and guidance to their customers (LO3)

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Cambridge **TECHNICALS LEVEL 3**

IT

Cambridge TECHNICALS 2016

Unit 21

Web design and prototyping

A/507/5024

Guided learning hours: 60

Version 2 - revised May 2016

*changes indicated by black vertical line

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LEVEL 3

UNIT 21: Web design and prototyping

A/507/5024

Guided learning hours: 60

Essential resources required for this unit: Access to web authoring software that allows learners to write HTML, Javascript (or equivalent language), CSS and PHP. Learners may need access to a server and database that they can connect through using any appropriate MySQL software.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Organisations are increasingly reliant on their websites to market goods or services and interact with clients and customers. As technology develops, so does the scope of functionality of websites and the importance of an effective design that meets the needs of the organisation. In this unit you will research, design and produce an interactive, responsive website that is specific to a client's needs, culminating in presenting the concept of the website using the prototype to the client. You will learn about the security risks in website design and how to minimise these threats. This unit will also allow you to incorporate existing interactive elements, as well as prototyping your own website.

This unit is optional within the Application Developer specialist pathway. Job roles within this pathway include web app developers as well as website designers, although these are different jobs with differing requirements, they do require similar skills, knowledge and understanding with respect to website creation and prototyping.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the fundamentals of web design	1.1 Components of web design, i.e.: • domain name • the purpose or a site map • the use of hyperlinks to join webpages • the position, structure and purpose of a navigation bar • page design (e.g. navigation bar, title, page content, a plan for content management) • designing for different browsers and devices o responsive design for mobile/tablet technology o how browser rendering can affect the layout of a webpage o an understanding of browser rendering engines as a key to performance o importance of design that takes this into account, what can happen when it is not considered • World Wide Web Consortium o open and closed standards in web development • storage of data (e.g. linking to a database) • methods of user interaction o different methods of interactivity o use of existing tools o use of languages to add interactivity(e.g. Javascript, Flash) 1.2 Security risks, i.e.: • the need for security • threats (e.g. SQL injection, interception of data, un-validated input) • prevention (e.g. validation, appropriate passwords, encryption)
2. Be able to plan the development of an interactive website for an identified client	2.1 Analysing needs, i.e.: • list of requirements • purpose of site • content • user interactivity required (e.g. submission of data, maps) • responsive design requirements

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	2.2 Produce a plan, i.e.: • site map • navigation bar • page structure • design (e.g. font, colour) • interactive elements to meet the client's needs • responsive design o for different devices, e.g. tablet, phone, PC o for different browsers, e.g. IE, Chrome, Safari
3. Be able to create prototype websites for an identified client	3.1 Creating a prototype, i.e.: • setting structure • adding content • inserting hyperlinks 3.2 Interactive elements, i.e.: • existing o copy HTML code from pre-existing components (e.g. social media feeds, videos, maps) o editing to make these appropriate for a webpage (e.g. making a map local) • language (e.g. Javascript, Flash) o writing code to perform a function o allowing for user interaction (e.g. changing an image based on a user selection, validating input) • database o creating a database on a server o connecting a webpage to a database using PHP o writing data from a user input (e.g. completion of a form) using MySQL • CSS o responsive design 3.3 Testing to include, i.e.: • checking content is appropriate, correct and complete • functionality o embedded components o programming language (e.g. Javascript, Flash) o database storage • comparison to requirements • responsive design o multiple devices o multiple browsers

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.4 Evaluate against client needs including, i.e.: • comparison against requirements • how the prototype meets the site purpose (e.g. if e-commerce, how this has been incorporated) • comparison of prototype against content required • comparison of interactivity built in, against the user's interactive requirements • comparison of responsive design to user's responsive design requirements • justify choices made in the development of the prototype against the original needs • improvements - suggest improvements to the design, structure or function of the prototype
4. Be able to present the interactive website concept to an identified client	4.1 Present the solution, i.e.: • choose a suitable method of presentation (e.g. live demonstration, use of presentation software, report) • plan a presentation to incorporate: o comparison of website against requirements o demonstration of functionality o demonstration of interactivity o demonstration of responsive design o justification of design choices • present the solution to the client 4.2 Future security and maintenance considerations, i.e.: • updating of content • training of staff • threats to information security • protection methods for securing personal data/information

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the fundamentals of web design	P1: Describe the key components of website construction	M1: Explain the security risks that must be considered when creating a website	
2. Be able to plan the development of an interactive website for an identified client	P2*: Determine a client's requirements for a website *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information)*		
	P3: Produce a plan for the website that meets the client's requirements		D1: Justify the use of components in the website design that meets the client's requirements
3. Be able to create prototype websites for an identified client	P4: Create a prototype of the website for the client	M2: Add interactive components to the prototype based upon the client's requirements	
	P5: Conduct testing of the prototype		D2: Evaluate the prototype against the client's requirements
4. Be able to present the interactive website concept to an identified client	P6: Create a presentation to demonstrate the prototype website to the client	M3: Communicate future website security and maintenance considerations to client	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the fundamentals of web design

This learning outcome is about the learners exploring the structure of a website and how components are combined to produce the finished product. It is important that the learners look at examples of websites to analyse how these are used effectively.
P1: The learners must describe the key components that make up website construction. Their descriptions must include web page and website structure, content and methods of user interaction including responsive design for a range of devices and browsers. Evidence can be in the form of a written report, presentation with detailed speaker notes, a video of presenting the information to an audience, or an information guide about designing a website.
M1: The learners must explain the security risks that must be considered when creating a website. They must identify threats to both accurate data storage and internal and external threats such as interception of a data transmission. Learners need to describe how prevention strategies can be integrated into a website design. This can build on the work in P1 with learners identifying components and website design elements that reduce security risks. The evidence can be in a written or verbal format e.g. a written report, presentation, audio or video recording of a presentation etc.

LO2 Be able to plan the development of an interactive website for an identified client

This learning outcome is about the learners investigating a client's requirements for a website. It is important that the scenario for the website allows learners to incorporate both interactivity and a responsive design, which will allow them to meet all of the assessment criteria.
P2: The learners must determine a client's needs for a website. They need to investigate the scenario and work with the client to determine their requirements. These should include items such as purpose, content (both text and image based), interactive elements (e.g. a form for submission of data, account to log into, purchase of items, a change in content based upon a user selection) and the increasing importance of responsive design to both different devices and browsers. The evidence can be in the form of a written report, presentation or recording of the requirements.
P3: The learners must produce a design plan for the website, based on the requirements they have identified in P2. The learners should produce a site map for the website and an annotated design template for the prototype that takes into account the required responsive design for the planned for browser and/or device. The learners will need to design the interactive elements, outlining how these will work. The evidence should be in the form of annotated drawings or electronic designs that identify the intended browser and device designed for. This plan would then enable a third party to produce the prototype
D1: The learners need to justify their design choices based upon the client's requirements. The learners need to compare the design produced in P3 to the requirements identified in P2. The evidence can be in the form of additions to P1 and P3, or as a separate written report, presentation or recording.

LO3 Be able to create prototype websites for an identified client

This learning outcome is about the learners producing and testing a prototype for the website design for LO2. This should include a number of webpages that are linked.
P4: The learners will need to create a prototype of the website for the client, which should have a number of linked webpages. The learners need to make sure this derives from the designs produced for LO2. Learners need to create a suitable structure, navigation, and content, including working hyperlinks. The evidence for this will be the prototype, either as a hosted website or as files that can be viewed and run locally.
M2: The learners are required to add different interactive components, as per their planned design. This can be added to the work produced for P4. The interactive elements need to link to the designs produced for LO2. The areas learners can add are; existing elements (pre-existing components that are inserted e.g. an interactive map, video, active weather forecast), JavaScript based (the learner may add code to perform specific functions, such as changing images, a submit button to e-mail a form), database (e.g. a log-in feature, storing data to a database) and CSS (e.g. creating a responsive website to different browsers or devices). The evidence for this will be the prototype produced and could be supported by a written report, presentation or video of the interactive elements.
P5: The learners are required to test the prototype. They need to ensure that all elements of the prototype work correctly, including; hyperlinks, interactive elements (if M3 is complete) and responsive design (if M3 is complete) as per their plan. The evidence for this should be in the form of their test plan.
D2: The learners are required to evaluate their prototype that they created for P4 and M2. The evaluation needs to include the bullet points listed in the teaching content. The evidence for this could be a documented evaluation in the form of a written report, presentation with detailed speaker notes, or a video of the learner presenting the results of their evaluation.

LO4 Be able to present the interactive website concept to an identified client

P6: The learners need to present the concept of the website, using the prototype, to an identified client. The learners must prepare a suitable presentation including the bullet points listed in the teaching content. They then need to present their presentation to the client. The evidence for this will be found in the resources created for the presentation (video, presentation with detailed speaker notes) and evidence of giving the presentation (e.g. a recording of the presentation).
M3: The learners are required to communicate future website security and maintenance considerations to their client. The learner should consider how the website will change over time and therefore it is important to update information, training needs and security measures on a regular basis. The evidence could be an extension of P6 and be in the form of a report or presentation with detailed speaker notes.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P2, P6
Problem solving	P2, P3, M2, P4, M3,
Time management	P1, P2, P3, P4, P5, P6, M1, M2, M3, D1, D2
Critical thinking	P5, P6, M1, D1, D2
Decision making	P2, P3, P4, P5, P6, M2, M3

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners could create a website for the employer whilst on a work-placement. The centre would need to support the technical skills, and documentation required at each stage, but a live website could be implemented for the employer.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	You could set a project with a local employer – the employer could be the client for the website they could be asked to provide details of their website requirements and sample content for the learners to work from.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	A web development employer could teach the learners the technical elements of this unit, for example JavaScript or SQL/PHP programming.

To find out more
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or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**

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Cambridge **TECHNICALS LEVEL 3**

IT

Unit 22

Big Data analytics

F/507/5025

Guided learning hours: 60

Version 1 September 2015



LEVEL 3

UNIT 22: Big Data analytics

F/507/5025

Guided learning hours: 60

Essential resources required for this unit: The centre will need to have the tools to allow for data to be analysed and the storage capacity for a large quantity of data for analysis.

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Data is all around us and the amount of data being gathered is growing. Big Data deals with extremely large data sets that can be analysed computationally to reveal patterns, trends and associations. This is especially the case in relation to human behaviour and interactions. In this unit you will learn what Big Data is, how it can be gathered, analysed and used by businesses. You will also look at how a company could use big data for planning for the future.

Due to the increased use of Big Data this is an optional unit in the Emerging Digital Technology Practitioner, Application Developer and Data Analyst specialist pathways.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes The Learner will:	Teaching content Learners must be taught:
1. Understand the scope of Big Data	1.1 What is Big Data, i.e.: • The Big Four o volume (i.e. scale of data) o variety (i.e. different forms of data) o velocity (i.e. analysis of streaming data) o veracity (i.e. uncertainty of data) 1.2 Use of Big Data, e.g.: • uncovering hidden patterns • unknown correlations, market trends • customer preferences 1.3 Impact on organisations, i.e.: • competitive advantage through Big Data • predictive analytics, e.g. determining patterns and forecasting future trends 1.4 Infrastructure challenges posed by Big Data, i.e.: • capacity (e.g. processing and storage, disk storage (arrays), cloud storage) • virtualisation • capacity allocation • distributed analytics • analytical software • scalability
2. Be able to process Big Data for business purposes	2.1 Big Data sources, e.g.: • social media content and social network activity reports • text from customer emails and survey responses • web server logs • Internet clickstream data • machine data captured by sensors connected to the Internet of Things

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	2.2 Big Data Risks, i.e.: • data cleansing • data validity • data reliability • legislative (e.g. data protection, freedom of information) • ethical (e.g. organisational and individual rights) 2.3 Preparing Big Data for analysis, e.g.: • cleansing (GIGO) • coding data • validity • integrity of the data 2.4 Processing Big Data, i.e.: • questioning the data (e.g. multiple tables, multiple criteria) • advanced formulaic functions • graphical information 2.5 Evaluating results, i.e.: • identified solution against requirements • meeting the customer needs • strengths of the Big Data Analytic • improvements to the Big Data Analytics
3. Be able to provide information resulting from processing Big Data	3.1 Presenting results, i.e.: • presentation methods, e.g. o video o presentation o formal report o visualisation (e.g. graphical, diagrammatical) • target audience consideration, i.e.: o technical o non-technical o location (e.g. same location, different location etc.) o size (e.g. number of people in the room) • Big Data recommendations, e.g.: o regional increases (marketing/sales efforts) o customer base (e.g. current and possible future customers/clients) o increased customer/client loyalty o patterns of distribution (e.g. current and future); to include both graphical and tabular information

GRADING CRITERIA

LO	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Understand the scope and challenges of Big Data	P1: Outline what is meant by Big Data	M1: Explain how Big Data can be used	D1: Evaluate the infrastructure challenges posed by Big Data
	P2: Explain the impacts that the use of Big Data could have on different organisations		
2. Be able to process Big Data for business purposes	P3: Select appropriate sources of Big Data for a business purpose providing a rationale for the choices made		D2: Evaluate the outcomes based on the processing of Big Data for the business purpose
	P4*: Suggest possible risks associated with the selected Big Data sources *(*Synoptic assessment from Unit 1 Fundamentals of IT, Unit 2 Global information and Unit 3 Cyber security)*	M2: Make recommendations for eradicating risks associated with the selected Big Data sources	
	P5: Process Big Data to obtain results for the business purpose		
3. Be able to provide information resulting from processing Big Data	P6: Present outcomes from processing Big Data to the business	M3: Make recommendations to the business arising from the processing of Big Data	

SYNOPTIC ASSESSMENT

When learners are taking an assessment task, or series of tasks, for this unit they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

ASSESSMENT GUIDANCE

LO1 Understand the scope and challenges of Big Data

P1: Learners are required to outline the term 'Big Data'. They should describe the Big Four identified in the teaching content setting out the main characteristics/points of each. This could be evidenced in the form of a presentation with detailed speaker notes, video of the learner delivering the presentation or an information sheet.
M1: Learners are required to explain how Big Data can be used. The learners should provide descriptive examples of the uses and include the purpose and/or reasons for the use. The evidence could be in the form of a report, a presentation with detailed speaker notes which could be used to present information to businesses, a tutor resource e.g. handout for learners or case studies.
P2: Learners are required to explain the impact that the use of Big Data could have on different organisations. This could be an extension of M1. The explanations should contain a description of the uses of Big Data as well as what the impacts are, with a further explanation as to why these uses results in these impacts. This could be presented as a report, presentation with detailed speaker notes, video of the learner delivering the presentation or an information guide for businesses.
D1: Learners are required to evaluate the infrastructure challenges faced by organisations that use and process Big Data. Learners should consider the infrastructure challenges as listed in the teaching content which are applicable to the organisations they have considered. The evidence could be in the form of a report, presentation with detailed speaker notes, information guide for businesses.

LO2 Be able to process Big Data for business purposes

It is useful if learners are provided with a scenario that will enable them to prepare, process and produce outcomes that can form the basis for their report in LO3.
P3: Learners are required to select sources of Big Data for a specified business purpose. Learners should provide a rationale for the choices they have made. The evidence could be in the form of a report or a presentation with detailed speaker notes.
P4: Learners are required to suggest the possible risks associated with the selected data sources from P3. They should consider the different risks involved explaining what they are and how they could be reduced. As this is an extension of P3, learners may well present evidence that combines the two assessment criteria into one report or presentation.

M2: Learners are required to make recommendations for how risks associated with the selected Big Data sources identified in P4 can be eradicated. The recommendations should focus on how the data should be managed from selection through to processing. The evidence would be most suited to a report but could also be a presentation with detailed speaker notes.
P5: Learners are required to process Big Data from the selected sources to meet the purpose of the business. Learners should demonstrate the preparation and processing techniques used to obtain results. The evidence could be the initial data prior to preparation, the data after preparation has taken place and then the results of the processing of the data. It does not have to be a step-by-step screenshot of every action taken.
D2: Learners are required to evaluate the outcomes based on the preparation and processing of the Big Data analytic carried out in P5. The evaluation should show that the learner has made qualitative judgements taking into account different factors and using available knowledge, experience and evidence. The evidence could be collated in preparation for the evidence for P6 and take the form of a report or presentation with any visual representations suitable for presenting the outcomes. It should be noted that where graphs/charts are used, the labelling must be accurate and clear.

LO3 Be able to provide information resulting from processing Big Data

P6: Learners are required to present information as a result of processing Big Data for a business purpose. The format of the evidence could be a video, report, presentation with detailed speaker notes or any form of visualisation. If graphs/charts are used, the labelling and content must be accurate, clear and suitable. The learners should state who the audience is and it must be clear from their style of presentation that they have taken the needs of the audience into consideration.
M3: Learners are required to make recommendations to the business arising from the processing of Big Data. The recommendations should reflect improvements to the business processes and/or changes to the focus for future activities. Learners could include these recommendations as part of their evidence for P6, or they could create a separate report or presentation with detailed speaker notes.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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EMPLOYABILITY SKILLS

Employability skills	Learning outcome
Communication	P1, P2, P3, P4, P5, P6, M1, M2, M3, D1, D2
Problem solving/decision making	P3, P4, P5, P6, M2, D1, D2
Time management	P3, P4, P5, P6, M2, D2, M3
Critical thinking	M1, M2, M3, D1, D2

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units..

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer involvement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work-experience or work-placements that develop skills and knowledge relevant to the qualification.	Learners may be able to work with sales and marketing in an organisation looking at the data that they gather from their sales in order to best target future marketing.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Businesses who use Big Data could give sessions on how Big Data is used within their organisations and the risks involved.

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

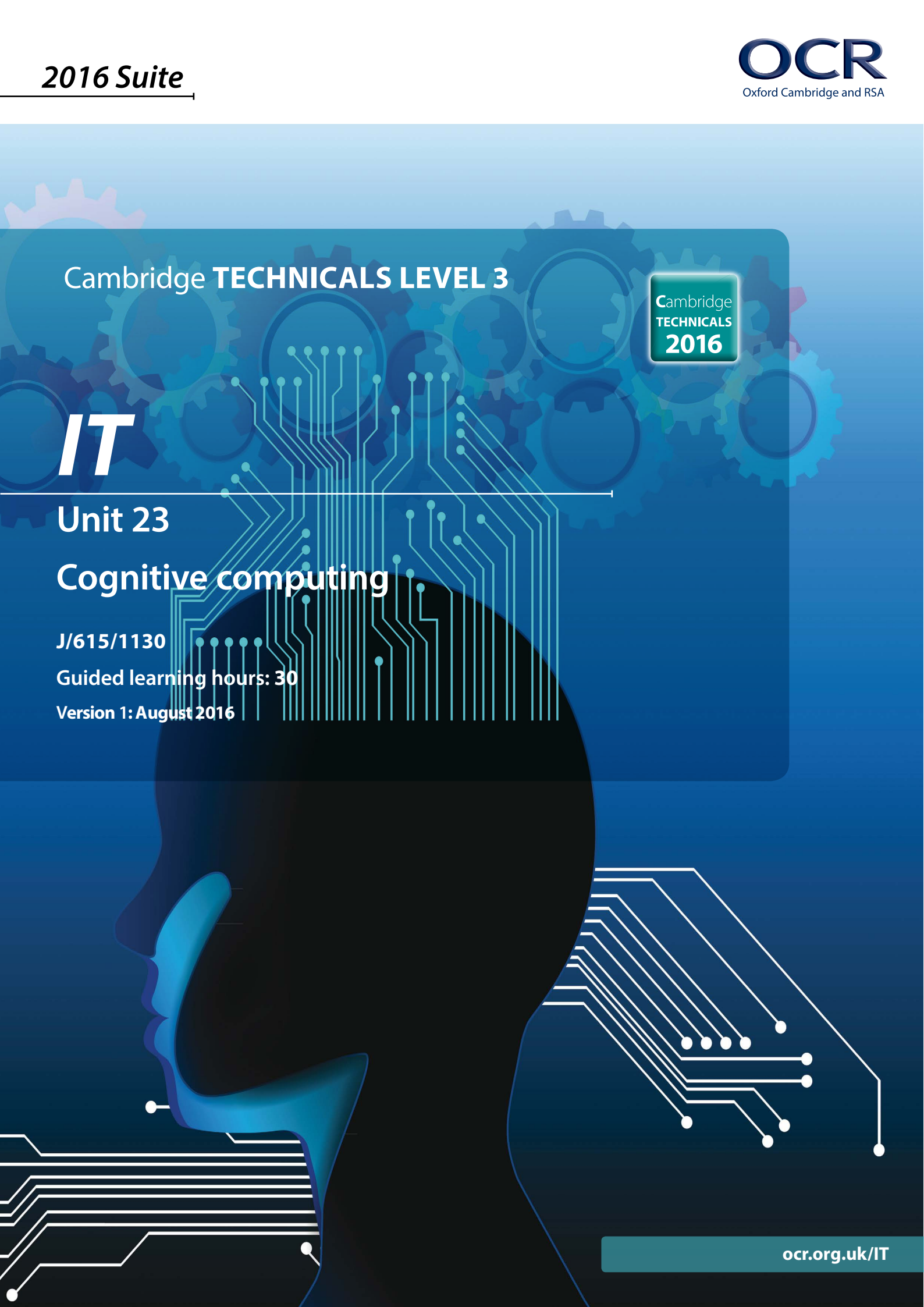
Unit 23

Cognitive computing

J/615/1130

Guided learning hours: 30

Version 1: August 2016



ocr.org.uk/IT

LEVEL 3

UNIT 23: Cognitive computing

J/615/1130

Guided learning hours: 30

Essential resources required for this unit: None

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Cognitive computing and artificial intelligence are where human thought processes and activities are simulated in a computerised model. This unit will provide you with the opportunity to gain a better understanding of cognitive computing and the role it has to play in current technologies and emerging technologies. Using your knowledge and understanding of cognitive computing you will consider how cognitive computing can be used on a global scale in the future and you will develop a business proposal for its use and implementation.

This unit is optional in both specialist pathways in the Level 3 Extended Diploma.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Know how cognitive computing is used in business	1.1 Current use of cognitive computing e.g.: • healthcare (e.g. analysing big data, suggesting diagnosis, recommending treatments) • retail (e.g. enhanced customer experience, optimisation of store operations, improved supply chain management, identify potential markets) • human resources (e.g. increase employee productivity, expedite time consuming business processes, recruitment) • procurement (e.g. improve customer services) • travel industry (e.g. personalised travel advice) • facial recognition (e.g. for use by security services, access to security systems) • education (e.g. personalise learning requirements, identify development needs) • manufacturing (e.g. operations management, controlling systems etc) 1.2 Business stakeholders using cognitive computing e.g. • individuals • corporate • SMEs (Small Medium Enterprises) 1.3 Drawbacks of cognitive computing applications i.e. • safeguarding of data (e.g. access to large data will have an impact on security of data) • employment security (e.g. will job roles become redundant?) • transform how we live, work and think

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to investigate opportunities for the positive application of cognitive computing	2.1 Investigating cognitive computing opportunities i.e. • purpose (e.g. what does the business want to achieve?) • how will it achieve its purpose? (e.g. tapping into large volumes of unstructured data or data that was previously ignored) • how would cognitive computing help? i.e. ○ rapidly innovate (e.g. cognitive technology is trained not programmed and therefore improves over time) ○ make more money with improved insights (e.g. enables businesses and individuals to make better decisions with more in-depth knowledge) ○ gain a competitive edge (e.g. enabling a business to offer capabilities their competitors cannot)
3. Be able to generate business proposals for an identified application of cognitive computing	3.1 Business proposal i.e. • overview of the business • target audience • problem statement i.e. what are the business needs • proposed solution i.e. detail explanation as to how each of the business needs could be addressed • technology application • risks • benefits • costs • legal and ethical considerations e.g. ○ replacing humans with machines (cognitive computing can mutually support the work of humans e.g. providing information, delivering training) ○ malicious human interaction (protecting the data and protecting the technology) ○ education e.g. ▪ by believing a computer performs a job better that a human can lower attention and standards ▪ care must be taken not lose the human ability to learn ▪ cannot replace the human urge of achieving the unknown ○ health e.g. improved diagnostic accuracy and speed ○ justice e.g. cannot replace the legal system in court cases and the law requires interpretation and knowledge to apply to each individual case 3.2 Minimise risks e.g. • combine humans and machines • use cognitive computing to support professionals e.g. health care, security, customer support, business intelligence • ensure data does not fall into the wrong hands (data protection and security) 3.3 Repurposing (to adapt for use for a different purpose)

GRADING CRITERIA

Learning Outcome (LO)	Pass	Merit	Distinction
	The assessment criteria are the pass requirements for this unit.	To achieve a merit the evidence must show that, in addition to the pass criteria, the candidate is able to:	To achieve a distinction the evidence must show that, in addition to the pass and merit criteria, the candidate is able to:
1. Know how cognitive computing is used in business	P1: Describe the current application of cognitive computing in business **Synoptic with Units 1, 2 and 3*	M1: Describe the drawbacks of an identified cognitive computing application	
2. Be able to investigate opportunities for the positive application of cognitive computing	P2: Research opportunities across business sectors where the application of cognitive computing could improve business activities **Synoptic with Units 1, 2 and 3*		
3. Be able to generate business proposals for an identified application of cognitive computing	P3: Prepare an outline business proposal for the development of a cognitive computing solution in a given sector **Synoptic with Units 1, 2 and 3*	M2: Minimise the social, moral and ethical implications of your proposed solution	D1: Evaluate the potential repurposing of your solution to other business sectors

*SYNOPTIC ASSESSMENT AND LINKS BETWEEN UNITS

When learners are taking an assessment task, or series of tasks, for this unit, they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the grading criteria (shown with an asterisk). Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

Links between this unit and other units

This unit and specific LO	Name of other unit and related LO
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive application of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 1: Fundamentals of IT** LO1: Understand computer hardware LO2: Understand computer software LO3: Understand business IT systems LO4: Understand ethical and operational issues and threats to computer systems
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive application of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 2: Global Information** LO1: Understand where information is held globally and how it is transmitted LO2: Understand the styles, classification and the management of global information LO3: Understand the use of global information and the benefits to individuals and organisations LO4: Understand the legal and regulatory framework governing the storage and use of global information LO5: Understand the principles of information security
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive application of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 3: Cyber Security** LO1: Understand what is meant by cyber security LO2: Understand the issues surrounding cyber security LO3: Understand measures used to protect against cyber security incidents
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive applications of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 5: Virtual and Augmented Reality** LO1: Understand virtual and augmented reality and how they may be used LO4: Be able to predict future applications for virtual or augmented reality
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive applications of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 16: Developing a Smarter Planet** LO1: Understand what is meant by a Smarter Planet LO2: Be able to propose ways to extend the scope of the Smarter Planet LO3: Be able to present, refine and evaluate Smarter Planet concepts
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive applications of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 17: Internet of Everything** LO1: Understand what is meant by the IoE LO2: Be able to repurpose technologies to extend the scope of the IoE LO3: Be able to present concept ideas for repurposed developments

This unit and specific LO	Name of other unit and related LO
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive applications of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 22: Big Data Analytics** LO1: Understand the scope of big data LO2: Be able to process Big Data for business purposes LO3: Be able to provide information resulting from processing Big Data
LO1: Know how cognitive computing is used in business LO2: Be able to investigate opportunities for the positive applications of cognitive computing LO3: Be able to generate business proposals for an identified application of cognitive computing	**Unit 24: Enterprise Computing** LO1: Understand the concept of enterprise computing systems LO2: Be able to investigate business requirements for an enterprise computer solution LO3: Be able to develop enterprise computing solutions to meet business requirements LO4: Be able to review the enterprise computing solution with stakeholders

ASSESSMENT GUIDANCE

LO1 Know how cognitive computing is used in business

P1: Learners must describe the current application of cognitive computing in business. This can be a report, presentation (with detailed speaker notes), a recorded discussion or information guide.

M1: The learners are required to describe the drawbacks of an identified cognitive computing application. It can be an extension of the evidence used for

LO2 Be able to investigate opportunities for the positive application of cognitive computing

P2: Learners must research opportunities across business sectors where the application of cognitive computing could improve business activities. This can be in the form of a report, a visualisation, diagram, illustrations or animation. The evidence should show that the learner has researched opportunities which can be used for the basis for the business case they will propose in LO3 along with the costing of the business benefits.

LO3 Be able to generate business proposals for an identified application of cognitive computing

P3: Learners must prepare an outline business proposal for the development of a cognitive computing solution in a given sector. The proposal can be in the form of a report or a presentation to potential stakeholders. The presentation can be video or the presentation slides with detailed speaker notes.

M2: Learners are required to show how they could minimise the social, moral and ethical implications of the proposed solution for P3. This can be included as part of the report or presentation produced for P3.

D1: Learners are required to evaluate the potential of repurposing their solution from P3 to other business sectors. The evaluation should show that the learner has made a qualitative judgements taking into account different factors and using available knowledge and evidence. This could be presented as a report, or presentation.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

For more information about internal assessment, including feedback, authentication and plagiarism, see the centre handbook. Information about how to reference is in the OCR *Guide to Referencing* available on our website: http://www.ocr.org.uk/i-want-to/skills-guides/.

EMPLOYABILITY SKILLS

Employability Skills	Learning outcome/Assessment criteria
Communication	P1, P2, P3, M1, D1
Problem solving/decision making	P3, D1
Critical thinking	M1, D1
Analytical skills	P2, P3

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Extended Diploma (Tech Level) qualification

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units. This unit is optional but we encourage you to involve employers where possible.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer engagement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work experience or work placements that develop skills and knowledge relevant to the qualification.	Learners could work in a sector where they have the opportunity to analyse how cognitive computing could be implemented into the working environment or if already implemented analyse how it is enhancing business processes and functions etc.
2. Learners undertake a project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Centres could work with industry practitioners to create business scenarios which could be given to the learners to support their evidence for LO3 - Be able to propose potential uses of cognitive computing to support global business needs.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Businesses who use cognitive computing could deliver sessions on how it is used within their organisation, future considerations for further development within this area and the risks and benefits involved.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	If learners are working within an appropriate working environment and are able to collect evidence from the workplace, industry practitioners within the workplace could provide expert witness testimonies to confirm a) the work is that of the learner and b) that their evidence is technically accurate.

You can find further information on employer involvement in the delivery of qualifications in the following documents:

- Employer involvement in the delivery and assessment of vocational qualifications
- DfE work experience guidance

To find out more
ocr.org.uk/IT
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit 24

Enterprise computing

L/615/1131

Guided learning hours: 60

Version 2: September 2016



ocr.org.uk/IT

LEVEL 3

UNIT 24: Enterprise computing

L/615/1131

Guided learning hours: 60

Essential resources required for this unit: None

This unit is internally assessed and externally moderated by OCR.

UNIT AIM

Enterprise computing is sold to business users as an entire solution that can be applied broadly across an organisation and then further customised by users within each business function. This means the analytics, reporting, database management and other applications are standard across the system, whilst the application packages used and the data accessed in each business function will be different. In this sense, enterprise computing is a departure from finding single software solutions to specific business problems, such as inventory or accounting software. Instead, enterprise computing is intended to offer integrated solutions to these problems. Enterprise computing is a concept for software and hardware solutions designed to meet the needs of large global organisations.

This unit will require you to carry out research based on a scenario. Your research will incorporate the knowledge, skills and understanding that you have obtained within your selected pathway i.e. Digital Technician or Application Data Technician. You will be evaluating a global organisation and making recommendations on how enterprise computing could be used to support their business needs. Therefore, it is important that this unit is carried out synoptically when you have secured the relevant knowledge, skills and understanding from other units.

This unit is mandatory in both specialist pathways in the Level 3 Extended Diploma.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades.

Anything which follows an i.e. details what must be taught as part of that area of content. Anything which follows an e.g. is illustrative. It should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, although these do not need to be the same ones specified in the unit content.

For internally assessed units you need to ensure that any assignments you create, or any modifications you make to an assignment, do not expect the learner to do more than they have been taught, but must enable them to access the full range of grades as described in the grading criteria.

Learning outcomes **The Learner will:**	**Teaching content** **Learners must be taught:**
1. Understand the concept of enterprise computing systems	1.1 Enterprise computing i.e. • business-oriented information technology solution • software and hardware solutions designed for global application • highly secure with multi-layered access requirements • easily customised platform • flexible access for user and organisations to IT assets • integrated solutions to business problems 1.2 Requirements and purpose i.e. • high level of reliability and functionality o including built-in redundancies (business can still operate should one system fail) • high level of security i.e. o ability to set carried access profiles for specific multi-users o must include: ▪ authentication ▪ access control ▪ data integrity ▪ data privacy ▪ non-repudiation (mechanism to prove user performance that cannot be denied by the user at a later stage) ▪ auditing • central data storage o collects and organises data from entire organisation regardless of location o access control according to security protocols • high level of scalability and adaptability e.g. ability to add and customise applications as and when required • high level of accessibility (e.g. users are not kept waiting for access) • potential platform and software independence

Learning outcomes The Learner will:	Teaching content Learners must be taught:
2. Be able to investigate business requirements for an enterprise computer solution	2.1 Business requirements i.e. • purpose • intended outcome (to include intended locations of access) • intended users • personalised user experience (e.g. represent business concepts to any user) i.e. • the users i.e. o local o international o global • types of interface e.g. o another application within the same business o interface for business clients o Web interface for customers • an appropriate window for each audience i.e. o without duplication of: ▪ systems ▪ effort to capture the business rules
3. Be able to develop enterprise computing solutions to meet business requirements	3.1 Outline scope to include: • objectives • goals • sub-phases • tasks • resources • budget • schedule • advantages/benefits • disadvantages 3.2 Logical design i.e. • platform independent • inputs • outputs • processes • data Physical design i.e. • platform dependent • hardware, e.g. Intel, AMD, HP, Sun, Apple, IBM • software e.g. operating systems, Java, .Net Framework • implementation environment, e.g. web browser

Learning outcomes The Learner will:	Teaching content Learners must be taught:
	3.3 Business benefits i.e. • operational efficiency (through virtualisation, workload management and larger I/O throughput) • systems' utilisation • greener computing (highly efficient designs deliver highly scalable service from a single system) • reduction in production time (can link to improved customer satisfaction) • increased productivity • reduced operational cost (less staff may be required due to automated systems) • enables growth and supports the competitive market • improved customer satisfaction 3.4 Security characteristics i.e. • **authentication** – the means by which communicating entities prove to one another that they are acting on behalf of specific identities • **access control for sensitive resources** – the means by which interactions with vital resources are limited to collections of users or programs for the purpose of enforcing integrity, confidentiality, or availability constraints • **data integrity** – the means used to confirm that information has not been tampered with by a third party. A recipient of data sent over an open network must be able to check whether the data gets altered on its way • **data privacy** – the means used to ensure that information is made available only to users who are authorised to access it • **non-repudiation** – the means used to prove that a user performed some transaction such that the user cannot deny it later • **auditing** – the means used to capture a tamper-resistant record of security-related events for the purpose of being able to evaluate the effectiveness of security policies and mechanisms
4. Be able to review the enterprise computing solution with stakeholders	4.1 Presentation of solution e.g. • reflects all aspects of design • meets business needs • understood by audience (e.g. level of technical jargon used) • format (e.g. report or presentation supported by diagrams or models) 4.2 Evaluation of design solution (e.g. meets business requirements, achievable, manageable, extendable, secure, customisable, user personalisation)

GRADING CRITERIA

Learning Outcome (LO)	Pass	Merit	Distinction
	The assessment criteria are the Pass requirements for this unit.	To achieve a Merit the evidence must show that, in addition to the Pass criteria, the learner is able to:	To achieve a Distinction the evidence must show that, in addition to the pass and merit criteria, the learner is able to:
1. Understand the concept of enterprise computing systems	P1: Explain the term enterprise computing	M1: Explain the requirements of an enterprise computing solution	
2. Be able to investigate business requirements for an enterprise computer solution	P2: Recommend an enterprise computing solution for business requirements		
3. Be able to develop enterprise computing solutions to meet business requirements	P3: Create an outline scope for a proposed enterprise solution to meet identified business requirements	M2: Justify the business benefits from the proposed solution	D1: Analyse the security characteristics that the proposed solution must include
	P4: Produce a logical or physical design for the scoped enterprise solution		
4. Be able to review the enterprise computing solution with stakeholders	P5: Present the proposed enterprise solution to stakeholders	M3: Refine the proposed solution based on stakeholder feedback	D2: Evaluate the proposed solution against the original specified business requirements

ASSESSMENT GUIDANCE

LO1 Understand the concept of enterprise computing systems

To aid the assessment of this particular learning outcome, it is important that learners have had the opportunity to look at a variety of businesses where enterprise computing has been or could be implemented. This should help them to use examples to support their descriptions and explanations.

P1: Learners must explain the term Enterprise Computing. What does it mean and how does it work?

M1: Learners must explain the requirements of an enterprise computing solution. There are certain requirements that must be met when considering the development and deployment of enterprise computing. It is important that the learners show that they have a clear understanding of these by explaining what they are and their purpose.

LO2 Be able to investigate business requirements for an enterprise computer solution

To aid the assessment of this particular learning outcome, it is important that learners investigate the information provided about the business that is considering enterprise computing.

P2: Learners must recommend an enterprise computing solution for business requirements. They should consider all aspects of a business's requirements. They will then be better informed when providing an overview of how enterprise computing will benefit the business and support its users. It is important that they include all potential users whether they are internal or external to a business.

LO3 Be able to develop enterprise computing solutions to meet business requirements

To aid the assessment of this particular learning outcome, it is important that learners refer back to the investigation that they carried out in LO2.

P3: Learners must create an outline scope for a proposed enterprise solution to meet identified business requirements. Learners should consider whether they want to produce a broad scope with less detail or a narrow scope with greater detail.

P4: Learners must produce a logical or physical design for the scoped enterprise solution. This follows on from P3 and the learners have a choice of whether they produce a logical design or a physical design of the enterprise solution for which they provided an outline scope in P2. Learners should consider in detail all criteria for the proposed design type.

M2: Learners must justify the business benefits from the proposed solution, this can be as an extension to P2 and/or P3. The justification should provide the business with a reasoned case as to how the proposed solution will be of benefit to the business. Learners should consider the full range of benefits.

D1: Learners must analyse the security characteristics that the proposed solution must include. The learners should include all aspects of security and be able to refer to the advantages and disadvantages of each characteristic and make a reasoned comment as to why it is important that they are included or excluded.

LO4 Be able to review the enterprise computing solution with stakeholders

P5: Learners must present the proposed enterprise solution to stakeholders. It could be a formal presentation, and could potentially be videoed, with a copy of the presentation and speaker notes provided. Alternatively, the learner could produce a formal report. Learners should have had plenty of experience of presenting their enterprise solutions from other units where they have presented their designs and/or prototypes etc.

M3: Learners must refine the proposed solution based on stakeholder feedback. The evidence should include feedback provided by the stakeholders. (It should be noted that stakeholders should be people who have a good understanding of business requirements and enterprise computing. Stakeholders should not be other members of the learner's group.) The learner is also required to show how they have refined their proposed solution based on the feedback. The refinements should include the changes made to the logical or physical design that they produced.

D2: Learners must evaluate the proposed solution against the original specified business requirements identified in P2. The evaluation should include qualitative judgments based on the learner taking different factors into account (e.g. is the proposed solution extendable, manageable and achievable as well as the other obvious factors identified by the business). This could be in the format of a report of presentation.

Feedback to learners: you can discuss work-in-progress towards summative assessment with learners to make sure it's being done in a planned and timely manner. It also provides an opportunity for you to check the authenticity of the work. You must intervene if you feel there's a health and safety risk.

Learners should use their own words when producing evidence of their knowledge and understanding. When learners use their own words it reduces the possibility of learners' work being identified as plagiarised. If a learner does use someone else's words and ideas in their work, they must acknowledge it, and this is done through referencing. Just quoting and referencing someone else's work will not show that the learner knows or understands it. It has to be clear in the work how the learner is using the material they have referenced **to inform their** thoughts, ideas or conclusions.

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SYNOPTIC ASSESSMENT AND LINKS BETWEEN UNITS

When learners are taking an assessment task, or series of tasks, for this unit, they will have opportunities to draw on relevant, appropriate knowledge, understanding and skills that they will have developed through other units. We've identified those opportunities in the table below. Learners should be encouraged to consider for themselves which skills/knowledge/understanding are most relevant to apply where we have placed an asterisk.

This unit and specific LO	Name of other unit and related LO
LO1: Understand the concept of enterprise computing systems	Unit 1: Fundamentals of IT – LO1, LO2, LO3, LO5 Unit 2: Global Information – LO1, LO2, LO3, LO4, LO5, LO6 Unit 4: Computer Networks – LO1 Unit 18: Computer Systems – Hardware – LO1 Unit 19: Computer Systems Software – LO1 Unit 20: IT Technical Support – LO1 Unit 22: Big Data Analytics – LO1 Unit 23: Cognitive Computing – LO1
LO2: Be able to investigate business requirements for an enterprise computer solution	Unit 2: Global Information – LO1, LO2, LO3, LO4, LO5, LO5 Unit 3: Cyber Security – LO1, LO2, LO3, LO4 Unit 4: Computer Networks – LO1 Unit 5: Augmented and Virtual Reality – LO1 Unit 6: Application Design – LO1, LO2 Unit 7: Data Analysis and Design – LO1, LO2 Unit 9: Product Development – LO1 Unit 10: Business Computing – LO2 Unit 11: Systems Analysis and Design – LO1, LO2 Unit 12: Mobile Technology – LO1, LO2 Unit 13: Social Media and Digital Marketing – LO1, LO2 Unit 14: Software Engineering for Business – LO2 Unit 15: Games Design and Prototyping – LO1 Unit 16: Developing a Smarter Planet – LO1 Unit 17: Internet of Everything – LO1 Unit 18: Computer Systems – Hardware – LO1 Unit 19 – Computer Systems Software – LO1, LO2 Unit 20 – IT Technical Support – LO1 Unit 21 – Web Design & Prototyping – LO1, LO2 Unit 22: Big Data Analytics – LO1 Unit 23: Cognitive Computing – LO2

This unit and specific LO	Name of other unit and related LO
LO3: Be able to develop enterprise computing solutions to meet business requirements	Unit 4: Computer Networks – LO2, LO4 Unit 5: Augmented and Virtual Reality – LO2 Unit 6: Application Design – LO3 Unit 7: Data Analysis and Design – LO3 Unit 9: Product Development – LO2, LO3 Unit 10: - Business Computing – LO3 Unit 11: Systems Analysis and Design – LO3, LO4 Unit 12: Mobile Technology – LO3 Unit 13: Social Media and Digital Marketing – LO3, LO4 Unit 14: Software Engineering for Business – LO3 Unit 15: Games Design and Prototyping – LO2, LO3 Unit 17: Internet of Everything – LO2 Unit 18: Computer Systems – Hardware – LO1, LO2 Unit 20 – IT Technical Support – LO1 Unit 21 – Web Design & Prototyping – LO1, LO2, LO3 Unit 22: Big Data Analytics – LO2, LO3 Unit 23: Cognitive Computing – LO2, LO3
LO4: Be able to review the enterprise computing solution with stakeholders	Unit 4: Computer Networks – LO3 Unit 5: Augmented and Virtual Reality – LO3, LO4 Unit 6: Application Design – LO4 Unit 7: Data Analysis and Design – LO4 Unit 9: Product Development - LO4 Unit 10: Business Computing – LO4 Unit 11: Systems Analysis and Design – LO4 Unit 12: Mobile Technology – LO4 Unit 14: Software Engineering for Business – LO4 Unit 15: Games Design and Prototyping – LO4 Unit 16: Developing a Smarter Planet – LO2, LO3 Unit 17: Internet of Everything – LO3 Unit 18: Computer Systems – Hardware – LO2 Unit 20 – IT Technical Support – LO1 Unit 21 – Web Design & Prototyping – LO4 Unit 23: Cognitive Computing – LO2, LO3

Synoptic assessment grid

Core unit	Core taught content	Assessment criteria in this unit
Unit 1 – Fundamentals of Computing	LO1 – Understand computer hardware	P1, P2, P3, P4, M1, M2
Unit 1 – Fundamentals of Computing	LO2 – Understand computer software	P1, P2, P3, P4, M1, M2
Unit 1 – Fundamentals of Computing	LO3 – Understand business IT systems	P2, P3, P4, M1, M2, M3
Unit 1 – Fundamentals of Computing	LO4 – Understand employability and communication skills used in an IT environment	P4, P5 P56, M2, M3, D2
Unit 1 – Fundamentals of Computing	LO5 – Understand ethical and operations issues and threats to computer systems	P1, P2, P3, P4, P5, M1, M2, M3, D1
Unit 2 – Global Information	LO1 – Understand where information is held globally and how it is transmitted	P1, M1
Unit 2 – Global Information	LO2 – Understand the styles, classification and the management of global information	P3, P4, P5, M2, M3
Unit 2 – Global Information	LO3 – Understand the use of global information and the benefits to individuals and organisations	P1, P2, M1, M2, M3, D2
Unit 2 – Global Information	LO4 – Understand the legal and regulatory framework governing the storage and use of global information	P1, M1, D1
Unit 2 – Global Information	LO5 – Understand the process flow of information	P2, P3, P4, P5, M2, M3
Unit 2 – Global Information	LO6 – Understand the principles of information security	M1, D1
Unit 3 – Cyber Security	LO1 – Understand what is meant by cyber security	P1, P2, P3, P4, P5, M1, M2, D1
Unit 3 – Cyber Security	LO2 – Understand the issues surrounding cyber security	P1, P2, P3, P4, P5, M1, M2, D1
Unit 3 – Cyber Security	LO3 – Understand measures used to protect against cyber security incidents	P1, P2, P3, P4, P5, M1, M2, D1
Unit 3 – Cyber Security	LO4 – Understand how to manage cyber security incidents	P1, P2, P3, P4, P5, M1, M2, D1
Unit CC* – Cloud Technology	LO1 – Know the characteristics and context of the use of cloud computing	P1, P2, P3, P4, M1, M2, D2
Unit CC* – Cloud Technology	LO2 – Know about cloud services	P1, P2, P3, P4, M1, M2, D2

Core unit	Core taught content	Assessment criteria in this unit
Unit CC* – Cloud Technology	LO3 – Know about the business benefits of the cloud	P5, M2, M3
Unit CC* – Cloud Technology	LO4 – Know what to consider when deploying cloud-based services for an organisation	P1, P2, P3, P4, P5, M1, M3, D2
Unit CC* – Cloud Technology	LO5 – Know regulatory issues that impact cloud computing	P2, P3, P4, P5, M1, M2, D1, D2
Unit CC* – Cloud Technology	LO6 – Know the impact, risks and security issues related to using cloud computing	P2, P3, P4, D1
Unit CC* – Cloud Technology	Know about cloud storage	P1, P2, P3, P4, P5, M1, M2, M3, D1, D2

*Formerly Unit 25

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Extended Diploma (Tech Level) qualifications

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory units (this unit is a mandatory unit in the Digital Technician and Application Data Technician pathways), although we encourage you to find ways to engage with employers for other units as well.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information including a list of activities that are not considered to meet this requirement.

Meaningful employer engagement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work experience or work placements that develop skills and knowledge relevant to the qualification.	
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Students could research a number of scenarios where enterprise computing could be implemented. Industry practitioners who already work in this area of IT would be able to provide support on the scope of the scenarios.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Students could have teaching sessions (specifically for LO1) about the meaning of enterprise computing, how it works and what is required.
4. Industry practitioners operate as 'expert witnesses' and contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Students could present their enterprise solutions (for LO4) to industry specialists and receive valuable feedback on their solutions.

You can find further information on employer involvement in the delivery of qualifications in the following documents:

- Employer involvement in the delivery and assessment of vocational qualifications
- DfE work experience guidance

To find out more
ocr.org.uk/IT
or call our Customer Contact Centre on **02476 851509**

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Cambridge **TECHNICALS LEVEL 3**

Cambridge TECHNICALS 2016

IT

Unit CC*

(*Formerly Unit 25)

Cloud technology

R/615/1132

Guided learning hours: 90

Version 2: September 2016

ocr.org.uk/IT

LEVEL 3

UNIT CC*: Cloud technology
***Formerly known as Unit 25**

R/615/1132

Guided learning hours: 90

Essential resources required for this unit:
Access to cloud-based services:

- IaaS/PaaS: Microsoft Azure, Amazon Web Services (AWS) and Google Compute
- SaaS: Microsoft Office 365 (admin portal), Google Apps, Microsoft Office Online
- Mobile Device Management: Microsoft Intune
- Cloud storage: OneDrive, Box.com, Google Drive

Most Cloud Technology providers have a free trial period or can provide extended educational trials.

This unit is externally assessed by an OCR- set and marked examination.

UNIT AIM

The way in which we interact with IT systems has changed. The explosion of mobile devices such as phones and tablets has led to the need for an IT infrastructure that can support these technologies, i.e. the cloud.

In this unit you will learn the basic concepts of cloud technology as it exists in an international setting. You will know the specific terminology and its application in the continued shift into the cloud, where costs are billed like electricity and reflect monthly usage levels rather than the traditional upfront cost of new servers and storage for a data centre. Migration into the cloud also poses issues for business culture, legal requirements and security.

This unit is mandatory in both specialist pathways in the Level 3 Extended Diploma. This unit is integral to any unit where you are expected to demonstrate your knowledge and understanding of cloud technologies and how this infrastructure supports IT-related activities.

TEACHING CONTENT

The teaching content in every unit states what has to be taught to ensure that learners are able to access the highest grades. Anything that follows an i.e. details what must be taught as part of that area of content. Anything that follows an e.g. is illustrative.

For externally assessed units, where the content contains i.e. and e.g. under specific areas of content, the following rules will be adhered to when we set questions for an exam:

- a direct question may be asked about unit content that follows an i.e.
- where unit content follows an e.g. a direct question will not be asked about that example.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
1. Understand the characteristics and context of cloud technology and why it is used	1.1 Cloud technology characteristics and terms, i.e. • Elasticity • Ubiquitous access • Rapid deployment • On-demand self-service • Resource pooling • Pay-as-you-grow • Multi-tenancy • Automation • Chargeback • Cloud bursting • Minimal management effort	Learners need to have an understanding of the terms used within cloud technology and the characteristics associated with them. An example would be the term 'elasticity'; characteristics of elasticity could include: • adaptable to changing workloads • provisions and de-provisions of resources – available resources match the demand as closely as possible.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	1.2 Cloud models, i.e. • Private • Public • Hybrid • Community Cloud models functionality, i.e. • A private cloud is used by a specific organisation; can be hosted at the organisation's premises or at a remote location (or even a third party) • Public clouds are used by multiple organisations and are typically hosted by a particular cloud vendor • Hybrid cloud is where an organisation utilises certain services on the public cloud, but has a private cloud for sensitive services/data • Community clouds are designed for organisations working in a particular industry (e.g. financial) in order to take advantage of services that would benefit them all (e.g. credit checking) Location of each cloud model, i.e. • On-premises • Off-premises Technical differences between private and public types of clouds, i.e. • Private cloud requires in-house infrastructure including hardware, software and networking components; a suite of software known as a 'Stack' would be required to operate it • Public cloud would just require a good internet connection and devices to access the service(s)	Learners need to be able to identify and understand the different cloud models currently available. Learners may find it beneficial to use diagrams to illustrate the different models that demonstrate who has access to them. For each of the cloud models, learners need to understand how it functions and which types of organisation would utilise it. They would also need to consider why different organisations would use specific cloud models. It is also important that learners understand where each of the cloud models are located, i.e. on or off the premises of the user. Technical differences between private and public cloud types is also an area that learners need to understand, including how they are structured and whether or not there are any considerations that need to be given. The combination of the above areas will ensure that the learners have a good understanding of the characteristics and context of cloud technology and why it is used.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	1.3 Types of organisation that would benefit from cloud technology, i.e. • Start-ups with little capital • Small businesses • Organisations with varying levels of workforce, e.g. the company may take on more employees at peak times of the year (such as Christmas) to handle the increased workload • Mobile/global workforce • Online retail • Marketing agency Reasons, i.e. • Lack of capital • Need to be flexible as business growth is uncertain • The work that the organisation does would be seasonal, hence the varying user population • The organisation's workforce would be spread globally and need to collaborate on joint projects • The organisation was 'born online' and thus has no fixed premises on which to host IT infrastructures 1.4 Types of organisation that may not benefit from cloud technology, i.e. • Military • Intelligence services • Health service • Research and development facility Reasons, i.e. • Security of information • Sensitivity of information	Learners need to be able to consider the benefits to different organisations of using cloud technology. In order to do this, they need to consider the different types of organisation that could embrace cloud technology and the reasons that they would use it. These reasons will link to the possible benefits felt by different organisation types. It is important that learners also understand that there are some organisations that would not benefit from using cloud technology, and this is usually based on the issues surrounding the security and senility of information should a breach occur. Learners should consider the different organisational types and be able to provide valid reasons for cloud technology not to be used. This could link with the cyber

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	• Potential disclosure of customer data • Lack of willingness to change • Internal processes do not align with cloud services • Perceived lack of technical knowhow • Already have considerable investment in own traditional IT infrastructure 1.5 Organisational roles and responsibilities when using cloud technology, i.e. • Capacity planners (responsible for costing the resource allocations with the cloud service assessments in order to apply chargeback costs to businesses) • Network operation centre staff (responsible for monitoring and managing cloud resources) • Vendor management staff (negotiate service-level contracts with the cloud vendor, update and extend them to meet changes over time) • Support desk staff (aid users during transfer from traditional IT to cloud services, provide client support and manage incidents and problems) • Cloud architect (expert responsible for the transition of organisations from using traditional IT to cloud technology, highly technical across several areas, e.g. networking, virtualisation, sound understanding of legislation requirements) • Cloud service manager (involved in service retirement, renewal, ordering	security unit where they consider what the issues could be. Learners should be able to identify the organisational roles involved in using cloud services, as well as explain the responsibilities of each role.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	and requesting procedures as well as tracking total cost of cloud ownership)	
2. Understand the business benefits of cloud services	2.1 Similarities and differences between cloud technology and outsourcing, i.e. • Ownership • Usage • Service levels • Costs	Learners need to understand that utilising cloud technology in some ways is similar to outsourcing your IT service. You don't own the servers, platforms or even the technicians that operate it. However, with outsourcing you don't have the flexibility that cloud technology brings and would be paying a fixed monthly cost, whereas with cloud technology you have a 'pay as you grow' model. Learners need to be able to compare cloud technology with outsourcing and consider the benefits and limitations of each.
	2.2 Business considerations when implementing cloud services, i.e. • Scalability • Security • Hardware independence • Variable cost • Time to market • Distribution over the internet (ease of access)	Learners need to understand that businesses must consider a number of factors before implementing cloud services. They need to understand why businesses are obliged to make these considerations.
	2.3 How cloud technology enhances business value, i.e. • Upfront (capital) expenditure is minimised • Variable (operating) cost is predictable • Reduces integrity risks – data backup is available • Ability to monitor access and data usage • Ease of providing access to users and partners to data in the cloud • Redundancy links that will allow for increased	Learners need to understand how cloud technology can reduce the pain of using IT for business, allowing them to concentrate on using IT effectively to enhance their business and meet their customer expectations.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	availability 2.4 Business benefits of service models, i.e. • Improvement of speed of internal development and deployment (developers do not need to worry about the platform or infrastructure prior to development) • Reduce reliance on internal technical skills (deployment is on demand and does not require consideration being given to procurement of hardware/software and installation/commissioning) • Simplified interface for user groups • Service designed around the needs of users (self-dash service models enable user experience provision and used to be considered when determining which platform to adopt)	Learners need to understand the benefits of different service models to businesses. They need to be able to identify the different benefits and explain how these are advantageous to businesses.
3. Understand the requirements of cloud services	3.1 Cloud service models, i.e. • Software-as-a-Service (SaaS) (e.g. Microsoft Office 365, Google Apps for Work/Google Docs, Zoho, Microsoft Dynamics CRM) • Platform-as-a-Service (PaaS) (e.g. Google App Engine, Force.com, Microsoft Azure) • Infrastructure-as-a-Service (IaaS) (e.g. Microsoft Azure, Amazon Web Services, Google Compute) • Communication-as-a-Service (CaaS) (e.g. Microsoft Skype, Google Voice) • Anything (or everything)-as-a-Service (Xaas)	Note: some cloud platforms such as Microsoft Azure work across the different service models. New platforms are emerging all the time so use the ones that are current.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	3.2 Accountabilities and responsibilities based on cloud service model, i.e. • IaaS – Organisation is responsible for virtual server creation, deploying updates and managing the cloud infrastructure. Cloud vendor is responsible for underlying hardware and physical datacentre infrastructure. From this the organisation can create their applications and deploy related services. • PaaS – The cloud vendor provides the platform to develop on and the organisation is responsible for developing their applications and deploying their services. • SaaS – The cloud vendor provides the applications and services that the organisation consumes as is. 3.3 Organisations' internal skills requirements to support their cloud model (SaaS, PaaS, IaaS), i.e. • Technical Skills • Project management skills • Vendor management skills • Data integration skills • Business and financial skills • Security and compliance skills	This builds on 3.1 where learners focus on what the cloud provider is responsible for and what the business is responsible for, e.g. updates to cloud-based servers within an IaaS environment could be the responsibility of the business (dependent on the service agreement). In a SaaS model the cloud vendor takes care of hardware and operating system maintenance. It is important that learners can explain the responsibilities and accountabilities for each service model. Learners need to be able to identify and describe the internal organisational skills required to support the selected cloud model. Examples are as follows: Technical skills are required to move away from 'installing' to 'provisioning'. This means that an initial set of information is provided (e.g. drive space, memory, credentials and OS) from which the service is created (e.g. virtual server). Needed with all cloud management/support roles, however, the cloud architect is the overall expert. Project management skills are required to deliver the

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
		transition from using traditional IT to using cloud services. These are skills that the cloud architect, the cloud service manager and the capacity planner should all have. Vendor management skills are required for the understanding of SLAs and negotiation contracts. These are skills that the cloud service manager and any vendor management staff should have. Data integration requires high level skills to take data from disparate sources and consolidate them into one database (e.g. big data). These are skills that the cloud architect should have. Business and financial skills are the lifeblood of how to use the cloud, as there are always on-going and potentially increasing costs as the business grows. These are skills that capacity planners, vendor management and cloud service management staff should all have. Security and compliance skills involve ensuring that the cloud service and the way in which the organisation interacts with it does not compromise data security in terms of any legal or compliance regulations. At various levels, these are skills that all staff should have.
	3.4 Technology that make cloud services possible, i.e. • Hardware (required to make the cloud operate), i.e. o Servers	Learners should understand that the cloud does not operate in isolation and there is underlying technology that powers the various cloud services. Learners need to be able to identify the technology required and explain

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	o Components, e.g. ▪ Hard drive/solid state drive (SSD) ▪ Central processing unit (CPU) ▪ Random access memory (RAM) o Storage, e.g. ▪ Network attached storage (NAS) ▪ Storage area network (SAN) ▪ Just a bunch of disks (JBOD) ▪ Redundant array of independent disks (RAID) • Networking (allows cloud availability), i.e. o Devices, e.g. ▪ Switch ▪ Router o Protocols, e.g. ▪ Internet Protocol Suite (TCP/IP) o Services, e.g. ▪ Domain Name System (DNS) ▪ Dynamic Host Configuration Protocol (DHCP) o Availability, e.g. ▪ Bandwidth ▪ Internet access ▪ Quality of service (QoS) • Software (drives the cloud enabling users to access the service), i.e. o Operating systems (e.g. Linux, Windows) o Databases (e.g. MySQL) o Applications (e.g. web browser, word processing, graphic design, customer relationship manager (CRM)) o Authentication, e.g.	how they support cloud operation. The cloud operates across multiple geo-located datacentres. These need to be powered appropriately to keep the cloud running and cooled to ensure that they don't overheat. Datacentre power distribution units and elements such as hot and cold aisles should be mentioned.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	▪ User account credentials ▪ Multifactor authentication (MFA) • Virtualisation (the building block of cloud services enabling scale and growth), i.e. o Hypervisor o Virtual machines (VM) o Network virtualisation • Orchestration (ensures continuation of service), i.e. o Managing the starting and stopping of server clusters • Datacentre components, i.e. o Power (e.g. hot and cold aisles) o Cooling	
4. Understand the features of cloud storage	4.1 Cloud storage services, i.e. • Consumer storage service options, i.e. o OneDrive o Dropbox o Google Drive • Corporate storage services, i.e. o One Drive for Business o Box.com Synchronisation/upload tools, i.e. • Desktop downloads • Dedicated app on mobile device Benefits, i.e. • Ability to roll back to previous version • Track editing/access	Learners need to understand cloud storage services, and that they should only store company data into the service provided for them. Learners need to be able to identify the different storage services and who would use them, as well as how data is synchronised/uploaded and the data available. They should also be able to describe the benefits of cloud storage. Cloud storage solutions have a background running service that enables data on your device to be automatically synchronised to the cloud. If synchronisation is interrupted, it can cause a failure of upload. This requires troubleshooting. Tools are built into the cloud storage service such as 'version history', which allows you to track who made edits to the file and enables you to roll back to a previous version if needed.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	4.2 Integration of SaaS with storage, i.e. • Cloud vendor storage services are often integrated with their SaaS productivity suite, which allows the file to be edited via the browser version or the desktop/mobile application • If the provider does not have its own productivity suite, it may have an add-on that allows you to use another vendor's SaaS application	Learners need to understand how SaaS integrates with storage (e.g. the ability to edit files on Google Drive with Google Apps, or how Box.com integrates with Microsoft Office Online and with an add-on can integrate with desktop applications too).
	4.3 Permission levels, i.e. • None (private) • Public • Shared o Edit o View	It is important that learners understand that permission and access levels must be assigned to users on a cloud service. Learners should consider the different permission/access levels and in what circumstances they would be used.
	4.4 Data deduplication to reduce capacity usage, e.g. • A lot of data on cloud services is replicated, such as emails that have been forwarded to several people in an organisation; data deduplication enables the removal of duplicated files and thus decreases storage space needed	Learners need to know that data duplication can reduce storage capacity, and this needs to be considered as part of cloud storage use. Learners also need to consider the different situations that can cause duplication of data.
	4.5 Migrating/integrating data hosted in-house to a cloud-based service, i.e. • Software tools o Command line utilities (requires expertise) (e.g. AzCopy) o Command line interface (CLI) (can be automated and highly repeatable) (e.g. PowerShell cmdlets/scripts)	Learners need to know that there are vendor and third party software tools that can be used to upload data into the cloud services. Learners need to be able to decide which approach is best for different organisations and their associated scenarios. Learners would benefit from researching the different methods available and comparing the advantages and

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	○ File transfer protocol (FTP) ○ Graphical utilities (e.g. Storage explorer) ○ Cloud portal site (can be cumbersome and is a manual task) (e.g. Azure portal page) • Send hard drive/data physically to the cloud vendor's local data centre (used when an organisation wishes to absolve themselves of the responsibility of uploading their own data): ○ Azure import export service ○ AWS import/export snowball	disadvantages of each.
5. Understand the deployment requirements for cloud-based services for organisations	5.1 Considerations for adopting cloud services, i.e. • Whether or not the service model is appropriate for the organisation's needs • Whether or not the vendor's cloud deployment model meets the organisation's needs and regulatory requirements • Whether or not change in security from organisational to cloud data centre hosting has any impact on legal or regulatory requirements • Whether or not the cloud vendor has an established track record in providing the service with a guaranteed uptime • Can the cloud service scale to meet the organisation's future expansion plans • Whether or not the cloud vendor has sufficient redundancy • Can the performance metrics of the cloud service be monitored by the business • Will the cloud service be affordable during both peak and off-peak hours	Learners need to understand when adopting cloud services a number of considerations should be taken into account. This is particularly important where an external provider is being used. They need to understand that moving to the cloud changes the way a business operates and brings with it an extra layer of complexity related to security, compliance, and availability. Within the considerations, an exit plan should also be discussed as a way to back out if the cloud vendor does not meet its side of the contractual obligations.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	• Whether or not the cloud vendor's systems are protected in a secure and stable facility • Whether or not the organisation's infrastructure is suitable to allow reliable access to cloud-based services • Whether or not there is a sound exit plan if the cloud vendor fails to uphold the contract 5.2 Cloud readiness assessment, i.e. • Set up a pilot programme • Identify cloud service opportunities based on business needs • Review results from pilot • Modify any technical and financial elements to enable adoption 5.3 Role of IT department when choosing cloud services, i.e. • Advise on application lifecycle • Help to understand performance metrics • Help to understand the provider's quality of service	Learners need to understand what is meant by 'cloud readiness' and its importance prior to going all-in to a cloud service. They need to know the process to follow and what should be considered at each stage, and also how an organisation may have to react to the outcome of the assessment. It is useful for them to understand that applications that are easy to migrate and have some business value are the most suitable for a cloud technology pilot. Following the pilot, the organisation may have to change their financial planning model, as the focus will be on operating expenditure. IT will also transform, as they would be concerned more with supporting and managing the service rather than dealing with maintenance. Learners need to understand that the most significant change associated with cloud is that IT becomes a consultant to the business to help it make the right decisions when taking up the cloud service.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
		This would involve the application lifecycle; an example could be that Office is on a 3-year refresh cycle, so the organisation after 3 years will be forced to upgrade to the latest version. The cloud service brings its own jargon when talking about availability and performance. IT can decipher this information and help the business to understand how well the cloud service will perform in their organisation. Related to performance metrics, the quality of service (QoS) that the cloud provider is able to deliver can also be demystified by the IT department.
	5.4 Service Level Agreements (SLA) A breakdown of services provided and excluded, and it should include: • Costs for services • Duration of the agreement • Responsibilities of the customer and the service provider • Availability and performance requirements • Service monitoring and reporting • Remediation and liability for service disruption • Dispute resolution procedures • A mechanism for reviewing and updating the SLA, including a change control process Cloud SLA should also include: • Data location • Service multitenancy	Learners need to understand that SLAs are basically contracts between the provider and the subscriber. As well as knowing what should be included in SLAs, they should also understand the following: • ICO guidance on usage of cloud services states that the written terms and conditions need to be obtained by the subscriber. This helps to ensure that both sides know and understand their responsibilities. • With the cloud, the SLA needs to be further enhanced to focus on issues such as location of data, which could have consequences for the subscriber if data is transmitted outside of the EEA.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	• Transparency (data breach notifications) • Disaster process recovery notification • Legal data release notification • Data ownership	
6. Know regulatory issues that impact cloud technology	6.1 Data transference, i.e. • Data Protection Act and its bearing on cloud technology • Transfer of data within the European Economic Area (EEA) • Transfer of data to the United States • Transfer of data to other parts of the world • Subject to local regulations • Privacy and Electronic Communications Regulations • Consequences of a data breach	It is important that learners know that the transfer of data must comply with regulatory requirements, and how these can impact on cloud technology. Learners need to realise that depending on where the data is being transferred to, the regulatory requirements within different countries must also be considered. The following are examples of types of regulatory requirements: • The Data Protection Act dictates that personal data 'shall not be transferred to any country or territory outside the European Economic Area (EEA) unless that country or territory ensures an adequate level of protection for the rights and freedoms of data subjects in relation to the processing of personal data.' • Transfer of data to the US was previously under the self-regulated Safe Harbour scheme, which was invalidated by the Court of Justice of the European Union (CJEU) in October 2015. A new EU-US privacy shield framework is due to be released. In the interim period, the ICO is allowing the use of Safe Harbour. • Local regulations: If the data is transferred to a

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
		cloud vendor's US data centre, then it could be subject to the US Patriot Act.
	6.2 Information Security i.e. • Computer Misuse Act • Official Secrets Act	Learners need to know about the legislation relating to information security and how this can impact on how information is handled via cloud services. For example, under the Computer Misuse Act, information should not be tampered with by cloud services and any changes that are not configurable from the interface should be flagged up to the cloud vendor. Another example of legislation is the Official Secrets Act, which focuses on stopping the disclosure of sensitive data (e.g. warship designs, location of military installations). A cloud-based system would need to ensure that these secrets are not leaked deliberately or accidentally.
7. Know about impact, risks and security issues related to cloud technology	7.1 Impact of cloud adoption on business processes, i.e. • Financial model transition from capital expenditure (Capex) to operating expenditure (Opex) • Data ownership • Skills required within the organisation (potential for employee redundancies) • Responsibilities when outages are experienced • Protection against external exposure • Employee access and storage of company data on personal devices, including use of MDM (Mobile Device Management) • Managing multiple cloud service providers, including	Learners need to understand that business processes must be adapted for the cloud. Learners need to know that with the cloud service taking away the physical infrastructure of servers, there is a potential for the 'middle tier' of IT to be made redundant. There will be a need for frontline support staff and the cloud architecture but the server administrator role will continue to diminish. Learners need to be aware that bring your own device (BYOD) is a concern in terms of how data on a cloud service could lead to a data breach. Learners would benefit from having knowledge of Mobile Device Management (MDM) solutions such as Microsoft Intune,

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	the use of SSO (Single-Sign-On) • Managing users and their rights across services • Privacy 7.2 Performance of the service, i.e. • Network latency • Available bandwidth • User access method (i.e. wireless) 7.3 Risks, i.e. • Customer data being exposed • Data integrity (e.g. data that is transmitted down a channel can sometimes be corrupted; this could be due to incomplete transmission, by accident or maliciously) • Lock in with a cloud provider (e.g. there could be increased reliance and if the vendor increases prices the organisation may feel obliged to pay it as they don't think they can leave) • Providers going out of business • Data ownership (e.g. it needs to be made clear in the SLA that the data, although hosted by a cloud vendor, belongs entirely to the organisation)	which allows the management of company data on users' personal devices. Learners need to know that if the use of several cloud-based services from different vendors is taken up, this leads to a sprawl of user identifies and necessitates tracking of what they have access to. Learners would benefit from knowing about Single-Sign-On (SSO) and how this could aid an organisation. It is important that learners know that the performance of the service should be regularly monitored if service degradation is experienced. This could be due to network links on the subscriber side or the provider side. It is therefore important to check the latency, bandwidth and access method on the subscriber side prior to contacting the provider. Learners need to understand that with cloud services offering ubiquitous access (anytime, anyplace, anywhere), the points of exposure are increased. It is important for learners to consider the risks associated with cloud services and the impact that these may have on organisations.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	• Service Level Agreements • Interfaces change (e.g. cloud services are in constant flux so there is a risk that the interface could change on a daily basis; having a clear road map from the vendor and a transitional phase (both are available simultaneously) between the interfaces can help to alleviate this issue • Availability (unforeseen downtime) • Lack of internet connection • Insider threats (e.g. disgruntled or opportunistic employees (that may be leaving the organisation) who can access and download data from the cloud service from anywhere) • External threats (i.e. hacking) • Accidental exposure via multitenancy (e.g. hacking or data leaks via multitenancy issues) • Shadow IT (e.g. departments within a business bypass their IT department and use cloud-based services instead) 7.4 Security, i.e. • Data encryption (in transit/in rest) • User account management • User permissions management • User devices/points of access • Validating security of provider • Multiple data access points (i.e. mobile devices) • User accessing organisational services using public places/public Wi-Fi • Shoulder surfing when users are accessing data from the cloud	This is about cloud security, although it will overlap with general cyber-security. The emphasis here must be on security in relation to the cloud as opposed to general security of the system.

Learning outcomes The learner will:	Teaching content Learners must be taught:	Exemplification
	Security differences between cloud models (public, private, hybrid, community), i.e. • Multitenancy issues • Data segregation • Network isolation • Laws and regulations 7.5 Incident management principles, i.e. • First response • Identify • Report to responsible person • Data/device preservation • Documentation	Learners need to understand that these are important issues and they have an impact on ensuring that data from one cloud tenant doesn't leak into another tenant's service. Learners need to know that if an incident occurs with the cloud service, it is most likely that the support desk staff will become aware first. They need to follow an established set of steps to gather information and ensure that the responsible person is aware. Additionally, any data or device involved needs to be isolated and preserved. Documentation needs to be kept and updated to track the incident from first contact to resolution.

LEARNING OUTCOME (LO) WEIGHTINGS

Each Learning Outcome in this unit has been given a percentage weighting. This reflects the size and demand of the content you need to cover and its contribution to the overall understanding of this unit. See table below:

LO1	17-21%
LO2	14-17%
LO3	24-30%
LO4	8-10%
LO5	13-16%
LO6	3-4%
LO7	4-5%

ASSESSMENT GUIDANCE

All Learning Outcomes are assessed through externally set written examination papers, worth a maximum of 70 marks and 1 hour 30 minutes in duration.

This is not a knowledge-based assessment where learners are only expected to recall information. There is an expectation that learners will have a thorough understanding of the unit content and how it is applied throughout industry practice across all sectors and sizes of organisations. Unlike many other multiple choice models, there will be questions in the summative assessment that require the demonstration of knowledge as well as the application of understanding in a variety of familiar and unfamiliar contexts. It is therefore important that your delivery model for the unit supports the level of knowledge and understanding required from the learners, as well as the different styles of multiple choice questions they will need to answer.

SYNOPTIC ASSESSMENT

For the assessment of this unit, learners will be required to draw on knowledge and understanding from units from across the complete range of units in the pathway that they have followed. The knowledge and understanding required to be drawn from other units is listed below.

Links between this unit and other units

This unit and specific LO	Name of other unit and related LO
LO1: Understand the characteristics and context of cloud technology and why it is used	*Unit 4 Computer Networks* LO1 Understand the concept of networks LO2 Be able to plan computer networks to meet client needs LO3 Be able to present network solutions to clients *Unit 5 Virtual and Augmented Reality* LO1 Understand virtual and augmented reality and how they may be used *Unit 6 Application Design* LO1 Understand how applications are designed *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 17 Internet of Everything* LO1 Understand what is meant by the Internet of Everything (IoE) *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 20 IT Technical Support* LO1 Understand the role of technical support *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO2: Understand the business benefits of cloud services	*Unit 4 Computer Networks* LO1 Understand the concept of networks LO2 Be able to plan computer networks to meet client needs LO3 Be able to present network solutions to clients *Unit 5 Virtual and Augmented Reality* LO1 Understand virtual and augmented reality and how they may be used *Unit 6 Application Design* LO1 Understand how applications are designed LO2 Be able to investigate potential solutions for application developments *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 17 Internet of Everything* LO1 Understand what is meant by the Internet of Everything (IoE) *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 20 IT Technical Support* LO1 Understand the role of technical support *Unit 21 Big Data Analytics* LO1 Understand the scope of Big Data *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO3: Understand the requirements of cloud services	*Unit 4 Computer Networks* LO1 Understand the concept of networks LO2 Be able to plan computer networks to meet client needs LO3 Be able to present network solutions to clients *Unit 6 Application Design* LO1 Understand how applications are designed LO2 Be able to investigate potential solutions for application developments *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 17 Internet of Everything* LO1 Understand what is meant by the Internet of Everything (IoE) *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO4: Understand the features of cloud storage	*Unit 7 Data Analysis and Design* LO1 Understand the purpose and stages of data analysis and *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 10 Business Computing* LO2 Be able to capture and store data for analysis *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 17 Internet of Everything* LO1 Understand what is meant by the Internet of Everything (IoE) *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 21 Big Data Analytics* LO1 Understand the scope of Big Data *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO5: Understand the deployment requirements for cloud-based services for organisations	*Unit 7 Data Analysis and Design* LO4 Be able to present data analysis design solutions to stakeholders *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 10 Business Computing* LO2 Be able to capture and store data for analysis *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 17 Internet of Everything* LO1 Understand what is meant by the Internet of Everything (IoE) *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations sand their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO6: Know regulatory issues that impact on cloud technology	*Unit 7 Data Analysis and Design* LO4 Be able to present data analysis design solutions to stakeholders *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 10 Business Computing* LO4 Be able to present data analysis outcomes *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

This unit and specific LO	Name of other unit and related LO
LO7: Know about impact, risks and security issues related to cloud technology	*Unit 7 Data Analysis and Design* LO4 Be able to present data analysis design solutions to stakeholders *Unit 9 Product Development* LO2 Be able to design products that meet identified client requirements *Unit 10 Business Computing* LO4 Be able to present data analysis outcomes *Unit 11 Systems Analysis and Design* LO4 Be able to create logical and physical designs for specified business systems *Unit 12 Mobile Technology* LO2 Be able to determine solutions for the use of mobile technologies *Unit 16 Developing a Smarter Planet* LO1 Understand what is meant by a Smarter Planet LO2 Be able to propose ways to extend the scope of the Smarter Planet *Unit 18 Computer Systems – Hardware* LO2 Be able to propose a computer system for identified business requirements *Unit 19 Computer Systems – Software* LO1 Understand different software installations and their purpose LO2 Be able to implement software installations and upgrades to meet specified user requirements *Unit 24 Enterprise Computing* LO1 Understand the concept of enterprise computing systems LO2 Be able to investigate business requirements for an enterprise computer solution LO3 Be able to develop enterprise computing solutions to meet business requirements

Synoptic assessment grid

Core unit	Core taught content	Assessment criteria in this unit
Unit 1 Fundamentals of IT	LO3 Understand business IT systems	LO1 Understand the characteristics and context of cloud technology and why it is used LO2 Understand the business benefits of cloud services LO3 Understand the requirements of cloud services LO4 Understand the features of cloud storage LO5 Understand the deployment requirements for cloud-based services for organisations
Unit 1 Fundamentals of IT	LO5 Understand ethical and operational issues and threats to computer systems	LO5 Understand the deployment requirements for cloud-based services for organisations LO6 Know the regulatory issues that impact on cloud technology LO7 Know about impact, risks and security measures related to cloud technology
Unit 2 Global Information	LO1 Understand where information is held globally and how it is transmitted	LO1 Understand the characteristics and context of cloud technology and why it is used LO2 Understand the business benefits of cloud services LO3 Understand the requirements of cloud services LO4 Understand the features of cloud storage
Unit 2 Global Information	LO3 Understand the use of global information and the benefits to individuals and organisations	LO1 Understand the characteristics and context of cloud technology and why it is used LO2 Understand the business benefits of cloud services LO3 Understand the requirements of cloud services LO4 Understand the features of cloud storage
Unit 2 Global Information	LO4 Understand the legal and regulatory framework governing the storage and use of global information	LO5 Understand the deployment requirements for cloud-based services for organisations LO6 Know the regulatory issues that impact on cloud technology LO7 Know about impact, risks and security measures related to cloud technology
Unit 3 Cyber Security	LO1 Understand what is meant by cyber security LO2 Understand the issues surrounding cyber security LO3 Understand measures used to protect against cyber security incidents	LO5 Understand the deployment requirements for cloud-based services for organisations LO6 Know the regulatory issues that impact on cloud technology LO7 Know about impact, risks and security measures related to cloud technology

MEANINGFUL EMPLOYER INVOLVEMENT - a requirement for the Extended Diploma (Tech Level) qualification

The 'Diploma' qualifications have been designed to be recognised as Tech Levels in performance tables in England. It is a requirement of these qualifications for centres to secure for every learner employer involvement through delivery and/or assessment of these qualifications.

The minimum amount of employer involvement must relate to at least one or more of the elements of the mandatory content (this unit is a mandatory unit in the Digital Technician and Application Data Technician pathways), although we encourage you to find ways in which to engage with employers for other units as well.

Eligible activities and suggestions/ideas that may help you in securing meaningful employer involvement for this unit are given in the table below.

Please refer to the *Qualification Handbook* for further information, including a list of activities that are not considered to meet this requirement.

Meaningful employer engagement	Suggestion/ideas for centres when delivering this unit
1. Learners undertake structured work experience or work placements that develop skills and knowledge relevant to the qualification.	Learners could be on work placement with an organisation that uses cloud services. This could be from a technical or an application point of view.
2. Learners undertake project(s), exercises(s) and/or assessments/examination(s) set with input from industry practitioner(s).	Centres could have support from industry practitioners in the development of learners' knowledge of how cloud services are deployed in industry.
3. Learners take one or more units delivered or co-delivered by an industry practitioner(s). This could take the form of master classes or guest lectures.	Industry practitioners could be invited to aspects of the unit content.
4. Industry practitioners operating as 'expert witnesses' that contribute to the assessment of a learner's work or practice, operating within a specified assessment framework. This may be a specific project(s), exercise(s) or examination(s), or all assessments for a qualification.	Not applicable for an examined unit unless incorporated into a project delivery approach.

You can find further information on employer involvement in the delivery of qualifications in the following documents:

- Employer involvement in the delivery and assessment of vocational qualifications
- DfE work experience guidance.

To find out more
ocr.org.uk/IT
or call our Customer Contact Centre on **02476 851509**

Alternatively, you can email us on **vocational.qualifications@ocr.org.uk**





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